K-FoldBERT_Multiclassification

May 3, 2026

[2]:
```python
# ============================================================
# BERT (base-uncased) - Streamlined Pipeline
# 1. Split: Train (85%) + Test (15%) ← test frozen
# 2. 5-Fold CV on Train (85%) for performance estimate
# 3. Final retrain on full oversampled Train (85%)
# 4. Evaluate once on Test (15%)
# ============================================================

import os
import random
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

import torch
from torch.utils.data import Dataset, DataLoader
from torch.optim import AdamW

from sklearn.model_selection import train_test_split, StratifiedKFold
from sklearn.metrics import classification_report, confusion_matrix
from imblearn.over_sampling import RandomOverSampler

from transformers import (
    BertTokenizerFast,
    BertForSequenceClassification,
    get_linear_schedule_with_warmup
)

# ============================================================
# Reproducibility
# ============================================================

SEED = 42

random.seed(SEED)
np.random.seed(SEED)
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torch.manual_seed(SEED)

if torch.cuda.is_available():
    torch.cuda.manual_seed_all(SEED)

# ============================================================
# Device
# ============================================================

if torch.cuda.is_available():
    device = torch.device("cuda")
elif torch.backends.mps.is_available() and torch.backends.mps.is_built():
    device = torch.device("mps")
else:
    device = torch.device("cpu")

print(f"Using device: {device}")

# ============================================================
# Load Dataset
# ============================================================

dataset_path = 'annotated_dataset_for_multi_classifications.csv'  # Define the dataset path

df = pd.read_csv(dataset_path, encoding='utf-8-sig')  # Use the defined dataset path

print("  Dataset loaded successfully")
print("Columns in dataset:", df.columns.tolist())

# ============================================================
# Basic Column Check
# ============================================================

required_columns = ["Base_Reviews", "category"]

for col in required_columns:
    if col not in df.columns:
        raise ValueError(f"Missing required column: {col}")

df = df.dropna(subset=["Base_Reviews", "category"]).copy()

# ============================================================
# Label Mapping
# IMPORTANT:
# Your dataframe category is converted to lowercase.
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# Therefore, all keys in y_dict must also be lowercase.
# The previous error was caused by "Bug" instead of "bug".
# ============================================================

y_dict = {
    "functionality issue": 0,
    "user experience issue": 1,
    "performance issue": 2,
    "stability issue": 3,
    "bug": 4,
    "device issue": 5,
    "customer support issue": 6,
    "privacy issue": 7,
    "user interface issue": 8,
    "compatibility issue": 9,
    "installation issue": 10,
    "security issue": 11,
    "network issue": 12,
}

class_labels = list(y_dict.keys())
all_labels = list(range(len(class_labels)))
n_classes = len(class_labels)

# ============================================================
# Clean and Filter Dataset
# ============================================================

df["category"] = df["category"].astype(str).str.lower().str.strip()

df = df[df["category"].isin(y_dict.keys())].copy()

df["label_id"] = df["category"].map(y_dict).astype(int)

print("\nDataset size after filtering:", len(df))
print("\nClass distribution:")
print(df["category"].value_counts())

if len(df) == 0:
    raise ValueError("Dataset is empty after filtering. Please check category␣
↪names.")

# ============================================================
# Split Train/Test
# ============================================================

texts = df["Base_Reviews"].astype(str).values
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y_int = df["label_id"].values

X_train_raw, X_test_raw, y_train_raw, y_test = train_test_split(
    texts,
    y_int,
    test_size=0.15,
    random_state=SEED,
    stratify=y_int
)

print("\nTrain size:", len(X_train_raw))
print("Test size:", len(X_test_raw))

print("\nTrain label distribution:")
print(pd.Series(y_train_raw).value_counts().sort_index())

print("\nTest label distribution:")
print(pd.Series(y_test).value_counts().sort_index())

# ============================================================
# Tokenizer
# ============================================================

tokenizer = BertTokenizerFast.from_pretrained("bert-base-uncased")

def tokenize_texts(texts, max_len):
    return tokenizer(
        [str(t) for t in texts],
        padding="max_length",
        truncation=True,
        max_length=max_len,
        return_tensors="pt",
        return_attention_mask=True
    )

# ============================================================
# Dataset Class
# ============================================================

class TextDS(Dataset):
    def __init__(self, toks, labels):
        self.ids = toks["input_ids"]
        self.msk = toks["attention_mask"]
        self.y = torch.tensor(labels, dtype=torch.long)

    def __len__(self):
        return len(self.y)
```

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    def __getitem__(self, i):
        return {
            "input_ids": self.ids[i],
            "attention_mask": self.msk[i],
            "labels": self.y[i]
        }

# ============================================================
# Training Function
# ============================================================

def train_one_run(train_loader, val_loader, epochs, lr, weight_decay=0.01):
    model = BertForSequenceClassification.from_pretrained(
        "bert-base-uncased",
        num_labels=n_classes
    ).to(device)

    optimizer = AdamW(
        model.parameters(),
        lr=lr,
        weight_decay=weight_decay
    )

    total_steps = len(train_loader) * epochs

    scheduler = get_linear_schedule_with_warmup(
        optimizer,
        num_warmup_steps=0,
        num_training_steps=total_steps
    )

    best_val_acc = -1.0
    best_state = None

    history = {
        "train_loss": [],
        "val_loss": [],
        "train_acc": [],
        "val_acc": []
    }

    for ep in range(epochs):
        print(f"\nEpoch {ep + 1}/{epochs}")

        # -------------------------
        # Training
```

```python
        # -------------------------
        model.train()

        tr_loss = 0.0
        tr_correct = 0
        tr_seen = 0

        for batch_idx, batch in enumerate(train_loader):
            optimizer.zero_grad()

            inputs = batch["input_ids"].to(device)
            masks = batch["attention_mask"].to(device)
            labels = batch["labels"].to(device)

            outputs = model(
                inputs,
                attention_mask=masks,
                labels=labels
            )

            loss = outputs.loss
            loss.backward()

            torch.nn.utils.clip_grad_norm_(model.parameters(), max_norm=1.0)

            optimizer.step()
            scheduler.step()

            tr_loss += loss.item()

            preds = torch.argmax(outputs.logits, dim=1)

            tr_correct += (preds == labels).sum().item()
            tr_seen += labels.size(0)

            if (batch_idx + 1) % 100 == 0:
                current_loss = tr_loss / (batch_idx + 1)
                current_acc = tr_correct / max(1, tr_seen)

                print(
                    f"Batch {batch_idx + 1}/{len(train_loader)} "
                    f"- Loss: {current_loss:.4f} "
                    f"Acc: {current_acc:.4f}"
                )

        avg_tr_loss = tr_loss / max(1, len(train_loader))
        avg_tr_acc = tr_correct / max(1, tr_seen)
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        history["train_loss"].append(avg_tr_loss)
        history["train_acc"].append(avg_tr_acc)

        # -------------------------
        # Validation
        # -------------------------
        model.eval()

        val_loss = 0.0
        val_correct = 0
        val_seen = 0

        with torch.no_grad():
            for batch in val_loader:
                inputs = batch["input_ids"].to(device)
                masks = batch["attention_mask"].to(device)
                labels = batch["labels"].to(device)

                outputs = model(
                    inputs,
                    attention_mask=masks,
                    labels=labels
                )

                val_loss += outputs.loss.item()

                preds = torch.argmax(outputs.logits, dim=1)

                val_correct += (preds == labels).sum().item()
                val_seen += labels.size(0)

        avg_val_loss = val_loss / max(1, len(val_loader))
        avg_val_acc = val_correct / max(1, val_seen)

        history["val_loss"].append(avg_val_loss)
        history["val_acc"].append(avg_val_acc)

        print(
            f"Epoch {ep + 1} "
            f"- Train Loss: {avg_tr_loss:.4f} "
            f"Train Acc: {avg_tr_acc:.4f} "
            f"Val Loss: {avg_val_loss:.4f} "
            f"Val Acc: {avg_val_acc:.4f}"
        )

        # Save best model according to validation accuracy
```

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        if avg_val_acc > best_val_acc + 1e-6:
            best_val_acc = avg_val_acc
            best_state = {
                k: v.detach().cpu().clone()
                for k, v in model.state_dict().items()
            }

    if best_state is not None:
        model.load_state_dict({
            k: v.to(device)
            for k, v in best_state.items()
        })

    return model, best_val_acc, history

# ============================================================
# Evaluation Function
# ============================================================

def evaluate_on_loader(model, data_loader):
    model.eval()

    y_true = []
    y_pred = []
    total_loss = 0.0

    with torch.no_grad():
        for batch in data_loader:
            inputs = batch["input_ids"].to(device)
            masks = batch["attention_mask"].to(device)
            labels = batch["labels"].to(device)

            outputs = model(
                inputs,
                attention_mask=masks,
                labels=labels
            )

            total_loss += outputs.loss.item()

            preds = torch.argmax(outputs.logits, dim=1)

            y_true.extend(labels.cpu().numpy())
            y_pred.extend(preds.cpu().numpy())

    y_true = np.array(y_true)
    y_pred = np.array(y_pred)
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    acc = (y_true == y_pred).mean()
    avg_loss = total_loss / max(1, len(data_loader))

    return acc, avg_loss, y_true, y_pred

# ============================================================
# Hyperparameters
# ============================================================

BEST_LR = 3e-5
BEST_BS = 8
BEST_EPOCHS = 8
MAXLEN = 128

print(
    f"\n  Using hyperparameters: "
    f"lr={BEST_LR}, bs={BEST_BS}, epochs={BEST_EPOCHS}, max_len={MAXLEN}"
)

# ============================================================
# 5-Fold Cross Validation
# ============================================================

print("\nStarting 5-Fold CV")

skf = StratifiedKFold(
    n_splits=5,
    shuffle=True,
    random_state=SEED
)

cv_results = []

for fold, (tr_idx, val_idx) in enumerate(
    skf.split(X_train_raw, y_train_raw),
    1
):
    print(f"\n==============================")
    print(f"Fold {fold}")
    print(f"==============================")

    X_tr_raw = X_train_raw[tr_idx]
    X_v_raw = X_train_raw[val_idx]

    y_tr_raw = y_train_raw[tr_idx]
    y_v_raw = y_train_raw[val_idx]
```

```
    # Oversample only training part of this fold
    ros = RandomOverSampler(random_state=SEED)

    X_tr_res, y_tr_res = ros.fit_resample(
        X_tr_raw.reshape(-1, 1),
        y_tr_raw
    )

    X_tr_res = X_tr_res.flatten()

    tok_tr = tokenize_texts(X_tr_res, MAXLEN)
    tok_v = tokenize_texts(X_v_raw, MAXLEN)

    train_ds = TextDS(tok_tr, y_tr_res)
    val_ds = TextDS(tok_v, y_v_raw)

    train_loader = DataLoader(
        train_ds,
        batch_size=BEST_BS,
        shuffle=True
    )

    val_loader = DataLoader(
        val_ds,
        batch_size=BEST_BS,
        shuffle=False
    )

    model, val_acc, _ = train_one_run(
        train_loader,
        val_loader,
        epochs=BEST_EPOCHS,
        lr=BEST_LR
    )

    print(f"→ Fold {fold} Best Val Acc: {val_acc:.4f}")

    cv_results.append({
        "fold": fold,
        "val_acc": val_acc
    })

    del model
    torch.cuda.empty_cache()

# ============================================================
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# CV Summary
# ============================================================

cv_df = pd.DataFrame(cv_results)

mean_cv = cv_df["val_acc"].mean()
std_cv = cv_df["val_acc"].std()

print("\n==============================")
print("5-Fold CV Results")
print("==============================")
print(cv_df)
print(f"\n  5-Fold CV Accuracy: {mean_cv:.4f} ± {std_cv:.4f}")

# ============================================================
# Final Model Training on Full Train Set
# ============================================================

print("\nRetraining final BERT on full TRAIN set")

ros_final = RandomOverSampler(random_state=SEED)

X_train_res, y_train_res = ros_final.fit_resample(
    X_train_raw.reshape(-1, 1),
    y_train_raw
)

X_train_res = X_train_res.flatten()

tok_train_final = tokenize_texts(X_train_res, MAXLEN)
tok_test_final = tokenize_texts(X_test_raw, MAXLEN)

train_ds_final = TextDS(tok_train_final, y_train_res)
test_ds_final = TextDS(tok_test_final, y_test)

train_loader_final = DataLoader(
    train_ds_final,
    batch_size=BEST_BS,
    shuffle=True
)

test_loader_final = DataLoader(
    test_ds_final,
    batch_size=BEST_BS,
    shuffle=False
)
```

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# ------------------------------------------------------------
# Validation loader for final training
# Here we use a small fixed subset from original train only
# for monitoring training.
# It is NOT the final test set.
# ------------------------------------------------------------

val_size = min(500, len(X_train_raw))

tok_val_monitor = tokenize_texts(X_train_raw[:val_size], MAXLEN)

val_monitor_ds = TextDS(
    tok_val_monitor,
    y_train_raw[:val_size]
)

val_monitor_loader = DataLoader(
    val_monitor_ds,
    batch_size=BEST_BS,
    shuffle=False
)

final_model, _, final_history = train_one_run(
    train_loader_final,
    val_monitor_loader,
    epochs=BEST_EPOCHS,
    lr=BEST_LR
)

# ============================================================
# Save Final Model
# ============================================================

SAVE_DIR = "./bert_best_final_model"

os.makedirs(SAVE_DIR, exist_ok=True)

final_model.save_pretrained(SAVE_DIR)
tokenizer.save_pretrained(SAVE_DIR)

print(f"\n  Saved BERT model to: {SAVE_DIR}")

# ============================================================
# Final Evaluation on Frozen Test Set
# ============================================================

test_acc, test_loss, y_true_test, y_pred_test = evaluate_on_loader(
```

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    final_model,
    test_loader_final
)

print("\n===============================")
print("FINAL TEST RESULTS")
print("===============================")
print(f"  FINAL TEST ACCURACY: {test_acc:.4f}")
print(f"  FINAL TEST LOSS: {test_loss:.4f}")

# ============================================================
# Classification Report
# FIXED:
# labels=all_labels is added to avoid mismatch error
# even if one class is absent from y_true or y_pred.
# ============================================================

print("\n=== FINAL CLASSIFICATION REPORT ===")

print(
    classification_report(
        y_true_test,
        y_pred_test,
        labels=all_labels,
        target_names=class_labels,
        digits=4,
        zero_division=0
    )
)

# ============================================================
# Confusion Matrix
# FIXED:
# labels=all_labels is added to force 13x13 matrix.
# ============================================================

cm = confusion_matrix(
    y_true_test,
    y_pred_test,
    labels=all_labels
)

plt.figure(figsize=(12, 10))

sns.heatmap(
    cm,
    annot=True,
```

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    fmt="d",
    cmap="Blues",
    xticklabels=class_labels,
    yticklabels=class_labels
)

plt.title("BERT Confusion Matrix - Test")
plt.xlabel("Predicted")
plt.ylabel("True")
plt.xticks(rotation=45, ha="right")
plt.yticks(rotation=0)
plt.tight_layout()
plt.show()

# ============================================================
# Save Results to CSV
# ============================================================

results_df = pd.DataFrame({
    "true_label_id": y_true_test,
    "pred_label_id": y_pred_test,
    "true_label": [class_labels[i] for i in y_true_test],
    "pred_label": [class_labels[i] for i in y_pred_test]
})

results_df.to_csv("bert_test_predictions.csv", index=False)

cv_df.to_csv("bert_cv_results.csv", index=False)

print("\n Test predictions saved to: bert_test_predictions.csv")
print("  CV results saved to: bert_cv_results.csv")

# ============================================================
# Save Classification Report as CSV
# ============================================================

report_dict = classification_report(
    y_true_test,
    y_pred_test,
    labels=all_labels,
    target_names=class_labels,
    digits=4,
    zero_division=0,
    output_dict=True
)

report_df = pd.DataFrame(report_dict).transpose()
```

```python
report_df.to_csv("bert_classification_report.csv", index=True)

print("  Classification report saved to: bert_classification_report.csv")

# ============================================================
# Optional: Plot Training Curves for Final Model
# ============================================================

plt.figure(figsize=(8, 6))
plt.plot(final_history["train_loss"], label="Train Loss")
plt.plot(final_history["val_loss"], label="Validation Loss")
plt.title("BERT Training and Validation Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.legend()
plt.tight_layout()
plt.show()

plt.figure(figsize=(8, 6))
plt.plot(final_history["train_acc"], label="Train Accuracy")
plt.plot(final_history["val_acc"], label="Validation Accuracy")
plt.title("BERT Training and Validation Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.legend()
plt.tight_layout()
plt.show()

print("\n  Full pipeline completed successfully.")
```

```
Using device: mps
 Dataset loaded successfully
Columns in dataset: ['Rating_Star', 'Headings', 'Issue', 'Base_Reviews',
'category', 'Issue Details']

Dataset size after filtering: 16034

Class distribution:
category
functionality issue       5313
user experience issue     2066
performance issue         1928
stability issue           1887
bug                       1098
device issue               937
customer support issue     714
privacy issue              643
```

```
user interface issue       516
compatibility issue        298
installation issue         217
security issue             215
network issue              202
Name: count, dtype: int64

Train size: 13628
Test size: 2406

Train label distribution:
0     4516
1     1756
2     1639
3     1604
4      933
5      796
6      607
7      546
8      439
9      253
10     184
11     183
12     172
Name: count, dtype: int64

Test label distribution:
0     797
1     310
2     289
3     283
4     165
5     141
6     107
7      97
8      77
9      45
10     33
11     32
12     30
Name: count, dtype: int64

 Using hyperparameters: lr=3e-05, bs=8, epochs=8, max_len=128

Starting 5-Fold CV

==============================
Fold 1
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```
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Some weights of BertForSequenceClassification were not initialized from the
model checkpoint at bert-base-uncased and are newly initialized:
['classifier.bias', 'classifier.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.


Epoch 1/8
Batch 100/5872 - Loss: 2.5293 Acc: 0.1150
Batch 200/5872 - Loss: 2.2892 Acc: 0.2263
Batch 300/5872 - Loss: 2.0331 Acc: 0.3262
Batch 400/5872 - Loss: 1.8407 Acc: 0.3928
Batch 500/5872 - Loss: 1.7088 Acc: 0.4370
Batch 600/5872 - Loss: 1.5882 Acc: 0.4800
Batch 700/5872 - Loss: 1.4809 Acc: 0.5164
Batch 800/5872 - Loss: 1.4005 Acc: 0.5447
Batch 900/5872 - Loss: 1.3305 Acc: 0.5669
Batch 1000/5872 - Loss: 1.2664 Acc: 0.5891
Batch 1100/5872 - Loss: 1.2142 Acc: 0.6050
Batch 1200/5872 - Loss: 1.1608 Acc: 0.6225
Batch 1300/5872 - Loss: 1.1121 Acc: 0.6383
Batch 1400/5872 - Loss: 1.0742 Acc: 0.6516
Batch 1500/5872 - Loss: 1.0358 Acc: 0.6643
Batch 1600/5872 - Loss: 1.0010 Acc: 0.6759
Batch 1700/5872 - Loss: 0.9729 Acc: 0.6853
Batch 1800/5872 - Loss: 0.9411 Acc: 0.6960
Batch 1900/5872 - Loss: 0.9153 Acc: 0.7040
Batch 2000/5872 - Loss: 0.8877 Acc: 0.7133
Batch 2100/5872 - Loss: 0.8611 Acc: 0.7222
Batch 2200/5872 - Loss: 0.8399 Acc: 0.7293
Batch 2300/5872 - Loss: 0.8203 Acc: 0.7364
Batch 2400/5872 - Loss: 0.8014 Acc: 0.7432
Batch 2500/5872 - Loss: 0.7832 Acc: 0.7495
Batch 2600/5872 - Loss: 0.7649 Acc: 0.7558
Batch 2700/5872 - Loss: 0.7479 Acc: 0.7615
Batch 2800/5872 - Loss: 0.7331 Acc: 0.7671
Batch 2900/5872 - Loss: 0.7175 Acc: 0.7725
Batch 3000/5872 - Loss: 0.7012 Acc: 0.7780
Batch 3100/5872 - Loss: 0.6883 Acc: 0.7829
Batch 3200/5872 - Loss: 0.6744 Acc: 0.7874
Batch 3300/5872 - Loss: 0.6633 Acc: 0.7913
Batch 3400/5872 - Loss: 0.6504 Acc: 0.7958
Batch 3500/5872 - Loss: 0.6395 Acc: 0.7996
Batch 3600/5872 - Loss: 0.6293 Acc: 0.8033
Batch 3700/5872 - Loss: 0.6193 Acc: 0.8068
Batch 3800/5872 - Loss: 0.6099 Acc: 0.8102
Batch 3900/5872 - Loss: 0.6004 Acc: 0.8135
```

```
Batch 4000/5872 - Loss: 0.5907 Acc: 0.8168
Batch 4100/5872 - Loss: 0.5824 Acc: 0.8197
Batch 4200/5872 - Loss: 0.5740 Acc: 0.8226
Batch 4300/5872 - Loss: 0.5651 Acc: 0.8256
Batch 4400/5872 - Loss: 0.5569 Acc: 0.8285
Batch 4500/5872 - Loss: 0.5492 Acc: 0.8311
Batch 4600/5872 - Loss: 0.5434 Acc: 0.8331
Batch 4700/5872 - Loss: 0.5353 Acc: 0.8358
Batch 4800/5872 - Loss: 0.5279 Acc: 0.8384
Batch 4900/5872 - Loss: 0.5209 Acc: 0.8406
Batch 5000/5872 - Loss: 0.5149 Acc: 0.8427
Batch 5100/5872 - Loss: 0.5089 Acc: 0.8449
Batch 5200/5872 - Loss: 0.5023 Acc: 0.8470
Batch 5300/5872 - Loss: 0.4963 Acc: 0.8489
Batch 5400/5872 - Loss: 0.4906 Acc: 0.8509
Batch 5500/5872 - Loss: 0.4842 Acc: 0.8529
Batch 5600/5872 - Loss: 0.4788 Acc: 0.8548
Batch 5700/5872 - Loss: 0.4741 Acc: 0.8566
Batch 5800/5872 - Loss: 0.4700 Acc: 0.8583
Epoch 1 - Train Loss: 0.4663 Train Acc: 0.8594 Val Loss: 0.7727 Val Acc: 0.8555

Epoch 2/8
Batch 100/5872 - Loss: 0.1075 Acc: 0.9788
Batch 200/5872 - Loss: 0.1258 Acc: 0.9750
Batch 300/5872 - Loss: 0.1178 Acc: 0.9762
Batch 400/5872 - Loss: 0.1222 Acc: 0.9734
Batch 500/5872 - Loss: 0.1254 Acc: 0.9722
Batch 600/5872 - Loss: 0.1320 Acc: 0.9717
Batch 700/5872 - Loss: 0.1313 Acc: 0.9707
Batch 800/5872 - Loss: 0.1368 Acc: 0.9689
Batch 900/5872 - Loss: 0.1439 Acc: 0.9674
Batch 1000/5872 - Loss: 0.1454 Acc: 0.9663
Batch 1100/5872 - Loss: 0.1450 Acc: 0.9659
Batch 1200/5872 - Loss: 0.1445 Acc: 0.9658
Batch 1300/5872 - Loss: 0.1473 Acc: 0.9650
Batch 1400/5872 - Loss: 0.1463 Acc: 0.9650
Batch 1500/5872 - Loss: 0.1472 Acc: 0.9650
Batch 1600/5872 - Loss: 0.1435 Acc: 0.9657
Batch 1700/5872 - Loss: 0.1438 Acc: 0.9655
Batch 1800/5872 - Loss: 0.1434 Acc: 0.9656
Batch 1900/5872 - Loss: 0.1400 Acc: 0.9663
Batch 2000/5872 - Loss: 0.1405 Acc: 0.9659
Batch 2100/5872 - Loss: 0.1406 Acc: 0.9657
Batch 2200/5872 - Loss: 0.1404 Acc: 0.9657
Batch 2300/5872 - Loss: 0.1390 Acc: 0.9660
Batch 2400/5872 - Loss: 0.1392 Acc: 0.9657
Batch 2500/5872 - Loss: 0.1375 Acc: 0.9661
Batch 2600/5872 - Loss: 0.1401 Acc: 0.9654
```

```
Batch 2700/5872 - Loss: 0.1392 Acc: 0.9656
Batch 2800/5872 - Loss: 0.1388 Acc: 0.9656
Batch 2900/5872 - Loss: 0.1382 Acc: 0.9658
Batch 3000/5872 - Loss: 0.1391 Acc: 0.9657
Batch 3100/5872 - Loss: 0.1398 Acc: 0.9654
Batch 3200/5872 - Loss: 0.1385 Acc: 0.9655
Batch 3300/5872 - Loss: 0.1386 Acc: 0.9656
Batch 3400/5872 - Loss: 0.1375 Acc: 0.9657
Batch 3500/5872 - Loss: 0.1367 Acc: 0.9658
Batch 3600/5872 - Loss: 0.1352 Acc: 0.9661
Batch 3700/5872 - Loss: 0.1358 Acc: 0.9660
Batch 3800/5872 - Loss: 0.1350 Acc: 0.9663
Batch 3900/5872 - Loss: 0.1345 Acc: 0.9663
Batch 4000/5872 - Loss: 0.1341 Acc: 0.9665
Batch 4100/5872 - Loss: 0.1327 Acc: 0.9669
Batch 4200/5872 - Loss: 0.1334 Acc: 0.9668
Batch 4300/5872 - Loss: 0.1328 Acc: 0.9670
Batch 4400/5872 - Loss: 0.1333 Acc: 0.9668
Batch 4500/5872 - Loss: 0.1342 Acc: 0.9666
Batch 4600/5872 - Loss: 0.1336 Acc: 0.9665
Batch 4700/5872 - Loss: 0.1338 Acc: 0.9665
Batch 4800/5872 - Loss: 0.1334 Acc: 0.9666
Batch 4900/5872 - Loss: 0.1326 Acc: 0.9669
Batch 5000/5872 - Loss: 0.1321 Acc: 0.9670
Batch 5100/5872 - Loss: 0.1318 Acc: 0.9671
Batch 5200/5872 - Loss: 0.1315 Acc: 0.9672
Batch 5300/5872 - Loss: 0.1310 Acc: 0.9672
Batch 5400/5872 - Loss: 0.1305 Acc: 0.9674
Batch 5500/5872 - Loss: 0.1294 Acc: 0.9675
Batch 5600/5872 - Loss: 0.1284 Acc: 0.9676
Batch 5700/5872 - Loss: 0.1278 Acc: 0.9678
Batch 5800/5872 - Loss: 0.1271 Acc: 0.9679
Epoch 2 - Train Loss: 0.1266 Train Acc: 0.9680 Val Loss: 0.8606 Val Acc: 0.8742

Epoch 3/8
Batch 100/5872 - Loss: 0.0914 Acc: 0.9762
Batch 200/5872 - Loss: 0.1083 Acc: 0.9719
Batch 300/5872 - Loss: 0.0995 Acc: 0.9733
Batch 400/5872 - Loss: 0.1025 Acc: 0.9744
Batch 500/5872 - Loss: 0.1027 Acc: 0.9748
Batch 600/5872 - Loss: 0.0968 Acc: 0.9762
Batch 700/5872 - Loss: 0.0931 Acc: 0.9766
Batch 800/5872 - Loss: 0.0915 Acc: 0.9766
Batch 900/5872 - Loss: 0.0891 Acc: 0.9767
Batch 1000/5872 - Loss: 0.0888 Acc: 0.9768
Batch 1100/5872 - Loss: 0.0894 Acc: 0.9770
Batch 1200/5872 - Loss: 0.0891 Acc: 0.9775
Batch 1300/5872 - Loss: 0.0889 Acc: 0.9775
```

```
Batch 1400/5872 - Loss: 0.0904 Acc: 0.9771
Batch 1500/5872 - Loss: 0.0907 Acc: 0.9768
Batch 1600/5872 - Loss: 0.0909 Acc: 0.9770
Batch 1700/5872 - Loss: 0.0900 Acc: 0.9771
Batch 1800/5872 - Loss: 0.0887 Acc: 0.9774
Batch 1900/5872 - Loss: 0.0889 Acc: 0.9776
Batch 2000/5872 - Loss: 0.0900 Acc: 0.9773
Batch 2100/5872 - Loss: 0.0898 Acc: 0.9770
Batch 2200/5872 - Loss: 0.0912 Acc: 0.9768
Batch 2300/5872 - Loss: 0.0910 Acc: 0.9767
Batch 2400/5872 - Loss: 0.0909 Acc: 0.9764
Batch 2500/5872 - Loss: 0.0915 Acc: 0.9761
Batch 2600/5872 - Loss: 0.0932 Acc: 0.9758
Batch 2700/5872 - Loss: 0.0923 Acc: 0.9758
Batch 2800/5872 - Loss: 0.0915 Acc: 0.9760
Batch 2900/5872 - Loss: 0.0905 Acc: 0.9764
Batch 3000/5872 - Loss: 0.0910 Acc: 0.9760
Batch 3100/5872 - Loss: 0.0905 Acc: 0.9762
Batch 3200/5872 - Loss: 0.0916 Acc: 0.9759
Batch 3300/5872 - Loss: 0.0924 Acc: 0.9755
Batch 3400/5872 - Loss: 0.0921 Acc: 0.9756
Batch 3500/5872 - Loss: 0.0926 Acc: 0.9756
Batch 3600/5872 - Loss: 0.0921 Acc: 0.9758
Batch 3700/5872 - Loss: 0.0916 Acc: 0.9758
Batch 3800/5872 - Loss: 0.0913 Acc: 0.9760
Batch 3900/5872 - Loss: 0.0914 Acc: 0.9758
Batch 4000/5872 - Loss: 0.0912 Acc: 0.9758
Batch 4100/5872 - Loss: 0.0913 Acc: 0.9757
Batch 4200/5872 - Loss: 0.0920 Acc: 0.9756
Batch 4300/5872 - Loss: 0.0919 Acc: 0.9756
Batch 4400/5872 - Loss: 0.0919 Acc: 0.9756
Batch 4500/5872 - Loss: 0.0914 Acc: 0.9757
Batch 4600/5872 - Loss: 0.0913 Acc: 0.9755
Batch 4700/5872 - Loss: 0.0908 Acc: 0.9756
Batch 4800/5872 - Loss: 0.0900 Acc: 0.9758
Batch 4900/5872 - Loss: 0.0903 Acc: 0.9758
Batch 5000/5872 - Loss: 0.0899 Acc: 0.9758
Batch 5100/5872 - Loss: 0.0895 Acc: 0.9759
Batch 5200/5872 - Loss: 0.0888 Acc: 0.9759
Batch 5300/5872 - Loss: 0.0891 Acc: 0.9759
Batch 5400/5872 - Loss: 0.0894 Acc: 0.9759
Batch 5500/5872 - Loss: 0.0891 Acc: 0.9759
Batch 5600/5872 - Loss: 0.0891 Acc: 0.9759
Batch 5700/5872 - Loss: 0.0890 Acc: 0.9759
Batch 5800/5872 - Loss: 0.0886 Acc: 0.9760
Epoch 3 - Train Loss: 0.0886 Train Acc: 0.9759 Val Loss: 0.7732 Val Acc: 0.8782

Epoch 4/8
```

```
Batch 100/5872 - Loss: 0.0616 Acc: 0.9775
Batch 200/5872 - Loss: 0.0506 Acc: 0.9838
Batch 300/5872 - Loss: 0.0586 Acc: 0.9817
Batch 400/5872 - Loss: 0.0595 Acc: 0.9812
Batch 500/5872 - Loss: 0.0711 Acc: 0.9800
Batch 600/5872 - Loss: 0.0704 Acc: 0.9794
Batch 700/5872 - Loss: 0.0674 Acc: 0.9791
Batch 800/5872 - Loss: 0.0661 Acc: 0.9797
Batch 900/5872 - Loss: 0.0655 Acc: 0.9797
Batch 1000/5872 - Loss: 0.0662 Acc: 0.9795
Batch 1100/5872 - Loss: 0.0655 Acc: 0.9795
Batch 1200/5872 - Loss: 0.0658 Acc: 0.9792
Batch 1300/5872 - Loss: 0.0693 Acc: 0.9782
Batch 1400/5872 - Loss: 0.0686 Acc: 0.9781
Batch 1500/5872 - Loss: 0.0694 Acc: 0.9781
Batch 1600/5872 - Loss: 0.0704 Acc: 0.9781
Batch 1700/5872 - Loss: 0.0699 Acc: 0.9782
Batch 1800/5872 - Loss: 0.0691 Acc: 0.9781
Batch 1900/5872 - Loss: 0.0686 Acc: 0.9783
Batch 2000/5872 - Loss: 0.0673 Acc: 0.9786
Batch 2100/5872 - Loss: 0.0687 Acc: 0.9783
Batch 2200/5872 - Loss: 0.0693 Acc: 0.9783
Batch 2300/5872 - Loss: 0.0687 Acc: 0.9784
Batch 2400/5872 - Loss: 0.0677 Acc: 0.9785
Batch 2500/5872 - Loss: 0.0683 Acc: 0.9782
Batch 2600/5872 - Loss: 0.0684 Acc: 0.9781
Batch 2700/5872 - Loss: 0.0681 Acc: 0.9781
Batch 2800/5872 - Loss: 0.0699 Acc: 0.9774
Batch 2900/5872 - Loss: 0.0696 Acc: 0.9775
Batch 3000/5872 - Loss: 0.0692 Acc: 0.9776
Batch 3100/5872 - Loss: 0.0695 Acc: 0.9774
Batch 3200/5872 - Loss: 0.0690 Acc: 0.9775
Batch 3300/5872 - Loss: 0.0694 Acc: 0.9775
Batch 3400/5872 - Loss: 0.0691 Acc: 0.9778
Batch 3500/5872 - Loss: 0.0687 Acc: 0.9781
Batch 3600/5872 - Loss: 0.0690 Acc: 0.9780
Batch 3700/5872 - Loss: 0.0690 Acc: 0.9780
Batch 3800/5872 - Loss: 0.0696 Acc: 0.9779
Batch 3900/5872 - Loss: 0.0698 Acc: 0.9778
Batch 4000/5872 - Loss: 0.0697 Acc: 0.9778
Batch 4100/5872 - Loss: 0.0696 Acc: 0.9779
Batch 4200/5872 - Loss: 0.0700 Acc: 0.9778
Batch 4300/5872 - Loss: 0.0693 Acc: 0.9781
Batch 4400/5872 - Loss: 0.0692 Acc: 0.9781
Batch 4500/5872 - Loss: 0.0703 Acc: 0.9778
Batch 4600/5872 - Loss: 0.0706 Acc: 0.9777
Batch 4700/5872 - Loss: 0.0710 Acc: 0.9776
Batch 4800/5872 - Loss: 0.0713 Acc: 0.9777
```

```
Batch 4900/5872 - Loss: 0.0721 Acc: 0.9774
Batch 5000/5872 - Loss: 0.0716 Acc: 0.9776
Batch 5100/5872 - Loss: 0.0716 Acc: 0.9777
Batch 5200/5872 - Loss: 0.0717 Acc: 0.9777
Batch 5300/5872 - Loss: 0.0714 Acc: 0.9779
Batch 5400/5872 - Loss: 0.0713 Acc: 0.9779
Batch 5500/5872 - Loss: 0.0715 Acc: 0.9779
Batch 5600/5872 - Loss: 0.0716 Acc: 0.9779
Batch 5700/5872 - Loss: 0.0714 Acc: 0.9779
Batch 5800/5872 - Loss: 0.0706 Acc: 0.9782
Epoch 4 - Train Loss: 0.0709 Train Acc: 0.9781 Val Loss: 0.8189 Val Acc: 0.8819

Epoch 5/8
Batch 100/5872 - Loss: 0.0594 Acc: 0.9775
Batch 200/5872 - Loss: 0.0516 Acc: 0.9794
Batch 300/5872 - Loss: 0.0653 Acc: 0.9767
Batch 400/5872 - Loss: 0.0618 Acc: 0.9788
Batch 500/5872 - Loss: 0.0669 Acc: 0.9762
Batch 600/5872 - Loss: 0.0676 Acc: 0.9750
Batch 700/5872 - Loss: 0.0690 Acc: 0.9748
Batch 800/5872 - Loss: 0.0658 Acc: 0.9753
Batch 900/5872 - Loss: 0.0670 Acc: 0.9754
Batch 1000/5872 - Loss: 0.0683 Acc: 0.9756
Batch 1100/5872 - Loss: 0.0671 Acc: 0.9762
Batch 1200/5872 - Loss: 0.0664 Acc: 0.9770
Batch 1300/5872 - Loss: 0.0647 Acc: 0.9777
Batch 1400/5872 - Loss: 0.0668 Acc: 0.9775
Batch 1500/5872 - Loss: 0.0673 Acc: 0.9774
Batch 1600/5872 - Loss: 0.0666 Acc: 0.9775
Batch 1700/5872 - Loss: 0.0654 Acc: 0.9779
Batch 1800/5872 - Loss: 0.0642 Acc: 0.9787
Batch 1900/5872 - Loss: 0.0633 Acc: 0.9789
Batch 2000/5872 - Loss: 0.0619 Acc: 0.9794
Batch 2100/5872 - Loss: 0.0614 Acc: 0.9795
Batch 2200/5872 - Loss: 0.0614 Acc: 0.9795
Batch 2300/5872 - Loss: 0.0618 Acc: 0.9797
Batch 2400/5872 - Loss: 0.0622 Acc: 0.9797
Batch 2500/5872 - Loss: 0.0619 Acc: 0.9798
Batch 2600/5872 - Loss: 0.0615 Acc: 0.9799
Batch 2700/5872 - Loss: 0.0613 Acc: 0.9801
Batch 2800/5872 - Loss: 0.0605 Acc: 0.9804
Batch 2900/5872 - Loss: 0.0618 Acc: 0.9803
Batch 3000/5872 - Loss: 0.0612 Acc: 0.9805
Batch 3100/5872 - Loss: 0.0617 Acc: 0.9804
Batch 3200/5872 - Loss: 0.0610 Acc: 0.9805
Batch 3300/5872 - Loss: 0.0605 Acc: 0.9807
Batch 3400/5872 - Loss: 0.0597 Acc: 0.9809
Batch 3500/5872 - Loss: 0.0604 Acc: 0.9809
```

```
Batch 3600/5872 - Loss: 0.0600 Acc: 0.9809
Batch 3700/5872 - Loss: 0.0602 Acc: 0.9808
Batch 3800/5872 - Loss: 0.0594 Acc: 0.9810
Batch 3900/5872 - Loss: 0.0592 Acc: 0.9811
Batch 4000/5872 - Loss: 0.0598 Acc: 0.9809
Batch 4100/5872 - Loss: 0.0599 Acc: 0.9808
Batch 4200/5872 - Loss: 0.0594 Acc: 0.9809
Batch 4300/5872 - Loss: 0.0599 Acc: 0.9808
Batch 4400/5872 - Loss: 0.0598 Acc: 0.9807
Batch 4500/5872 - Loss: 0.0599 Acc: 0.9808
Batch 4600/5872 - Loss: 0.0596 Acc: 0.9808
Batch 4700/5872 - Loss: 0.0599 Acc: 0.9807
Batch 4800/5872 - Loss: 0.0596 Acc: 0.9806
Batch 4900/5872 - Loss: 0.0594 Acc: 0.9807
Batch 5000/5872 - Loss: 0.0595 Acc: 0.9808
Batch 5100/5872 - Loss: 0.0593 Acc: 0.9808
Batch 5200/5872 - Loss: 0.0590 Acc: 0.9808
Batch 5300/5872 - Loss: 0.0590 Acc: 0.9809
Batch 5400/5872 - Loss: 0.0589 Acc: 0.9809
Batch 5500/5872 - Loss: 0.0587 Acc: 0.9809
Batch 5600/5872 - Loss: 0.0585 Acc: 0.9809
Batch 5700/5872 - Loss: 0.0587 Acc: 0.9809
Batch 5800/5872 - Loss: 0.0586 Acc: 0.9809
Epoch 5 - Train Loss: 0.0586 Train Acc: 0.9809 Val Loss: 0.8439 Val Acc: 0.8797

Epoch 6/8
Batch 100/5872 - Loss: 0.0372 Acc: 0.9825
Batch 200/5872 - Loss: 0.0508 Acc: 0.9812
Batch 300/5872 - Loss: 0.0538 Acc: 0.9800
Batch 400/5872 - Loss: 0.0509 Acc: 0.9809
Batch 500/5872 - Loss: 0.0492 Acc: 0.9822
Batch 600/5872 - Loss: 0.0493 Acc: 0.9821
Batch 700/5872 - Loss: 0.0499 Acc: 0.9816
Batch 800/5872 - Loss: 0.0487 Acc: 0.9822
Batch 900/5872 - Loss: 0.0466 Acc: 0.9832
Batch 1000/5872 - Loss: 0.0491 Acc: 0.9826
Batch 1100/5872 - Loss: 0.0500 Acc: 0.9825
Batch 1200/5872 - Loss: 0.0496 Acc: 0.9825
Batch 1300/5872 - Loss: 0.0475 Acc: 0.9833
Batch 1400/5872 - Loss: 0.0488 Acc: 0.9825
Batch 1500/5872 - Loss: 0.0503 Acc: 0.9821
Batch 1600/5872 - Loss: 0.0517 Acc: 0.9817
Batch 1700/5872 - Loss: 0.0509 Acc: 0.9821
Batch 1800/5872 - Loss: 0.0510 Acc: 0.9820
Batch 1900/5872 - Loss: 0.0514 Acc: 0.9818
Batch 2000/5872 - Loss: 0.0503 Acc: 0.9824
Batch 2100/5872 - Loss: 0.0506 Acc: 0.9825
Batch 2200/5872 - Loss: 0.0516 Acc: 0.9822
```

```
Batch 2300/5872 - Loss: 0.0518 Acc: 0.9821
Batch 2400/5872 - Loss: 0.0518 Acc: 0.9820
Batch 2500/5872 - Loss: 0.0531 Acc: 0.9820
Batch 2600/5872 - Loss: 0.0529 Acc: 0.9819
Batch 2700/5872 - Loss: 0.0526 Acc: 0.9819
Batch 2800/5872 - Loss: 0.0521 Acc: 0.9821
Batch 2900/5872 - Loss: 0.0522 Acc: 0.9821
Batch 3000/5872 - Loss: 0.0524 Acc: 0.9821
Batch 3100/5872 - Loss: 0.0519 Acc: 0.9823
Batch 3200/5872 - Loss: 0.0516 Acc: 0.9823
Batch 3300/5872 - Loss: 0.0517 Acc: 0.9824
Batch 3400/5872 - Loss: 0.0519 Acc: 0.9822
Batch 3500/5872 - Loss: 0.0520 Acc: 0.9821
Batch 3600/5872 - Loss: 0.0521 Acc: 0.9821
Batch 3700/5872 - Loss: 0.0532 Acc: 0.9817
Batch 3800/5872 - Loss: 0.0528 Acc: 0.9819
Batch 3900/5872 - Loss: 0.0529 Acc: 0.9817
Batch 4000/5872 - Loss: 0.0530 Acc: 0.9817
Batch 4100/5872 - Loss: 0.0526 Acc: 0.9819
Batch 4200/5872 - Loss: 0.0532 Acc: 0.9816
Batch 4300/5872 - Loss: 0.0527 Acc: 0.9817
Batch 4400/5872 - Loss: 0.0526 Acc: 0.9816
Batch 4500/5872 - Loss: 0.0527 Acc: 0.9816
Batch 4600/5872 - Loss: 0.0526 Acc: 0.9815
Batch 4700/5872 - Loss: 0.0523 Acc: 0.9816
Batch 4800/5872 - Loss: 0.0520 Acc: 0.9817
Batch 4900/5872 - Loss: 0.0518 Acc: 0.9817
Batch 5000/5872 - Loss: 0.0518 Acc: 0.9817
Batch 5100/5872 - Loss: 0.0516 Acc: 0.9818
Batch 5200/5872 - Loss: 0.0518 Acc: 0.9818
Batch 5300/5872 - Loss: 0.0519 Acc: 0.9817
Batch 5400/5872 - Loss: 0.0516 Acc: 0.9817
Batch 5500/5872 - Loss: 0.0515 Acc: 0.9817
Batch 5600/5872 - Loss: 0.0513 Acc: 0.9817
Batch 5700/5872 - Loss: 0.0514 Acc: 0.9817
Batch 5800/5872 - Loss: 0.0514 Acc: 0.9817
Epoch 6 - Train Loss: 0.0515 Train Acc: 0.9817 Val Loss: 0.8154 Val Acc: 0.8885

Epoch 7/8
Batch 100/5872 - Loss: 0.0496 Acc: 0.9775
Batch 200/5872 - Loss: 0.0594 Acc: 0.9769
Batch 300/5872 - Loss: 0.0624 Acc: 0.9758
Batch 400/5872 - Loss: 0.0581 Acc: 0.9772
Batch 500/5872 - Loss: 0.0583 Acc: 0.9782
Batch 600/5872 - Loss: 0.0575 Acc: 0.9777
Batch 700/5872 - Loss: 0.0545 Acc: 0.9784
Batch 800/5872 - Loss: 0.0534 Acc: 0.9786
Batch 900/5872 - Loss: 0.0526 Acc: 0.9788
```

```
Batch 1000/5872 - Loss: 0.0493 Acc: 0.9801
Batch 1100/5872 - Loss: 0.0493 Acc: 0.9805
Batch 1200/5872 - Loss: 0.0499 Acc: 0.9803
Batch 1300/5872 - Loss: 0.0493 Acc: 0.9806
Batch 1400/5872 - Loss: 0.0481 Acc: 0.9808
Batch 1500/5872 - Loss: 0.0473 Acc: 0.9813
Batch 1600/5872 - Loss: 0.0456 Acc: 0.9819
Batch 1700/5872 - Loss: 0.0457 Acc: 0.9821
Batch 1800/5872 - Loss: 0.0461 Acc: 0.9819
Batch 1900/5872 - Loss: 0.0455 Acc: 0.9820
Batch 2000/5872 - Loss: 0.0447 Acc: 0.9822
Batch 2100/5872 - Loss: 0.0455 Acc: 0.9820
Batch 2200/5872 - Loss: 0.0460 Acc: 0.9818
Batch 2300/5872 - Loss: 0.0456 Acc: 0.9818
Batch 2400/5872 - Loss: 0.0468 Acc: 0.9817
Batch 2500/5872 - Loss: 0.0472 Acc: 0.9817
Batch 2600/5872 - Loss: 0.0468 Acc: 0.9818
Batch 2700/5872 - Loss: 0.0463 Acc: 0.9820
Batch 2800/5872 - Loss: 0.0470 Acc: 0.9819
Batch 2900/5872 - Loss: 0.0467 Acc: 0.9819
Batch 3000/5872 - Loss: 0.0462 Acc: 0.9822
Batch 3100/5872 - Loss: 0.0463 Acc: 0.9823
Batch 3200/5872 - Loss: 0.0463 Acc: 0.9823
Batch 3300/5872 - Loss: 0.0464 Acc: 0.9824
Batch 3400/5872 - Loss: 0.0457 Acc: 0.9826
Batch 3500/5872 - Loss: 0.0457 Acc: 0.9827
Batch 3600/5872 - Loss: 0.0460 Acc: 0.9826
Batch 3700/5872 - Loss: 0.0463 Acc: 0.9825
Batch 3800/5872 - Loss: 0.0468 Acc: 0.9823
Batch 3900/5872 - Loss: 0.0468 Acc: 0.9822
Batch 4000/5872 - Loss: 0.0464 Acc: 0.9824
Batch 4100/5872 - Loss: 0.0465 Acc: 0.9824
Batch 4200/5872 - Loss: 0.0463 Acc: 0.9825
Batch 4300/5872 - Loss: 0.0465 Acc: 0.9825
Batch 4400/5872 - Loss: 0.0461 Acc: 0.9826
Batch 4500/5872 - Loss: 0.0456 Acc: 0.9829
Batch 4600/5872 - Loss: 0.0453 Acc: 0.9830
Batch 4700/5872 - Loss: 0.0462 Acc: 0.9826
Batch 4800/5872 - Loss: 0.0462 Acc: 0.9826
Batch 4900/5872 - Loss: 0.0467 Acc: 0.9825
Batch 5000/5872 - Loss: 0.0472 Acc: 0.9823
Batch 5100/5872 - Loss: 0.0474 Acc: 0.9823
Batch 5200/5872 - Loss: 0.0475 Acc: 0.9822
Batch 5300/5872 - Loss: 0.0473 Acc: 0.9822
Batch 5400/5872 - Loss: 0.0474 Acc: 0.9821
Batch 5500/5872 - Loss: 0.0471 Acc: 0.9822
Batch 5600/5872 - Loss: 0.0469 Acc: 0.9823
Batch 5700/5872 - Loss: 0.0470 Acc: 0.9822
```

```
Batch 5800/5872 - Loss: 0.0466 Acc: 0.9824
Epoch 7 - Train Loss: 0.0467 Train Acc: 0.9824 Val Loss: 0.8471 Val Acc: 0.8822

Epoch 8/8
Batch 100/5872 - Loss: 0.0504 Acc: 0.9788
Batch 200/5872 - Loss: 0.0493 Acc: 0.9806
Batch 300/5872 - Loss: 0.0436 Acc: 0.9821
Batch 400/5872 - Loss: 0.0403 Acc: 0.9828
Batch 500/5872 - Loss: 0.0413 Acc: 0.9825
Batch 600/5872 - Loss: 0.0404 Acc: 0.9831
Batch 700/5872 - Loss: 0.0423 Acc: 0.9820
Batch 800/5872 - Loss: 0.0408 Acc: 0.9825
Batch 900/5872 - Loss: 0.0418 Acc: 0.9818
Batch 1000/5872 - Loss: 0.0417 Acc: 0.9816
Batch 1100/5872 - Loss: 0.0411 Acc: 0.9818
Batch 1200/5872 - Loss: 0.0408 Acc: 0.9823
Batch 1300/5872 - Loss: 0.0422 Acc: 0.9817
Batch 1400/5872 - Loss: 0.0432 Acc: 0.9812
Batch 1500/5872 - Loss: 0.0430 Acc: 0.9813
Batch 1600/5872 - Loss: 0.0430 Acc: 0.9816
Batch 1700/5872 - Loss: 0.0425 Acc: 0.9818
Batch 1800/5872 - Loss: 0.0435 Acc: 0.9815
Batch 1900/5872 - Loss: 0.0423 Acc: 0.9819
Batch 2000/5872 - Loss: 0.0422 Acc: 0.9821
Batch 2100/5872 - Loss: 0.0412 Acc: 0.9826
Batch 2200/5872 - Loss: 0.0415 Acc: 0.9826
Batch 2300/5872 - Loss: 0.0420 Acc: 0.9826
Batch 2400/5872 - Loss: 0.0426 Acc: 0.9823
Batch 2500/5872 - Loss: 0.0428 Acc: 0.9822
Batch 2600/5872 - Loss: 0.0423 Acc: 0.9824
Batch 2700/5872 - Loss: 0.0424 Acc: 0.9824
Batch 2800/5872 - Loss: 0.0422 Acc: 0.9825
Batch 2900/5872 - Loss: 0.0422 Acc: 0.9825
Batch 3000/5872 - Loss: 0.0426 Acc: 0.9822
Batch 3100/5872 - Loss: 0.0431 Acc: 0.9820
Batch 3200/5872 - Loss: 0.0432 Acc: 0.9820
Batch 3300/5872 - Loss: 0.0431 Acc: 0.9820
Batch 3400/5872 - Loss: 0.0428 Acc: 0.9822
Batch 3500/5872 - Loss: 0.0431 Acc: 0.9821
Batch 3600/5872 - Loss: 0.0429 Acc: 0.9822
Batch 3700/5872 - Loss: 0.0434 Acc: 0.9821
Batch 3800/5872 - Loss: 0.0434 Acc: 0.9821
Batch 3900/5872 - Loss: 0.0434 Acc: 0.9821
Batch 4000/5872 - Loss: 0.0430 Acc: 0.9823
Batch 4100/5872 - Loss: 0.0431 Acc: 0.9823
Batch 4200/5872 - Loss: 0.0429 Acc: 0.9824
Batch 4300/5872 - Loss: 0.0429 Acc: 0.9823
Batch 4400/5872 - Loss: 0.0428 Acc: 0.9823
```

```
Batch 4500/5872 - Loss: 0.0427 Acc: 0.9824
Batch 4600/5872 - Loss: 0.0426 Acc: 0.9825
Batch 4700/5872 - Loss: 0.0427 Acc: 0.9824
Batch 4800/5872 - Loss: 0.0424 Acc: 0.9825
Batch 4900/5872 - Loss: 0.0423 Acc: 0.9826
Batch 5000/5872 - Loss: 0.0422 Acc: 0.9826
Batch 5100/5872 - Loss: 0.0420 Acc: 0.9827
Batch 5200/5872 - Loss: 0.0425 Acc: 0.9827
Batch 5300/5872 - Loss: 0.0430 Acc: 0.9826
Batch 5400/5872 - Loss: 0.0430 Acc: 0.9827
Batch 5500/5872 - Loss: 0.0431 Acc: 0.9827
Batch 5600/5872 - Loss: 0.0427 Acc: 0.9827
Batch 5700/5872 - Loss: 0.0429 Acc: 0.9827
Batch 5800/5872 - Loss: 0.0430 Acc: 0.9826
Epoch 8 - Train Loss: 0.0433 Train Acc: 0.9825 Val Loss: 0.8631 Val Acc: 0.8863
→ Fold 1 Best Val Acc: 0.8885

==============================
Fold 2
==============================

Some weights of BertForSequenceClassification were not initialized from the
model checkpoint at bert-base-uncased and are newly initialized:
['classifier.bias', 'classifier.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.


Epoch 1/8
Batch 100/5872 - Loss: 2.4844 Acc: 0.1437
Batch 200/5872 - Loss: 2.2382 Acc: 0.2656
Batch 300/5872 - Loss: 2.0268 Acc: 0.3471
Batch 400/5872 - Loss: 1.8367 Acc: 0.4100
Batch 500/5872 - Loss: 1.6881 Acc: 0.4580
Batch 600/5872 - Loss: 1.5731 Acc: 0.4948
Batch 700/5872 - Loss: 1.4736 Acc: 0.5284
Batch 800/5872 - Loss: 1.3936 Acc: 0.5534
Batch 900/5872 - Loss: 1.3215 Acc: 0.5767
Batch 1000/5872 - Loss: 1.2542 Acc: 0.5996
Batch 1100/5872 - Loss: 1.1972 Acc: 0.6160
Batch 1200/5872 - Loss: 1.1493 Acc: 0.6318
Batch 1300/5872 - Loss: 1.1044 Acc: 0.6471
Batch 1400/5872 - Loss: 1.0581 Acc: 0.6613
Batch 1500/5872 - Loss: 1.0167 Acc: 0.6747
Batch 1600/5872 - Loss: 0.9833 Acc: 0.6860
Batch 1700/5872 - Loss: 0.9510 Acc: 0.6971
Batch 1800/5872 - Loss: 0.9237 Acc: 0.7064
Batch 1900/5872 - Loss: 0.8991 Acc: 0.7148
Batch 2000/5872 - Loss: 0.8729 Acc: 0.7234
```

```
Batch 2100/5872 - Loss: 0.8516 Acc: 0.7306
Batch 2200/5872 - Loss: 0.8332 Acc: 0.7369
Batch 2300/5872 - Loss: 0.8117 Acc: 0.7440
Batch 2400/5872 - Loss: 0.7924 Acc: 0.7504
Batch 2500/5872 - Loss: 0.7747 Acc: 0.7564
Batch 2600/5872 - Loss: 0.7575 Acc: 0.7624
Batch 2700/5872 - Loss: 0.7395 Acc: 0.7681
Batch 2800/5872 - Loss: 0.7255 Acc: 0.7729
Batch 2900/5872 - Loss: 0.7125 Acc: 0.7773
Batch 3000/5872 - Loss: 0.6991 Acc: 0.7819
Batch 3100/5872 - Loss: 0.6839 Acc: 0.7867
Batch 3200/5872 - Loss: 0.6726 Acc: 0.7908
Batch 3300/5872 - Loss: 0.6609 Acc: 0.7949
Batch 3400/5872 - Loss: 0.6486 Acc: 0.7989
Batch 3500/5872 - Loss: 0.6391 Acc: 0.8023
Batch 3600/5872 - Loss: 0.6270 Acc: 0.8062
Batch 3700/5872 - Loss: 0.6174 Acc: 0.8096
Batch 3800/5872 - Loss: 0.6072 Acc: 0.8133
Batch 3900/5872 - Loss: 0.5968 Acc: 0.8166
Batch 4000/5872 - Loss: 0.5874 Acc: 0.8198
Batch 4100/5872 - Loss: 0.5784 Acc: 0.8228
Batch 4200/5872 - Loss: 0.5701 Acc: 0.8256
Batch 4300/5872 - Loss: 0.5645 Acc: 0.8280
Batch 4400/5872 - Loss: 0.5584 Acc: 0.8303
Batch 4500/5872 - Loss: 0.5495 Acc: 0.8331
Batch 4600/5872 - Loss: 0.5424 Acc: 0.8354
Batch 4700/5872 - Loss: 0.5368 Acc: 0.8376
Batch 4800/5872 - Loss: 0.5299 Acc: 0.8398
Batch 4900/5872 - Loss: 0.5239 Acc: 0.8418
Batch 5000/5872 - Loss: 0.5179 Acc: 0.8438
Batch 5100/5872 - Loss: 0.5113 Acc: 0.8459
Batch 5200/5872 - Loss: 0.5060 Acc: 0.8478
Batch 5300/5872 - Loss: 0.5003 Acc: 0.8496
Batch 5400/5872 - Loss: 0.4944 Acc: 0.8514
Batch 5500/5872 - Loss: 0.4882 Acc: 0.8533
Batch 5600/5872 - Loss: 0.4820 Acc: 0.8552
Batch 5700/5872 - Loss: 0.4770 Acc: 0.8571
Batch 5800/5872 - Loss: 0.4727 Acc: 0.8587
Epoch 1 - Train Loss: 0.4686 Train Acc: 0.8600 Val Loss: 0.6522 Val Acc: 0.8753

Epoch 2/8
Batch 100/5872 - Loss: 0.1105 Acc: 0.9700
Batch 200/5872 - Loss: 0.1498 Acc: 0.9663
Batch 300/5872 - Loss: 0.1456 Acc: 0.9663
Batch 400/5872 - Loss: 0.1445 Acc: 0.9666
Batch 500/5872 - Loss: 0.1429 Acc: 0.9667
Batch 600/5872 - Loss: 0.1367 Acc: 0.9683
Batch 700/5872 - Loss: 0.1345 Acc: 0.9689
```

```
Batch 800/5872 - Loss: 0.1387 Acc: 0.9680
Batch 900/5872 - Loss: 0.1348 Acc: 0.9678
Batch 1000/5872 - Loss: 0.1360 Acc: 0.9675
Batch 1100/5872 - Loss: 0.1369 Acc: 0.9670
Batch 1200/5872 - Loss: 0.1361 Acc: 0.9671
Batch 1300/5872 - Loss: 0.1355 Acc: 0.9674
Batch 1400/5872 - Loss: 0.1371 Acc: 0.9668
Batch 1500/5872 - Loss: 0.1398 Acc: 0.9662
Batch 1600/5872 - Loss: 0.1401 Acc: 0.9658
Batch 1700/5872 - Loss: 0.1405 Acc: 0.9656
Batch 1800/5872 - Loss: 0.1405 Acc: 0.9656
Batch 1900/5872 - Loss: 0.1396 Acc: 0.9655
Batch 2000/5872 - Loss: 0.1367 Acc: 0.9662
Batch 2100/5872 - Loss: 0.1357 Acc: 0.9664
Batch 2200/5872 - Loss: 0.1348 Acc: 0.9669
Batch 2300/5872 - Loss: 0.1330 Acc: 0.9673
Batch 2400/5872 - Loss: 0.1334 Acc: 0.9672
Batch 2500/5872 - Loss: 0.1318 Acc: 0.9677
Batch 2600/5872 - Loss: 0.1311 Acc: 0.9680
Batch 2700/5872 - Loss: 0.1299 Acc: 0.9681
Batch 2800/5872 - Loss: 0.1302 Acc: 0.9680
Batch 2900/5872 - Loss: 0.1287 Acc: 0.9682
Batch 3000/5872 - Loss: 0.1275 Acc: 0.9685
Batch 3100/5872 - Loss: 0.1274 Acc: 0.9685
Batch 3200/5872 - Loss: 0.1263 Acc: 0.9687
Batch 3300/5872 - Loss: 0.1256 Acc: 0.9690
Batch 3400/5872 - Loss: 0.1259 Acc: 0.9691
Batch 3500/5872 - Loss: 0.1260 Acc: 0.9691
Batch 3600/5872 - Loss: 0.1261 Acc: 0.9690
Batch 3700/5872 - Loss: 0.1260 Acc: 0.9688
Batch 3800/5872 - Loss: 0.1263 Acc: 0.9688
Batch 3900/5872 - Loss: 0.1263 Acc: 0.9688
Batch 4000/5872 - Loss: 0.1264 Acc: 0.9688
Batch 4100/5872 - Loss: 0.1253 Acc: 0.9689
Batch 4200/5872 - Loss: 0.1259 Acc: 0.9687
Batch 4300/5872 - Loss: 0.1259 Acc: 0.9686
Batch 4400/5872 - Loss: 0.1258 Acc: 0.9687
Batch 4500/5872 - Loss: 0.1258 Acc: 0.9688
Batch 4600/5872 - Loss: 0.1258 Acc: 0.9688
Batch 4700/5872 - Loss: 0.1254 Acc: 0.9689
Batch 4800/5872 - Loss: 0.1254 Acc: 0.9689
Batch 4900/5872 - Loss: 0.1259 Acc: 0.9688
Batch 5000/5872 - Loss: 0.1249 Acc: 0.9690
Batch 5100/5872 - Loss: 0.1255 Acc: 0.9688
Batch 5200/5872 - Loss: 0.1246 Acc: 0.9689
Batch 5300/5872 - Loss: 0.1249 Acc: 0.9689
Batch 5400/5872 - Loss: 0.1243 Acc: 0.9689
Batch 5500/5872 - Loss: 0.1236 Acc: 0.9690
```

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Batch 5600/5872 - Loss: 0.1228 Acc: 0.9692
Batch 5700/5872 - Loss: 0.1221 Acc: 0.9692
Batch 5800/5872 - Loss: 0.1224 Acc: 0.9692
Epoch 2 - Train Loss: 0.1225 Train Acc: 0.9692 Val Loss: 0.7731 Val Acc: 0.8709

Epoch 3/8
Batch 100/5872 - Loss: 0.0771 Acc: 0.9738
Batch 200/5872 - Loss: 0.0700 Acc: 0.9788
Batch 300/5872 - Loss: 0.0740 Acc: 0.9767
Batch 400/5872 - Loss: 0.0743 Acc: 0.9778
Batch 500/5872 - Loss: 0.0765 Acc: 0.9768
Batch 600/5872 - Loss: 0.0795 Acc: 0.9765
Batch 700/5872 - Loss: 0.0792 Acc: 0.9773
Batch 800/5872 - Loss: 0.0762 Acc: 0.9780
Batch 900/5872 - Loss: 0.0807 Acc: 0.9772
Batch 1000/5872 - Loss: 0.0769 Acc: 0.9781
Batch 1100/5872 - Loss: 0.0782 Acc: 0.9776
Batch 1200/5872 - Loss: 0.0777 Acc: 0.9778
Batch 1300/5872 - Loss: 0.0779 Acc: 0.9780
Batch 1400/5872 - Loss: 0.0806 Acc: 0.9769
Batch 1500/5872 - Loss: 0.0836 Acc: 0.9763
Batch 1600/5872 - Loss: 0.0833 Acc: 0.9765
Batch 1700/5872 - Loss: 0.0840 Acc: 0.9765
Batch 1800/5872 - Loss: 0.0837 Acc: 0.9761
Batch 1900/5872 - Loss: 0.0826 Acc: 0.9766
Batch 2000/5872 - Loss: 0.0821 Acc: 0.9769
Batch 2100/5872 - Loss: 0.0819 Acc: 0.9772
Batch 2200/5872 - Loss: 0.0825 Acc: 0.9772
Batch 2300/5872 - Loss: 0.0852 Acc: 0.9766
Batch 2400/5872 - Loss: 0.0850 Acc: 0.9768
Batch 2500/5872 - Loss: 0.0853 Acc: 0.9769
Batch 2600/5872 - Loss: 0.0858 Acc: 0.9768
Batch 2700/5872 - Loss: 0.0843 Acc: 0.9773
Batch 2800/5872 - Loss: 0.0849 Acc: 0.9772
Batch 2900/5872 - Loss: 0.0847 Acc: 0.9772
Batch 3000/5872 - Loss: 0.0841 Acc: 0.9774
Batch 3100/5872 - Loss: 0.0845 Acc: 0.9773
Batch 3200/5872 - Loss: 0.0837 Acc: 0.9774
Batch 3300/5872 - Loss: 0.0835 Acc: 0.9773
Batch 3400/5872 - Loss: 0.0831 Acc: 0.9774
Batch 3500/5872 - Loss: 0.0832 Acc: 0.9773
Batch 3600/5872 - Loss: 0.0822 Acc: 0.9775
Batch 3700/5872 - Loss: 0.0816 Acc: 0.9777
Batch 3800/5872 - Loss: 0.0813 Acc: 0.9776
Batch 3900/5872 - Loss: 0.0816 Acc: 0.9775
Batch 4000/5872 - Loss: 0.0819 Acc: 0.9774
Batch 4100/5872 - Loss: 0.0813 Acc: 0.9775
Batch 4200/5872 - Loss: 0.0814 Acc: 0.9775
```

```
Batch 4300/5872 - Loss: 0.0816 Acc: 0.9773
Batch 4400/5872 - Loss: 0.0815 Acc: 0.9773
Batch 4500/5872 - Loss: 0.0821 Acc: 0.9770
Batch 4600/5872 - Loss: 0.0822 Acc: 0.9770
Batch 4700/5872 - Loss: 0.0820 Acc: 0.9770
Batch 4800/5872 - Loss: 0.0821 Acc: 0.9770
Batch 4900/5872 - Loss: 0.0825 Acc: 0.9769
Batch 5000/5872 - Loss: 0.0825 Acc: 0.9768
Batch 5100/5872 - Loss: 0.0827 Acc: 0.9767
Batch 5200/5872 - Loss: 0.0825 Acc: 0.9768
Batch 5300/5872 - Loss: 0.0828 Acc: 0.9767
Batch 5400/5872 - Loss: 0.0827 Acc: 0.9767
Batch 5500/5872 - Loss: 0.0825 Acc: 0.9768
Batch 5600/5872 - Loss: 0.0831 Acc: 0.9767
Batch 5700/5872 - Loss: 0.0832 Acc: 0.9767
Batch 5800/5872 - Loss: 0.0839 Acc: 0.9765
Epoch 3 - Train Loss: 0.0841 Train Acc: 0.9764 Val Loss: 0.6367 Val Acc: 0.9043

Epoch 4/8
Batch 100/5872 - Loss: 0.0863 Acc: 0.9762
Batch 200/5872 - Loss: 0.0785 Acc: 0.9762
Batch 300/5872 - Loss: 0.0930 Acc: 0.9721
Batch 400/5872 - Loss: 0.0846 Acc: 0.9747
Batch 500/5872 - Loss: 0.0861 Acc: 0.9745
Batch 600/5872 - Loss: 0.0829 Acc: 0.9752
Batch 700/5872 - Loss: 0.0783 Acc: 0.9766
Batch 800/5872 - Loss: 0.0793 Acc: 0.9769
Batch 900/5872 - Loss: 0.0785 Acc: 0.9768
Batch 1000/5872 - Loss: 0.0752 Acc: 0.9776
Batch 1100/5872 - Loss: 0.0758 Acc: 0.9781
Batch 1200/5872 - Loss: 0.0760 Acc: 0.9778
Batch 1300/5872 - Loss: 0.0744 Acc: 0.9783
Batch 1400/5872 - Loss: 0.0725 Acc: 0.9788
Batch 1500/5872 - Loss: 0.0712 Acc: 0.9790
Batch 1600/5872 - Loss: 0.0706 Acc: 0.9791
Batch 1700/5872 - Loss: 0.0702 Acc: 0.9793
Batch 1800/5872 - Loss: 0.0706 Acc: 0.9792
Batch 1900/5872 - Loss: 0.0706 Acc: 0.9790
Batch 2000/5872 - Loss: 0.0689 Acc: 0.9796
Batch 2100/5872 - Loss: 0.0677 Acc: 0.9799
Batch 2200/5872 - Loss: 0.0676 Acc: 0.9799
Batch 2300/5872 - Loss: 0.0694 Acc: 0.9797
Batch 2400/5872 - Loss: 0.0692 Acc: 0.9797
Batch 2500/5872 - Loss: 0.0690 Acc: 0.9798
Batch 2600/5872 - Loss: 0.0685 Acc: 0.9798
Batch 2700/5872 - Loss: 0.0688 Acc: 0.9797
Batch 2800/5872 - Loss: 0.0689 Acc: 0.9796
Batch 2900/5872 - Loss: 0.0691 Acc: 0.9796
```

```
Batch 3000/5872 - Loss: 0.0700 Acc: 0.9792
Batch 3100/5872 - Loss: 0.0691 Acc: 0.9795
Batch 3200/5872 - Loss: 0.0688 Acc: 0.9795
Batch 3300/5872 - Loss: 0.0698 Acc: 0.9792
Batch 3400/5872 - Loss: 0.0691 Acc: 0.9794
Batch 3500/5872 - Loss: 0.0694 Acc: 0.9793
Batch 3600/5872 - Loss: 0.0694 Acc: 0.9793
Batch 3700/5872 - Loss: 0.0693 Acc: 0.9794
Batch 3800/5872 - Loss: 0.0691 Acc: 0.9793
Batch 3900/5872 - Loss: 0.0692 Acc: 0.9793
Batch 4000/5872 - Loss: 0.0709 Acc: 0.9790
Batch 4100/5872 - Loss: 0.0703 Acc: 0.9791
Batch 4200/5872 - Loss: 0.0708 Acc: 0.9790
Batch 4300/5872 - Loss: 0.0715 Acc: 0.9789
Batch 4400/5872 - Loss: 0.0716 Acc: 0.9788
Batch 4500/5872 - Loss: 0.0715 Acc: 0.9788
Batch 4600/5872 - Loss: 0.0715 Acc: 0.9788
Batch 4700/5872 - Loss: 0.0719 Acc: 0.9786
Batch 4800/5872 - Loss: 0.0715 Acc: 0.9787
Batch 4900/5872 - Loss: 0.0711 Acc: 0.9787
Batch 5000/5872 - Loss: 0.0711 Acc: 0.9787
Batch 5100/5872 - Loss: 0.0711 Acc: 0.9787
Batch 5200/5872 - Loss: 0.0716 Acc: 0.9785
Batch 5300/5872 - Loss: 0.0715 Acc: 0.9786
Batch 5400/5872 - Loss: 0.0714 Acc: 0.9786
Batch 5500/5872 - Loss: 0.0711 Acc: 0.9787
Batch 5600/5872 - Loss: 0.0709 Acc: 0.9787
Batch 5700/5872 - Loss: 0.0711 Acc: 0.9786
Batch 5800/5872 - Loss: 0.0707 Acc: 0.9787
Epoch 4 - Train Loss: 0.0709 Train Acc: 0.9786 Val Loss: 0.7361 Val Acc: 0.8855

Epoch 5/8
Batch 100/5872 - Loss: 0.0728 Acc: 0.9800
Batch 200/5872 - Loss: 0.0774 Acc: 0.9775
Batch 300/5872 - Loss: 0.0779 Acc: 0.9738
Batch 400/5872 - Loss: 0.0684 Acc: 0.9759
Batch 500/5872 - Loss: 0.0689 Acc: 0.9762
Batch 600/5872 - Loss: 0.0717 Acc: 0.9769
Batch 700/5872 - Loss: 0.0719 Acc: 0.9770
Batch 800/5872 - Loss: 0.0706 Acc: 0.9772
Batch 900/5872 - Loss: 0.0688 Acc: 0.9778
Batch 1000/5872 - Loss: 0.0683 Acc: 0.9779
Batch 1100/5872 - Loss: 0.0667 Acc: 0.9778
Batch 1200/5872 - Loss: 0.0667 Acc: 0.9779
Batch 1300/5872 - Loss: 0.0654 Acc: 0.9784
Batch 1400/5872 - Loss: 0.0673 Acc: 0.9779
Batch 1500/5872 - Loss: 0.0660 Acc: 0.9787
Batch 1600/5872 - Loss: 0.0664 Acc: 0.9786
```

```
Batch 1700/5872 - Loss: 0.0658 Acc: 0.9785
Batch 1800/5872 - Loss: 0.0648 Acc: 0.9788
Batch 1900/5872 - Loss: 0.0655 Acc: 0.9784
Batch 2000/5872 - Loss: 0.0656 Acc: 0.9786
Batch 2100/5872 - Loss: 0.0644 Acc: 0.9790
Batch 2200/5872 - Loss: 0.0645 Acc: 0.9790
Batch 2300/5872 - Loss: 0.0634 Acc: 0.9793
Batch 2400/5872 - Loss: 0.0633 Acc: 0.9792
Batch 2500/5872 - Loss: 0.0631 Acc: 0.9790
Batch 2600/5872 - Loss: 0.0632 Acc: 0.9790
Batch 2700/5872 - Loss: 0.0640 Acc: 0.9788
Batch 2800/5872 - Loss: 0.0636 Acc: 0.9791
Batch 2900/5872 - Loss: 0.0629 Acc: 0.9794
Batch 3000/5872 - Loss: 0.0626 Acc: 0.9795
Batch 3100/5872 - Loss: 0.0628 Acc: 0.9797
Batch 3200/5872 - Loss: 0.0622 Acc: 0.9798
Batch 3300/5872 - Loss: 0.0619 Acc: 0.9799
Batch 3400/5872 - Loss: 0.0621 Acc: 0.9799
Batch 3500/5872 - Loss: 0.0611 Acc: 0.9801
Batch 3600/5872 - Loss: 0.0612 Acc: 0.9802
Batch 3700/5872 - Loss: 0.0608 Acc: 0.9804
Batch 3800/5872 - Loss: 0.0607 Acc: 0.9804
Batch 3900/5872 - Loss: 0.0604 Acc: 0.9804
Batch 4000/5872 - Loss: 0.0604 Acc: 0.9805
Batch 4100/5872 - Loss: 0.0604 Acc: 0.9805
Batch 4200/5872 - Loss: 0.0603 Acc: 0.9805
Batch 4300/5872 - Loss: 0.0599 Acc: 0.9806
Batch 4400/5872 - Loss: 0.0595 Acc: 0.9807
Batch 4500/5872 - Loss: 0.0605 Acc: 0.9805
Batch 4600/5872 - Loss: 0.0602 Acc: 0.9806
Batch 4700/5872 - Loss: 0.0601 Acc: 0.9806
Batch 4800/5872 - Loss: 0.0598 Acc: 0.9807
Batch 4900/5872 - Loss: 0.0598 Acc: 0.9807
Batch 5000/5872 - Loss: 0.0596 Acc: 0.9807
Batch 5100/5872 - Loss: 0.0591 Acc: 0.9808
Batch 5200/5872 - Loss: 0.0592 Acc: 0.9808
Batch 5300/5872 - Loss: 0.0588 Acc: 0.9808
Batch 5400/5872 - Loss: 0.0589 Acc: 0.9808
Batch 5500/5872 - Loss: 0.0586 Acc: 0.9808
Batch 5600/5872 - Loss: 0.0585 Acc: 0.9809
Batch 5700/5872 - Loss: 0.0591 Acc: 0.9807
Batch 5800/5872 - Loss: 0.0588 Acc: 0.9809
Epoch 5 - Train Loss: 0.0588 Train Acc: 0.9809 Val Loss: 0.6935 Val Acc: 0.9039

Epoch 6/8
Batch 100/5872 - Loss: 0.0350 Acc: 0.9888
Batch 200/5872 - Loss: 0.0427 Acc: 0.9869
Batch 300/5872 - Loss: 0.0410 Acc: 0.9883
```

```
Batch 400/5872 - Loss: 0.0394 Acc: 0.9884
Batch 500/5872 - Loss: 0.0411 Acc: 0.9880
Batch 600/5872 - Loss: 0.0421 Acc: 0.9871
Batch 700/5872 - Loss: 0.0432 Acc: 0.9866
Batch 800/5872 - Loss: 0.0469 Acc: 0.9855
Batch 900/5872 - Loss: 0.0476 Acc: 0.9850
Batch 1000/5872 - Loss: 0.0479 Acc: 0.9849
Batch 1100/5872 - Loss: 0.0473 Acc: 0.9850
Batch 1200/5872 - Loss: 0.0488 Acc: 0.9847
Batch 1300/5872 - Loss: 0.0490 Acc: 0.9845
Batch 1400/5872 - Loss: 0.0491 Acc: 0.9844
Batch 1500/5872 - Loss: 0.0497 Acc: 0.9840
Batch 1600/5872 - Loss: 0.0499 Acc: 0.9841
Batch 1700/5872 - Loss: 0.0496 Acc: 0.9841
Batch 1800/5872 - Loss: 0.0514 Acc: 0.9835
Batch 1900/5872 - Loss: 0.0521 Acc: 0.9832
Batch 2000/5872 - Loss: 0.0523 Acc: 0.9831
Batch 2100/5872 - Loss: 0.0533 Acc: 0.9828
Batch 2200/5872 - Loss: 0.0527 Acc: 0.9830
Batch 2300/5872 - Loss: 0.0538 Acc: 0.9826
Batch 2400/5872 - Loss: 0.0532 Acc: 0.9829
Batch 2500/5872 - Loss: 0.0549 Acc: 0.9823
Batch 2600/5872 - Loss: 0.0534 Acc: 0.9828
Batch 2700/5872 - Loss: 0.0538 Acc: 0.9825
Batch 2800/5872 - Loss: 0.0543 Acc: 0.9823
Batch 2900/5872 - Loss: 0.0543 Acc: 0.9821
Batch 3000/5872 - Loss: 0.0553 Acc: 0.9818
Batch 3100/5872 - Loss: 0.0551 Acc: 0.9817
Batch 3200/5872 - Loss: 0.0545 Acc: 0.9820
Batch 3300/5872 - Loss: 0.0547 Acc: 0.9819
Batch 3400/5872 - Loss: 0.0543 Acc: 0.9819
Batch 3500/5872 - Loss: 0.0539 Acc: 0.9821
Batch 3600/5872 - Loss: 0.0536 Acc: 0.9822
Batch 3700/5872 - Loss: 0.0535 Acc: 0.9823
Batch 3800/5872 - Loss: 0.0535 Acc: 0.9823
Batch 3900/5872 - Loss: 0.0533 Acc: 0.9824
Batch 4000/5872 - Loss: 0.0535 Acc: 0.9823
Batch 4100/5872 - Loss: 0.0533 Acc: 0.9823
Batch 4200/5872 - Loss: 0.0529 Acc: 0.9824
Batch 4300/5872 - Loss: 0.0529 Acc: 0.9824
Batch 4400/5872 - Loss: 0.0525 Acc: 0.9824
Batch 4500/5872 - Loss: 0.0522 Acc: 0.9825
Batch 4600/5872 - Loss: 0.0523 Acc: 0.9825
Batch 4700/5872 - Loss: 0.0528 Acc: 0.9824
Batch 4800/5872 - Loss: 0.0531 Acc: 0.9823
Batch 4900/5872 - Loss: 0.0535 Acc: 0.9822
Batch 5000/5872 - Loss: 0.0535 Acc: 0.9821
Batch 5100/5872 - Loss: 0.0533 Acc: 0.9822
```

```
Batch 5200/5872 - Loss: 0.0533 Acc: 0.9822
Batch 5300/5872 - Loss: 0.0530 Acc: 0.9823
Batch 5400/5872 - Loss: 0.0529 Acc: 0.9824
Batch 5500/5872 - Loss: 0.0526 Acc: 0.9824
Batch 5600/5872 - Loss: 0.0523 Acc: 0.9825
Batch 5700/5872 - Loss: 0.0525 Acc: 0.9823
Batch 5800/5872 - Loss: 0.0527 Acc: 0.9822
Epoch 6 - Train Loss: 0.0526 Train Acc: 0.9822 Val Loss: 0.6902 Val Acc: 0.9035

Epoch 7/8
Batch 100/5872 - Loss: 0.0356 Acc: 0.9900
Batch 200/5872 - Loss: 0.0443 Acc: 0.9850
Batch 300/5872 - Loss: 0.0431 Acc: 0.9846
Batch 400/5872 - Loss: 0.0440 Acc: 0.9856
Batch 500/5872 - Loss: 0.0460 Acc: 0.9848
Batch 600/5872 - Loss: 0.0457 Acc: 0.9846
Batch 700/5872 - Loss: 0.0449 Acc: 0.9845
Batch 800/5872 - Loss: 0.0448 Acc: 0.9844
Batch 900/5872 - Loss: 0.0451 Acc: 0.9840
Batch 1000/5872 - Loss: 0.0441 Acc: 0.9842
Batch 1100/5872 - Loss: 0.0448 Acc: 0.9841
Batch 1200/5872 - Loss: 0.0448 Acc: 0.9842
Batch 1300/5872 - Loss: 0.0444 Acc: 0.9843
Batch 1400/5872 - Loss: 0.0450 Acc: 0.9841
Batch 1500/5872 - Loss: 0.0468 Acc: 0.9836
Batch 1600/5872 - Loss: 0.0466 Acc: 0.9838
Batch 1700/5872 - Loss: 0.0470 Acc: 0.9835
Batch 1800/5872 - Loss: 0.0469 Acc: 0.9835
Batch 1900/5872 - Loss: 0.0465 Acc: 0.9836
Batch 2000/5872 - Loss: 0.0458 Acc: 0.9838
Batch 2100/5872 - Loss: 0.0456 Acc: 0.9839
Batch 2200/5872 - Loss: 0.0455 Acc: 0.9838
Batch 2300/5872 - Loss: 0.0459 Acc: 0.9834
Batch 2400/5872 - Loss: 0.0450 Acc: 0.9836
Batch 2500/5872 - Loss: 0.0446 Acc: 0.9837
Batch 2600/5872 - Loss: 0.0443 Acc: 0.9838
Batch 2700/5872 - Loss: 0.0449 Acc: 0.9837
Batch 2800/5872 - Loss: 0.0459 Acc: 0.9833
Batch 2900/5872 - Loss: 0.0464 Acc: 0.9829
Batch 3000/5872 - Loss: 0.0461 Acc: 0.9831
Batch 3100/5872 - Loss: 0.0458 Acc: 0.9832
Batch 3200/5872 - Loss: 0.0456 Acc: 0.9833
Batch 3300/5872 - Loss: 0.0461 Acc: 0.9831
Batch 3400/5872 - Loss: 0.0462 Acc: 0.9833
Batch 3500/5872 - Loss: 0.0462 Acc: 0.9834
Batch 3600/5872 - Loss: 0.0461 Acc: 0.9834
Batch 3700/5872 - Loss: 0.0464 Acc: 0.9833
Batch 3800/5872 - Loss: 0.0465 Acc: 0.9833
```

```
Batch 3900/5872 - Loss: 0.0465 Acc: 0.9833
Batch 4000/5872 - Loss: 0.0466 Acc: 0.9833
Batch 4100/5872 - Loss: 0.0470 Acc: 0.9831
Batch 4200/5872 - Loss: 0.0470 Acc: 0.9831
Batch 4300/5872 - Loss: 0.0469 Acc: 0.9832
Batch 4400/5872 - Loss: 0.0470 Acc: 0.9832
Batch 4500/5872 - Loss: 0.0467 Acc: 0.9834
Batch 4600/5872 - Loss: 0.0469 Acc: 0.9833
Batch 4700/5872 - Loss: 0.0471 Acc: 0.9832
Batch 4800/5872 - Loss: 0.0471 Acc: 0.9833
Batch 4900/5872 - Loss: 0.0480 Acc: 0.9829
Batch 5000/5872 - Loss: 0.0477 Acc: 0.9830
Batch 5100/5872 - Loss: 0.0477 Acc: 0.9830
Batch 5200/5872 - Loss: 0.0477 Acc: 0.9830
Batch 5300/5872 - Loss: 0.0478 Acc: 0.9829
Batch 5400/5872 - Loss: 0.0474 Acc: 0.9830
Batch 5500/5872 - Loss: 0.0481 Acc: 0.9826
Batch 5600/5872 - Loss: 0.0480 Acc: 0.9826
Batch 5700/5872 - Loss: 0.0479 Acc: 0.9827
Batch 5800/5872 - Loss: 0.0476 Acc: 0.9828
Epoch 7 - Train Loss: 0.0477 Train Acc: 0.9827 Val Loss: 0.7081 Val Acc: 0.9072

Epoch 8/8
Batch 100/5872 - Loss: 0.0388 Acc: 0.9850
Batch 200/5872 - Loss: 0.0444 Acc: 0.9844
Batch 300/5872 - Loss: 0.0497 Acc: 0.9817
Batch 400/5872 - Loss: 0.0437 Acc: 0.9834
Batch 500/5872 - Loss: 0.0441 Acc: 0.9838
Batch 600/5872 - Loss: 0.0453 Acc: 0.9835
Batch 700/5872 - Loss: 0.0456 Acc: 0.9832
Batch 800/5872 - Loss: 0.0464 Acc: 0.9827
Batch 900/5872 - Loss: 0.0461 Acc: 0.9828
Batch 1000/5872 - Loss: 0.0445 Acc: 0.9834
Batch 1100/5872 - Loss: 0.0453 Acc: 0.9832
Batch 1200/5872 - Loss: 0.0453 Acc: 0.9832
Batch 1300/5872 - Loss: 0.0449 Acc: 0.9835
Batch 1400/5872 - Loss: 0.0431 Acc: 0.9840
Batch 1500/5872 - Loss: 0.0425 Acc: 0.9842
Batch 1600/5872 - Loss: 0.0437 Acc: 0.9838
Batch 1700/5872 - Loss: 0.0441 Acc: 0.9838
Batch 1800/5872 - Loss: 0.0438 Acc: 0.9839
Batch 1900/5872 - Loss: 0.0436 Acc: 0.9840
Batch 2000/5872 - Loss: 0.0429 Acc: 0.9843
Batch 2100/5872 - Loss: 0.0436 Acc: 0.9842
Batch 2200/5872 - Loss: 0.0436 Acc: 0.9843
Batch 2300/5872 - Loss: 0.0434 Acc: 0.9842
Batch 2400/5872 - Loss: 0.0434 Acc: 0.9842
Batch 2500/5872 - Loss: 0.0435 Acc: 0.9840
```

```
Batch 2600/5872 - Loss: 0.0432 Acc: 0.9842
Batch 2700/5872 - Loss: 0.0433 Acc: 0.9841
Batch 2800/5872 - Loss: 0.0437 Acc: 0.9840
Batch 2900/5872 - Loss: 0.0434 Acc: 0.9840
Batch 3000/5872 - Loss: 0.0438 Acc: 0.9839
Batch 3100/5872 - Loss: 0.0440 Acc: 0.9839
Batch 3200/5872 - Loss: 0.0438 Acc: 0.9839
Batch 3300/5872 - Loss: 0.0436 Acc: 0.9839
Batch 3400/5872 - Loss: 0.0434 Acc: 0.9840
Batch 3500/5872 - Loss: 0.0436 Acc: 0.9839
Batch 3600/5872 - Loss: 0.0434 Acc: 0.9839
Batch 3700/5872 - Loss: 0.0433 Acc: 0.9840
Batch 3800/5872 - Loss: 0.0432 Acc: 0.9840
Batch 3900/5872 - Loss: 0.0432 Acc: 0.9841
Batch 4000/5872 - Loss: 0.0430 Acc: 0.9842
Batch 4100/5872 - Loss: 0.0429 Acc: 0.9842
Batch 4200/5872 - Loss: 0.0433 Acc: 0.9840
Batch 4300/5872 - Loss: 0.0434 Acc: 0.9840
Batch 4400/5872 - Loss: 0.0437 Acc: 0.9838
Batch 4500/5872 - Loss: 0.0438 Acc: 0.9838
Batch 4600/5872 - Loss: 0.0441 Acc: 0.9837
Batch 4700/5872 - Loss: 0.0441 Acc: 0.9838
Batch 4800/5872 - Loss: 0.0443 Acc: 0.9837
Batch 4900/5872 - Loss: 0.0443 Acc: 0.9836
Batch 5000/5872 - Loss: 0.0440 Acc: 0.9838
Batch 5100/5872 - Loss: 0.0442 Acc: 0.9837
Batch 5200/5872 - Loss: 0.0441 Acc: 0.9837
Batch 5300/5872 - Loss: 0.0443 Acc: 0.9836
Batch 5400/5872 - Loss: 0.0442 Acc: 0.9837
Batch 5500/5872 - Loss: 0.0441 Acc: 0.9837
Batch 5600/5872 - Loss: 0.0444 Acc: 0.9835
Batch 5700/5872 - Loss: 0.0442 Acc: 0.9835
Batch 5800/5872 - Loss: 0.0447 Acc: 0.9834
Epoch 8 - Train Loss: 0.0446 Train Acc: 0.9834 Val Loss: 0.7212 Val Acc: 0.9043
→ Fold 2 Best Val Acc: 0.9072

==============================
Fold 3
==============================

Some weights of BertForSequenceClassification were not initialized from the
model checkpoint at bert-base-uncased and are newly initialized:
['classifier.bias', 'classifier.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.


Epoch 1/8
Batch 100/5870 - Loss: 2.4493 Acc: 0.1787
```

```
Batch 200/5870 - Loss: 2.1665 Acc: 0.3031
Batch 300/5870 - Loss: 1.9178 Acc: 0.3908
Batch 400/5870 - Loss: 1.7700 Acc: 0.4387
Batch 500/5870 - Loss: 1.6344 Acc: 0.4815
Batch 600/5870 - Loss: 1.5215 Acc: 0.5212
Batch 700/5870 - Loss: 1.4215 Acc: 0.5552
Batch 800/5870 - Loss: 1.3420 Acc: 0.5800
Batch 900/5870 - Loss: 1.2750 Acc: 0.6004
Batch 1000/5870 - Loss: 1.2108 Acc: 0.6195
Batch 1100/5870 - Loss: 1.1637 Acc: 0.6347
Batch 1200/5870 - Loss: 1.1177 Acc: 0.6491
Batch 1300/5870 - Loss: 1.0790 Acc: 0.6613
Batch 1400/5870 - Loss: 1.0382 Acc: 0.6739
Batch 1500/5870 - Loss: 1.0050 Acc: 0.6846
Batch 1600/5870 - Loss: 0.9695 Acc: 0.6958
Batch 1700/5870 - Loss: 0.9384 Acc: 0.7057
Batch 1800/5870 - Loss: 0.9097 Acc: 0.7149
Batch 1900/5870 - Loss: 0.8823 Acc: 0.7238
Batch 2000/5870 - Loss: 0.8577 Acc: 0.7319
Batch 2100/5870 - Loss: 0.8348 Acc: 0.7397
Batch 2200/5870 - Loss: 0.8115 Acc: 0.7477
Batch 2300/5870 - Loss: 0.7942 Acc: 0.7532
Batch 2400/5870 - Loss: 0.7784 Acc: 0.7583
Batch 2500/5870 - Loss: 0.7596 Acc: 0.7645
Batch 2600/5870 - Loss: 0.7443 Acc: 0.7698
Batch 2700/5870 - Loss: 0.7290 Acc: 0.7748
Batch 2800/5870 - Loss: 0.7151 Acc: 0.7797
Batch 2900/5870 - Loss: 0.7032 Acc: 0.7841
Batch 3000/5870 - Loss: 0.6899 Acc: 0.7885
Batch 3100/5870 - Loss: 0.6765 Acc: 0.7928
Batch 3200/5870 - Loss: 0.6636 Acc: 0.7971
Batch 3300/5870 - Loss: 0.6536 Acc: 0.8010
Batch 3400/5870 - Loss: 0.6425 Acc: 0.8048
Batch 3500/5870 - Loss: 0.6318 Acc: 0.8084
Batch 3600/5870 - Loss: 0.6209 Acc: 0.8120
Batch 3700/5870 - Loss: 0.6112 Acc: 0.8152
Batch 3800/5870 - Loss: 0.6016 Acc: 0.8185
Batch 3900/5870 - Loss: 0.5918 Acc: 0.8217
Batch 4000/5870 - Loss: 0.5837 Acc: 0.8246
Batch 4100/5870 - Loss: 0.5749 Acc: 0.8275
Batch 4200/5870 - Loss: 0.5668 Acc: 0.8301
Batch 4300/5870 - Loss: 0.5585 Acc: 0.8328
Batch 4400/5870 - Loss: 0.5506 Acc: 0.8352
Batch 4500/5870 - Loss: 0.5425 Acc: 0.8377
Batch 4600/5870 - Loss: 0.5358 Acc: 0.8400
Batch 4700/5870 - Loss: 0.5293 Acc: 0.8421
Batch 4800/5870 - Loss: 0.5218 Acc: 0.8444
Batch 4900/5870 - Loss: 0.5164 Acc: 0.8462
```

```
Batch 5000/5870 - Loss: 0.5098 Acc: 0.8485
Batch 5100/5870 - Loss: 0.5033 Acc: 0.8507
Batch 5200/5870 - Loss: 0.4974 Acc: 0.8527
Batch 5300/5870 - Loss: 0.4916 Acc: 0.8546
Batch 5400/5870 - Loss: 0.4860 Acc: 0.8565
Batch 5500/5870 - Loss: 0.4811 Acc: 0.8583
Batch 5600/5870 - Loss: 0.4763 Acc: 0.8598
Batch 5700/5870 - Loss: 0.4725 Acc: 0.8614
Batch 5800/5870 - Loss: 0.4684 Acc: 0.8627
Epoch 1 - Train Loss: 0.4648 Train Acc: 0.8637 Val Loss: 0.6981 Val Acc: 0.8595

Epoch 2/8
Batch 100/5870 - Loss: 0.1637 Acc: 0.9587
Batch 200/5870 - Loss: 0.1368 Acc: 0.9644
Batch 300/5870 - Loss: 0.1408 Acc: 0.9637
Batch 400/5870 - Loss: 0.1473 Acc: 0.9631
Batch 500/5870 - Loss: 0.1433 Acc: 0.9643
Batch 600/5870 - Loss: 0.1413 Acc: 0.9646
Batch 700/5870 - Loss: 0.1407 Acc: 0.9646
Batch 800/5870 - Loss: 0.1417 Acc: 0.9639
Batch 900/5870 - Loss: 0.1411 Acc: 0.9640
Batch 1000/5870 - Loss: 0.1424 Acc: 0.9635
Batch 1100/5870 - Loss: 0.1453 Acc: 0.9633
Batch 1200/5870 - Loss: 0.1463 Acc: 0.9629
Batch 1300/5870 - Loss: 0.1442 Acc: 0.9632
Batch 1400/5870 - Loss: 0.1443 Acc: 0.9633
Batch 1500/5870 - Loss: 0.1450 Acc: 0.9632
Batch 1600/5870 - Loss: 0.1459 Acc: 0.9631
Batch 1700/5870 - Loss: 0.1465 Acc: 0.9629
Batch 1800/5870 - Loss: 0.1466 Acc: 0.9636
Batch 1900/5870 - Loss: 0.1457 Acc: 0.9636
Batch 2000/5870 - Loss: 0.1445 Acc: 0.9641
Batch 2100/5870 - Loss: 0.1426 Acc: 0.9644
Batch 2200/5870 - Loss: 0.1426 Acc: 0.9644
Batch 2300/5870 - Loss: 0.1429 Acc: 0.9645
Batch 2400/5870 - Loss: 0.1420 Acc: 0.9648
Batch 2500/5870 - Loss: 0.1420 Acc: 0.9650
Batch 2600/5870 - Loss: 0.1427 Acc: 0.9648
Batch 2700/5870 - Loss: 0.1411 Acc: 0.9650
Batch 2800/5870 - Loss: 0.1398 Acc: 0.9653
Batch 2900/5870 - Loss: 0.1398 Acc: 0.9654
Batch 3000/5870 - Loss: 0.1383 Acc: 0.9658
Batch 3100/5870 - Loss: 0.1372 Acc: 0.9660
Batch 3200/5870 - Loss: 0.1378 Acc: 0.9661
Batch 3300/5870 - Loss: 0.1366 Acc: 0.9663
Batch 3400/5870 - Loss: 0.1375 Acc: 0.9663
Batch 3500/5870 - Loss: 0.1369 Acc: 0.9666
Batch 3600/5870 - Loss: 0.1368 Acc: 0.9666
```

```
Batch 3700/5870 - Loss: 0.1352 Acc: 0.9670
Batch 3800/5870 - Loss: 0.1338 Acc: 0.9673
Batch 3900/5870 - Loss: 0.1334 Acc: 0.9674
Batch 4000/5870 - Loss: 0.1336 Acc: 0.9673
Batch 4100/5870 - Loss: 0.1355 Acc: 0.9667
Batch 4200/5870 - Loss: 0.1341 Acc: 0.9670
Batch 4300/5870 - Loss: 0.1346 Acc: 0.9669
Batch 4400/5870 - Loss: 0.1348 Acc: 0.9669
Batch 4500/5870 - Loss: 0.1354 Acc: 0.9668
Batch 4600/5870 - Loss: 0.1354 Acc: 0.9667
Batch 4700/5870 - Loss: 0.1353 Acc: 0.9666
Batch 4800/5870 - Loss: 0.1362 Acc: 0.9664
Batch 4900/5870 - Loss: 0.1349 Acc: 0.9668
Batch 5000/5870 - Loss: 0.1344 Acc: 0.9668
Batch 5100/5870 - Loss: 0.1358 Acc: 0.9665
Batch 5200/5870 - Loss: 0.1347 Acc: 0.9667
Batch 5300/5870 - Loss: 0.1344 Acc: 0.9668
Batch 5400/5870 - Loss: 0.1341 Acc: 0.9669
Batch 5500/5870 - Loss: 0.1343 Acc: 0.9669
Batch 5600/5870 - Loss: 0.1343 Acc: 0.9669
Batch 5700/5870 - Loss: 0.1341 Acc: 0.9668
Batch 5800/5870 - Loss: 0.1332 Acc: 0.9670
Epoch 2 - Train Loss: 0.1337 Train Acc: 0.9668 Val Loss: 0.6949 Val Acc: 0.8841

Epoch 3/8
Batch 100/5870 - Loss: 0.0817 Acc: 0.9800
Batch 200/5870 - Loss: 0.0829 Acc: 0.9812
Batch 300/5870 - Loss: 0.0822 Acc: 0.9796
Batch 400/5870 - Loss: 0.0832 Acc: 0.9788
Batch 500/5870 - Loss: 0.0885 Acc: 0.9772
Batch 600/5870 - Loss: 0.0910 Acc: 0.9760
Batch 700/5870 - Loss: 0.0917 Acc: 0.9755
Batch 800/5870 - Loss: 0.0906 Acc: 0.9758
Batch 900/5870 - Loss: 0.0947 Acc: 0.9751
Batch 1000/5870 - Loss: 0.0928 Acc: 0.9751
Batch 1100/5870 - Loss: 0.0933 Acc: 0.9750
Batch 1200/5870 - Loss: 0.0909 Acc: 0.9756
Batch 1300/5870 - Loss: 0.0922 Acc: 0.9754
Batch 1400/5870 - Loss: 0.0929 Acc: 0.9752
Batch 1500/5870 - Loss: 0.0905 Acc: 0.9758
Batch 1600/5870 - Loss: 0.0909 Acc: 0.9757
Batch 1700/5870 - Loss: 0.0890 Acc: 0.9762
Batch 1800/5870 - Loss: 0.0897 Acc: 0.9756
Batch 1900/5870 - Loss: 0.0887 Acc: 0.9758
Batch 2000/5870 - Loss: 0.0868 Acc: 0.9762
Batch 2100/5870 - Loss: 0.0868 Acc: 0.9761
Batch 2200/5870 - Loss: 0.0878 Acc: 0.9761
Batch 2300/5870 - Loss: 0.0881 Acc: 0.9762
```

```
Batch 2400/5870 - Loss: 0.0884 Acc: 0.9760
Batch 2500/5870 - Loss: 0.0872 Acc: 0.9764
Batch 2600/5870 - Loss: 0.0872 Acc: 0.9761
Batch 2700/5870 - Loss: 0.0869 Acc: 0.9759
Batch 2800/5870 - Loss: 0.0876 Acc: 0.9756
Batch 2900/5870 - Loss: 0.0870 Acc: 0.9757
Batch 3000/5870 - Loss: 0.0866 Acc: 0.9759
Batch 3100/5870 - Loss: 0.0874 Acc: 0.9757
Batch 3200/5870 - Loss: 0.0885 Acc: 0.9755
Batch 3300/5870 - Loss: 0.0883 Acc: 0.9756
Batch 3400/5870 - Loss: 0.0892 Acc: 0.9755
Batch 3500/5870 - Loss: 0.0894 Acc: 0.9754
Batch 3600/5870 - Loss: 0.0895 Acc: 0.9753
Batch 3700/5870 - Loss: 0.0899 Acc: 0.9752
Batch 3800/5870 - Loss: 0.0905 Acc: 0.9750
Batch 3900/5870 - Loss: 0.0895 Acc: 0.9752
Batch 4000/5870 - Loss: 0.0891 Acc: 0.9753
Batch 4100/5870 - Loss: 0.0891 Acc: 0.9752
Batch 4200/5870 - Loss: 0.0880 Acc: 0.9755
Batch 4300/5870 - Loss: 0.0873 Acc: 0.9755
Batch 4400/5870 - Loss: 0.0873 Acc: 0.9757
Batch 4500/5870 - Loss: 0.0868 Acc: 0.9758
Batch 4600/5870 - Loss: 0.0871 Acc: 0.9758
Batch 4700/5870 - Loss: 0.0872 Acc: 0.9758
Batch 4800/5870 - Loss: 0.0868 Acc: 0.9757
Batch 4900/5870 - Loss: 0.0868 Acc: 0.9757
Batch 5000/5870 - Loss: 0.0860 Acc: 0.9758
Batch 5100/5870 - Loss: 0.0860 Acc: 0.9758
Batch 5200/5870 - Loss: 0.0852 Acc: 0.9760
Batch 5300/5870 - Loss: 0.0853 Acc: 0.9760
Batch 5400/5870 - Loss: 0.0851 Acc: 0.9759
Batch 5500/5870 - Loss: 0.0851 Acc: 0.9759
Batch 5600/5870 - Loss: 0.0852 Acc: 0.9759
Batch 5700/5870 - Loss: 0.0855 Acc: 0.9758
Batch 5800/5870 - Loss: 0.0854 Acc: 0.9758
Epoch 3 - Train Loss: 0.0854 Train Acc: 0.9758 Val Loss: 0.7692 Val Acc: 0.8888

Epoch 4/8
Batch 100/5870 - Loss: 0.1131 Acc: 0.9688
Batch 200/5870 - Loss: 0.0929 Acc: 0.9725
Batch 300/5870 - Loss: 0.0827 Acc: 0.9767
Batch 400/5870 - Loss: 0.0815 Acc: 0.9762
Batch 500/5870 - Loss: 0.0774 Acc: 0.9775
Batch 600/5870 - Loss: 0.0746 Acc: 0.9777
Batch 700/5870 - Loss: 0.0802 Acc: 0.9762
Batch 800/5870 - Loss: 0.0776 Acc: 0.9770
Batch 900/5870 - Loss: 0.0783 Acc: 0.9769
Batch 1000/5870 - Loss: 0.0783 Acc: 0.9770
```

```
Batch 1100/5870 - Loss: 0.0799 Acc: 0.9770
Batch 1200/5870 - Loss: 0.0825 Acc: 0.9760
Batch 1300/5870 - Loss: 0.0793 Acc: 0.9766
Batch 1400/5870 - Loss: 0.0786 Acc: 0.9767
Batch 1500/5870 - Loss: 0.0779 Acc: 0.9763
Batch 1600/5870 - Loss: 0.0789 Acc: 0.9766
Batch 1700/5870 - Loss: 0.0775 Acc: 0.9769
Batch 1800/5870 - Loss: 0.0771 Acc: 0.9769
Batch 1900/5870 - Loss: 0.0762 Acc: 0.9774
Batch 2000/5870 - Loss: 0.0766 Acc: 0.9774
Batch 2100/5870 - Loss: 0.0761 Acc: 0.9776
Batch 2200/5870 - Loss: 0.0754 Acc: 0.9776
Batch 2300/5870 - Loss: 0.0767 Acc: 0.9769
Batch 2400/5870 - Loss: 0.0758 Acc: 0.9773
Batch 2500/5870 - Loss: 0.0749 Acc: 0.9777
Batch 2600/5870 - Loss: 0.0749 Acc: 0.9776
Batch 2700/5870 - Loss: 0.0749 Acc: 0.9775
Batch 2800/5870 - Loss: 0.0748 Acc: 0.9775
Batch 2900/5870 - Loss: 0.0746 Acc: 0.9776
Batch 3000/5870 - Loss: 0.0753 Acc: 0.9775
Batch 3100/5870 - Loss: 0.0763 Acc: 0.9773
Batch 3200/5870 - Loss: 0.0759 Acc: 0.9773
Batch 3300/5870 - Loss: 0.0766 Acc: 0.9772
Batch 3400/5870 - Loss: 0.0763 Acc: 0.9774
Batch 3500/5870 - Loss: 0.0762 Acc: 0.9775
Batch 3600/5870 - Loss: 0.0759 Acc: 0.9777
Batch 3700/5870 - Loss: 0.0757 Acc: 0.9778
Batch 3800/5870 - Loss: 0.0764 Acc: 0.9775
Batch 3900/5870 - Loss: 0.0760 Acc: 0.9775
Batch 4000/5870 - Loss: 0.0766 Acc: 0.9774
Batch 4100/5870 - Loss: 0.0768 Acc: 0.9773
Batch 4200/5870 - Loss: 0.0767 Acc: 0.9774
Batch 4300/5870 - Loss: 0.0761 Acc: 0.9775
Batch 4400/5870 - Loss: 0.0758 Acc: 0.9775
Batch 4500/5870 - Loss: 0.0755 Acc: 0.9775
Batch 4600/5870 - Loss: 0.0754 Acc: 0.9774
Batch 4700/5870 - Loss: 0.0757 Acc: 0.9774
Batch 4800/5870 - Loss: 0.0758 Acc: 0.9773
Batch 4900/5870 - Loss: 0.0754 Acc: 0.9774
Batch 5000/5870 - Loss: 0.0754 Acc: 0.9774
Batch 5100/5870 - Loss: 0.0752 Acc: 0.9774
Batch 5200/5870 - Loss: 0.0750 Acc: 0.9774
Batch 5300/5870 - Loss: 0.0744 Acc: 0.9775
Batch 5400/5870 - Loss: 0.0739 Acc: 0.9777
Batch 5500/5870 - Loss: 0.0739 Acc: 0.9777
Batch 5600/5870 - Loss: 0.0738 Acc: 0.9778
Batch 5700/5870 - Loss: 0.0738 Acc: 0.9779
Batch 5800/5870 - Loss: 0.0739 Acc: 0.9778
```

```
Epoch 4 - Train Loss: 0.0737 Train Acc: 0.9779 Val Loss: 0.7258 Val Acc: 0.8910

Epoch 5/8
Batch 100/5870 - Loss: 0.0458 Acc: 0.9800
Batch 200/5870 - Loss: 0.0654 Acc: 0.9781
Batch 300/5870 - Loss: 0.0587 Acc: 0.9804
Batch 400/5870 - Loss: 0.0598 Acc: 0.9806
Batch 500/5870 - Loss: 0.0586 Acc: 0.9810
Batch 600/5870 - Loss: 0.0587 Acc: 0.9812
Batch 700/5870 - Loss: 0.0580 Acc: 0.9821
Batch 800/5870 - Loss: 0.0563 Acc: 0.9822
Batch 900/5870 - Loss: 0.0582 Acc: 0.9817
Batch 1000/5870 - Loss: 0.0587 Acc: 0.9814
Batch 1100/5870 - Loss: 0.0598 Acc: 0.9809
Batch 1200/5870 - Loss: 0.0619 Acc: 0.9801
Batch 1300/5870 - Loss: 0.0598 Acc: 0.9805
Batch 1400/5870 - Loss: 0.0595 Acc: 0.9804
Batch 1500/5870 - Loss: 0.0593 Acc: 0.9802
Batch 1600/5870 - Loss: 0.0586 Acc: 0.9802
Batch 1700/5870 - Loss: 0.0600 Acc: 0.9799
Batch 1800/5870 - Loss: 0.0631 Acc: 0.9789
Batch 1900/5870 - Loss: 0.0626 Acc: 0.9789
Batch 2000/5870 - Loss: 0.0636 Acc: 0.9784
Batch 2100/5870 - Loss: 0.0619 Acc: 0.9789
Batch 2200/5870 - Loss: 0.0628 Acc: 0.9789
Batch 2300/5870 - Loss: 0.0632 Acc: 0.9788
Batch 2400/5870 - Loss: 0.0618 Acc: 0.9794
Batch 2500/5870 - Loss: 0.0628 Acc: 0.9794
Batch 2600/5870 - Loss: 0.0616 Acc: 0.9797
Batch 2700/5870 - Loss: 0.0625 Acc: 0.9796
Batch 2800/5870 - Loss: 0.0615 Acc: 0.9800
Batch 2900/5870 - Loss: 0.0615 Acc: 0.9801
Batch 3000/5870 - Loss: 0.0616 Acc: 0.9802
Batch 3100/5870 - Loss: 0.0612 Acc: 0.9804
Batch 3200/5870 - Loss: 0.0617 Acc: 0.9801
Batch 3300/5870 - Loss: 0.0614 Acc: 0.9803
Batch 3400/5870 - Loss: 0.0613 Acc: 0.9801
Batch 3500/5870 - Loss: 0.0621 Acc: 0.9799
Batch 3600/5870 - Loss: 0.0618 Acc: 0.9799
Batch 3700/5870 - Loss: 0.0621 Acc: 0.9798
Batch 3800/5870 - Loss: 0.0617 Acc: 0.9798
Batch 3900/5870 - Loss: 0.0607 Acc: 0.9801
Batch 4000/5870 - Loss: 0.0604 Acc: 0.9802
Batch 4100/5870 - Loss: 0.0597 Acc: 0.9805
Batch 4200/5870 - Loss: 0.0598 Acc: 0.9805
Batch 4300/5870 - Loss: 0.0596 Acc: 0.9806
Batch 4400/5870 - Loss: 0.0600 Acc: 0.9805
Batch 4500/5870 - Loss: 0.0597 Acc: 0.9806
```

```
Batch 4600/5870 - Loss: 0.0598 Acc: 0.9807
Batch 4700/5870 - Loss: 0.0603 Acc: 0.9805
Batch 4800/5870 - Loss: 0.0603 Acc: 0.9806
Batch 4900/5870 - Loss: 0.0607 Acc: 0.9805
Batch 5000/5870 - Loss: 0.0607 Acc: 0.9805
Batch 5100/5870 - Loss: 0.0608 Acc: 0.9805
Batch 5200/5870 - Loss: 0.0607 Acc: 0.9805
Batch 5300/5870 - Loss: 0.0605 Acc: 0.9805
Batch 5400/5870 - Loss: 0.0605 Acc: 0.9806
Batch 5500/5870 - Loss: 0.0607 Acc: 0.9805
Batch 5600/5870 - Loss: 0.0607 Acc: 0.9805
Batch 5700/5870 - Loss: 0.0604 Acc: 0.9805
Batch 5800/5870 - Loss: 0.0605 Acc: 0.9805
Epoch 5 - Train Loss: 0.0604 Train Acc: 0.9805 Val Loss: 0.7641 Val Acc: 0.8918

Epoch 6/8
Batch 100/5870 - Loss: 0.0454 Acc: 0.9838
Batch 200/5870 - Loss: 0.0421 Acc: 0.9850
Batch 300/5870 - Loss: 0.0482 Acc: 0.9842
Batch 400/5870 - Loss: 0.0502 Acc: 0.9838
Batch 500/5870 - Loss: 0.0525 Acc: 0.9828
Batch 600/5870 - Loss: 0.0552 Acc: 0.9821
Batch 700/5870 - Loss: 0.0532 Acc: 0.9825
Batch 800/5870 - Loss: 0.0525 Acc: 0.9825
Batch 900/5870 - Loss: 0.0512 Acc: 0.9826
Batch 1000/5870 - Loss: 0.0514 Acc: 0.9824
Batch 1100/5870 - Loss: 0.0496 Acc: 0.9830
Batch 1200/5870 - Loss: 0.0502 Acc: 0.9825
Batch 1300/5870 - Loss: 0.0507 Acc: 0.9824
Batch 1400/5870 - Loss: 0.0509 Acc: 0.9825
Batch 1500/5870 - Loss: 0.0508 Acc: 0.9825
Batch 1600/5870 - Loss: 0.0517 Acc: 0.9821
Batch 1700/5870 - Loss: 0.0521 Acc: 0.9818
Batch 1800/5870 - Loss: 0.0523 Acc: 0.9817
Batch 1900/5870 - Loss: 0.0509 Acc: 0.9821
Batch 2000/5870 - Loss: 0.0500 Acc: 0.9824
Batch 2100/5870 - Loss: 0.0495 Acc: 0.9826
Batch 2200/5870 - Loss: 0.0497 Acc: 0.9824
Batch 2300/5870 - Loss: 0.0502 Acc: 0.9823
Batch 2400/5870 - Loss: 0.0494 Acc: 0.9826
Batch 2500/5870 - Loss: 0.0490 Acc: 0.9826
Batch 2600/5870 - Loss: 0.0498 Acc: 0.9825
Batch 2700/5870 - Loss: 0.0507 Acc: 0.9823
Batch 2800/5870 - Loss: 0.0505 Acc: 0.9823
Batch 2900/5870 - Loss: 0.0501 Acc: 0.9823
Batch 3000/5870 - Loss: 0.0505 Acc: 0.9821
Batch 3100/5870 - Loss: 0.0501 Acc: 0.9821
Batch 3200/5870 - Loss: 0.0502 Acc: 0.9820
```

```
Batch 3300/5870 - Loss: 0.0498 Acc: 0.9821
Batch 3400/5870 - Loss: 0.0493 Acc: 0.9824
Batch 3500/5870 - Loss: 0.0490 Acc: 0.9825
Batch 3600/5870 - Loss: 0.0496 Acc: 0.9824
Batch 3700/5870 - Loss: 0.0497 Acc: 0.9823
Batch 3800/5870 - Loss: 0.0499 Acc: 0.9823
Batch 3900/5870 - Loss: 0.0505 Acc: 0.9822
Batch 4000/5870 - Loss: 0.0507 Acc: 0.9821
Batch 4100/5870 - Loss: 0.0513 Acc: 0.9819
Batch 4200/5870 - Loss: 0.0514 Acc: 0.9820
Batch 4300/5870 - Loss: 0.0508 Acc: 0.9822
Batch 4400/5870 - Loss: 0.0510 Acc: 0.9822
Batch 4500/5870 - Loss: 0.0512 Acc: 0.9821
Batch 4600/5870 - Loss: 0.0510 Acc: 0.9823
Batch 4700/5870 - Loss: 0.0511 Acc: 0.9822
Batch 4800/5870 - Loss: 0.0518 Acc: 0.9820
Batch 4900/5870 - Loss: 0.0521 Acc: 0.9818
Batch 5000/5870 - Loss: 0.0520 Acc: 0.9819
Batch 5100/5870 - Loss: 0.0520 Acc: 0.9819
Batch 5200/5870 - Loss: 0.0521 Acc: 0.9818
Batch 5300/5870 - Loss: 0.0519 Acc: 0.9819
Batch 5400/5870 - Loss: 0.0517 Acc: 0.9820
Batch 5500/5870 - Loss: 0.0515 Acc: 0.9821
Batch 5600/5870 - Loss: 0.0513 Acc: 0.9821
Batch 5700/5870 - Loss: 0.0514 Acc: 0.9821
Batch 5800/5870 - Loss: 0.0512 Acc: 0.9822
Epoch 6 - Train Loss: 0.0511 Train Acc: 0.9822 Val Loss: 0.8257 Val Acc: 0.8892

Epoch 7/8
Batch 100/5870 - Loss: 0.0638 Acc: 0.9800
Batch 200/5870 - Loss: 0.0589 Acc: 0.9794
Batch 300/5870 - Loss: 0.0595 Acc: 0.9783
Batch 400/5870 - Loss: 0.0523 Acc: 0.9809
Batch 500/5870 - Loss: 0.0525 Acc: 0.9810
Batch 600/5870 - Loss: 0.0480 Acc: 0.9825
Batch 700/5870 - Loss: 0.0481 Acc: 0.9825
Batch 800/5870 - Loss: 0.0473 Acc: 0.9828
Batch 900/5870 - Loss: 0.0453 Acc: 0.9835
Batch 1000/5870 - Loss: 0.0467 Acc: 0.9830
Batch 1100/5870 - Loss: 0.0486 Acc: 0.9827
Batch 1200/5870 - Loss: 0.0474 Acc: 0.9830
Batch 1300/5870 - Loss: 0.0477 Acc: 0.9829
Batch 1400/5870 - Loss: 0.0478 Acc: 0.9829
Batch 1500/5870 - Loss: 0.0497 Acc: 0.9822
Batch 1600/5870 - Loss: 0.0504 Acc: 0.9818
Batch 1700/5870 - Loss: 0.0489 Acc: 0.9823
Batch 1800/5870 - Loss: 0.0484 Acc: 0.9826
Batch 1900/5870 - Loss: 0.0475 Acc: 0.9831
```

```
Batch 2000/5870 - Loss: 0.0476 Acc: 0.9830
Batch 2100/5870 - Loss: 0.0478 Acc: 0.9832
Batch 2200/5870 - Loss: 0.0486 Acc: 0.9830
Batch 2300/5870 - Loss: 0.0490 Acc: 0.9827
Batch 2400/5870 - Loss: 0.0498 Acc: 0.9824
Batch 2500/5870 - Loss: 0.0499 Acc: 0.9822
Batch 2600/5870 - Loss: 0.0496 Acc: 0.9823
Batch 2700/5870 - Loss: 0.0498 Acc: 0.9822
Batch 2800/5870 - Loss: 0.0501 Acc: 0.9821
Batch 2900/5870 - Loss: 0.0496 Acc: 0.9823
Batch 3000/5870 - Loss: 0.0493 Acc: 0.9823
Batch 3100/5870 - Loss: 0.0489 Acc: 0.9825
Batch 3200/5870 - Loss: 0.0487 Acc: 0.9826
Batch 3300/5870 - Loss: 0.0486 Acc: 0.9827
Batch 3400/5870 - Loss: 0.0485 Acc: 0.9826
Batch 3500/5870 - Loss: 0.0483 Acc: 0.9828
Batch 3600/5870 - Loss: 0.0482 Acc: 0.9827
Batch 3700/5870 - Loss: 0.0479 Acc: 0.9828
Batch 3800/5870 - Loss: 0.0480 Acc: 0.9828
Batch 3900/5870 - Loss: 0.0482 Acc: 0.9826
Batch 4000/5870 - Loss: 0.0485 Acc: 0.9826
Batch 4100/5870 - Loss: 0.0484 Acc: 0.9826
Batch 4200/5870 - Loss: 0.0482 Acc: 0.9826
Batch 4300/5870 - Loss: 0.0479 Acc: 0.9827
Batch 4400/5870 - Loss: 0.0481 Acc: 0.9827
Batch 4500/5870 - Loss: 0.0483 Acc: 0.9827
Batch 4600/5870 - Loss: 0.0482 Acc: 0.9827
Batch 4700/5870 - Loss: 0.0482 Acc: 0.9827
Batch 4800/5870 - Loss: 0.0484 Acc: 0.9826
Batch 4900/5870 - Loss: 0.0485 Acc: 0.9826
Batch 5000/5870 - Loss: 0.0483 Acc: 0.9827
Batch 5100/5870 - Loss: 0.0482 Acc: 0.9827
Batch 5200/5870 - Loss: 0.0479 Acc: 0.9828
Batch 5300/5870 - Loss: 0.0479 Acc: 0.9828
Batch 5400/5870 - Loss: 0.0476 Acc: 0.9829
Batch 5500/5870 - Loss: 0.0478 Acc: 0.9828
Batch 5600/5870 - Loss: 0.0483 Acc: 0.9826
Batch 5700/5870 - Loss: 0.0481 Acc: 0.9827
Batch 5800/5870 - Loss: 0.0485 Acc: 0.9825
Epoch 7 - Train Loss: 0.0488 Train Acc: 0.9824 Val Loss: 0.7819 Val Acc: 0.8966

Epoch 8/8
Batch 100/5870 - Loss: 0.0422 Acc: 0.9862
Batch 200/5870 - Loss: 0.0354 Acc: 0.9869
Batch 300/5870 - Loss: 0.0376 Acc: 0.9867
Batch 400/5870 - Loss: 0.0408 Acc: 0.9847
Batch 500/5870 - Loss: 0.0472 Acc: 0.9825
Batch 600/5870 - Loss: 0.0441 Acc: 0.9838
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```
Batch 700/5870 - Loss: 0.0453 Acc: 0.9834
Batch 800/5870 - Loss: 0.0447 Acc: 0.9836
Batch 900/5870 - Loss: 0.0448 Acc: 0.9833
Batch 1000/5870 - Loss: 0.0446 Acc: 0.9835
Batch 1100/5870 - Loss: 0.0441 Acc: 0.9836
Batch 1200/5870 - Loss: 0.0445 Acc: 0.9833
Batch 1300/5870 - Loss: 0.0450 Acc: 0.9832
Batch 1400/5870 - Loss: 0.0451 Acc: 0.9831
Batch 1500/5870 - Loss: 0.0448 Acc: 0.9832
Batch 1600/5870 - Loss: 0.0450 Acc: 0.9830
Batch 1700/5870 - Loss: 0.0439 Acc: 0.9835
Batch 1800/5870 - Loss: 0.0444 Acc: 0.9833
Batch 1900/5870 - Loss: 0.0438 Acc: 0.9838
Batch 2000/5870 - Loss: 0.0436 Acc: 0.9838
Batch 2100/5870 - Loss: 0.0432 Acc: 0.9840
Batch 2200/5870 - Loss: 0.0434 Acc: 0.9840
Batch 2300/5870 - Loss: 0.0432 Acc: 0.9840
Batch 2400/5870 - Loss: 0.0435 Acc: 0.9840
Batch 2500/5870 - Loss: 0.0441 Acc: 0.9839
Batch 2600/5870 - Loss: 0.0440 Acc: 0.9838
Batch 2700/5870 - Loss: 0.0445 Acc: 0.9836
Batch 2800/5870 - Loss: 0.0444 Acc: 0.9836
Batch 2900/5870 - Loss: 0.0447 Acc: 0.9835
Batch 3000/5870 - Loss: 0.0445 Acc: 0.9836
Batch 3100/5870 - Loss: 0.0444 Acc: 0.9837
Batch 3200/5870 - Loss: 0.0443 Acc: 0.9838
Batch 3300/5870 - Loss: 0.0447 Acc: 0.9836
Batch 3400/5870 - Loss: 0.0448 Acc: 0.9835
Batch 3500/5870 - Loss: 0.0454 Acc: 0.9833
Batch 3600/5870 - Loss: 0.0459 Acc: 0.9831
Batch 3700/5870 - Loss: 0.0457 Acc: 0.9832
Batch 3800/5870 - Loss: 0.0456 Acc: 0.9832
Batch 3900/5870 - Loss: 0.0462 Acc: 0.9829
Batch 4000/5870 - Loss: 0.0461 Acc: 0.9830
Batch 4100/5870 - Loss: 0.0462 Acc: 0.9829
Batch 4200/5870 - Loss: 0.0468 Acc: 0.9826
Batch 4300/5870 - Loss: 0.0467 Acc: 0.9826
Batch 4400/5870 - Loss: 0.0461 Acc: 0.9828
Batch 4500/5870 - Loss: 0.0462 Acc: 0.9828
Batch 4600/5870 - Loss: 0.0463 Acc: 0.9827
Batch 4700/5870 - Loss: 0.0463 Acc: 0.9828
Batch 4800/5870 - Loss: 0.0457 Acc: 0.9830
Batch 4900/5870 - Loss: 0.0453 Acc: 0.9831
Batch 5000/5870 - Loss: 0.0455 Acc: 0.9830
Batch 5100/5870 - Loss: 0.0452 Acc: 0.9831
Batch 5200/5870 - Loss: 0.0454 Acc: 0.9831
Batch 5300/5870 - Loss: 0.0455 Acc: 0.9830
Batch 5400/5870 - Loss: 0.0454 Acc: 0.9830
```

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Batch 5500/5870 - Loss: 0.0452 Acc: 0.9831
Batch 5600/5870 - Loss: 0.0456 Acc: 0.9830
Batch 5700/5870 - Loss: 0.0459 Acc: 0.9829
Batch 5800/5870 - Loss: 0.0458 Acc: 0.9829
Epoch 8 - Train Loss: 0.0459 Train Acc: 0.9828 Val Loss: 0.7916 Val Acc: 0.8962
→ Fold 3 Best Val Acc: 0.8966

==============================
Fold 4
==============================

Some weights of BertForSequenceClassification were not initialized from the
model checkpoint at bert-base-uncased and are newly initialized:
['classifier.bias', 'classifier.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.


Epoch 1/8
Batch 100/5872 - Loss: 2.4835 Acc: 0.1562
Batch 200/5872 - Loss: 2.2727 Acc: 0.2525
Batch 300/5872 - Loss: 2.0527 Acc: 0.3404
Batch 400/5872 - Loss: 1.8689 Acc: 0.4006
Batch 500/5872 - Loss: 1.7204 Acc: 0.4537
Batch 600/5872 - Loss: 1.6031 Acc: 0.4908
Batch 700/5872 - Loss: 1.4998 Acc: 0.5264
Batch 800/5872 - Loss: 1.4193 Acc: 0.5506
Batch 900/5872 - Loss: 1.3363 Acc: 0.5775
Batch 1000/5872 - Loss: 1.2681 Acc: 0.5994
Batch 1100/5872 - Loss: 1.2139 Acc: 0.6159
Batch 1200/5872 - Loss: 1.1621 Acc: 0.6326
Batch 1300/5872 - Loss: 1.1213 Acc: 0.6452
Batch 1400/5872 - Loss: 1.0784 Acc: 0.6584
Batch 1500/5872 - Loss: 1.0371 Acc: 0.6721
Batch 1600/5872 - Loss: 1.0024 Acc: 0.6830
Batch 1700/5872 - Loss: 0.9708 Acc: 0.6938
Batch 1800/5872 - Loss: 0.9441 Acc: 0.7024
Batch 1900/5872 - Loss: 0.9131 Acc: 0.7124
Batch 2000/5872 - Loss: 0.8897 Acc: 0.7209
Batch 2100/5872 - Loss: 0.8633 Acc: 0.7292
Batch 2200/5872 - Loss: 0.8418 Acc: 0.7365
Batch 2300/5872 - Loss: 0.8223 Acc: 0.7431
Batch 2400/5872 - Loss: 0.8029 Acc: 0.7496
Batch 2500/5872 - Loss: 0.7836 Acc: 0.7562
Batch 2600/5872 - Loss: 0.7666 Acc: 0.7617
Batch 2700/5872 - Loss: 0.7495 Acc: 0.7673
Batch 2800/5872 - Loss: 0.7313 Acc: 0.7733
Batch 2900/5872 - Loss: 0.7181 Acc: 0.7782
Batch 3000/5872 - Loss: 0.7036 Acc: 0.7830
```

```
Batch 3100/5872 - Loss: 0.6896 Acc: 0.7876
Batch 3200/5872 - Loss: 0.6799 Acc: 0.7913
Batch 3300/5872 - Loss: 0.6677 Acc: 0.7955
Batch 3400/5872 - Loss: 0.6559 Acc: 0.7994
Batch 3500/5872 - Loss: 0.6451 Acc: 0.8029
Batch 3600/5872 - Loss: 0.6355 Acc: 0.8061
Batch 3700/5872 - Loss: 0.6249 Acc: 0.8097
Batch 3800/5872 - Loss: 0.6165 Acc: 0.8126
Batch 3900/5872 - Loss: 0.6068 Acc: 0.8157
Batch 4000/5872 - Loss: 0.5983 Acc: 0.8185
Batch 4100/5872 - Loss: 0.5886 Acc: 0.8216
Batch 4200/5872 - Loss: 0.5798 Acc: 0.8245
Batch 4300/5872 - Loss: 0.5728 Acc: 0.8271
Batch 4400/5872 - Loss: 0.5654 Acc: 0.8296
Batch 4500/5872 - Loss: 0.5578 Acc: 0.8321
Batch 4600/5872 - Loss: 0.5508 Acc: 0.8346
Batch 4700/5872 - Loss: 0.5447 Acc: 0.8365
Batch 4800/5872 - Loss: 0.5383 Acc: 0.8389
Batch 4900/5872 - Loss: 0.5322 Acc: 0.8411
Batch 5000/5872 - Loss: 0.5267 Acc: 0.8430
Batch 5100/5872 - Loss: 0.5195 Acc: 0.8452
Batch 5200/5872 - Loss: 0.5130 Acc: 0.8473
Batch 5300/5872 - Loss: 0.5074 Acc: 0.8492
Batch 5400/5872 - Loss: 0.5014 Acc: 0.8512
Batch 5500/5872 - Loss: 0.4958 Acc: 0.8530
Batch 5600/5872 - Loss: 0.4907 Acc: 0.8547
Batch 5700/5872 - Loss: 0.4849 Acc: 0.8566
Batch 5800/5872 - Loss: 0.4796 Acc: 0.8583
Epoch 1 - Train Loss: 0.4762 Train Acc: 0.8594 Val Loss: 0.7417 Val Acc: 0.8558

Epoch 2/8
Batch 100/5872 - Loss: 0.1382 Acc: 0.9625
Batch 200/5872 - Loss: 0.1415 Acc: 0.9625
Batch 300/5872 - Loss: 0.1365 Acc: 0.9642
Batch 400/5872 - Loss: 0.1276 Acc: 0.9663
Batch 500/5872 - Loss: 0.1286 Acc: 0.9677
Batch 600/5872 - Loss: 0.1326 Acc: 0.9663
Batch 700/5872 - Loss: 0.1352 Acc: 0.9654
Batch 800/5872 - Loss: 0.1368 Acc: 0.9655
Batch 900/5872 - Loss: 0.1328 Acc: 0.9664
Batch 1000/5872 - Loss: 0.1356 Acc: 0.9670
Batch 1100/5872 - Loss: 0.1363 Acc: 0.9669
Batch 1200/5872 - Loss: 0.1366 Acc: 0.9668
Batch 1300/5872 - Loss: 0.1349 Acc: 0.9673
Batch 1400/5872 - Loss: 0.1357 Acc: 0.9673
Batch 1500/5872 - Loss: 0.1373 Acc: 0.9670
Batch 1600/5872 - Loss: 0.1372 Acc: 0.9670
Batch 1700/5872 - Loss: 0.1373 Acc: 0.9668
```

```
Batch 1800/5872 - Loss: 0.1402 Acc: 0.9661
Batch 1900/5872 - Loss: 0.1391 Acc: 0.9663
Batch 2000/5872 - Loss: 0.1392 Acc: 0.9662
Batch 2100/5872 - Loss: 0.1385 Acc: 0.9665
Batch 2200/5872 - Loss: 0.1403 Acc: 0.9662
Batch 2300/5872 - Loss: 0.1405 Acc: 0.9664
Batch 2400/5872 - Loss: 0.1412 Acc: 0.9662
Batch 2500/5872 - Loss: 0.1402 Acc: 0.9663
Batch 2600/5872 - Loss: 0.1396 Acc: 0.9666
Batch 2700/5872 - Loss: 0.1386 Acc: 0.9671
Batch 2800/5872 - Loss: 0.1391 Acc: 0.9670
Batch 2900/5872 - Loss: 0.1391 Acc: 0.9669
Batch 3000/5872 - Loss: 0.1366 Acc: 0.9672
Batch 3100/5872 - Loss: 0.1371 Acc: 0.9673
Batch 3200/5872 - Loss: 0.1371 Acc: 0.9673
Batch 3300/5872 - Loss: 0.1361 Acc: 0.9673
Batch 3400/5872 - Loss: 0.1350 Acc: 0.9673
Batch 3500/5872 - Loss: 0.1351 Acc: 0.9671
Batch 3600/5872 - Loss: 0.1348 Acc: 0.9672
Batch 3700/5872 - Loss: 0.1334 Acc: 0.9674
Batch 3800/5872 - Loss: 0.1340 Acc: 0.9672
Batch 3900/5872 - Loss: 0.1334 Acc: 0.9674
Batch 4000/5872 - Loss: 0.1342 Acc: 0.9673
Batch 4100/5872 - Loss: 0.1337 Acc: 0.9673
Batch 4200/5872 - Loss: 0.1323 Acc: 0.9675
Batch 4300/5872 - Loss: 0.1311 Acc: 0.9678
Batch 4400/5872 - Loss: 0.1316 Acc: 0.9678
Batch 4500/5872 - Loss: 0.1324 Acc: 0.9677
Batch 4600/5872 - Loss: 0.1315 Acc: 0.9678
Batch 4700/5872 - Loss: 0.1300 Acc: 0.9681
Batch 4800/5872 - Loss: 0.1308 Acc: 0.9679
Batch 4900/5872 - Loss: 0.1303 Acc: 0.9680
Batch 5000/5872 - Loss: 0.1301 Acc: 0.9681
Batch 5100/5872 - Loss: 0.1294 Acc: 0.9682
Batch 5200/5872 - Loss: 0.1297 Acc: 0.9682
Batch 5300/5872 - Loss: 0.1300 Acc: 0.9681
Batch 5400/5872 - Loss: 0.1302 Acc: 0.9682
Batch 5500/5872 - Loss: 0.1301 Acc: 0.9681
Batch 5600/5872 - Loss: 0.1297 Acc: 0.9682
Batch 5700/5872 - Loss: 0.1288 Acc: 0.9684
Batch 5800/5872 - Loss: 0.1280 Acc: 0.9686
Epoch 2 - Train Loss: 0.1279 Train Acc: 0.9687 Val Loss: 0.7071 Val Acc: 0.8866

Epoch 3/8
Batch 100/5872 - Loss: 0.1186 Acc: 0.9738
Batch 200/5872 - Loss: 0.1118 Acc: 0.9744
Batch 300/5872 - Loss: 0.1043 Acc: 0.9758
Batch 400/5872 - Loss: 0.1017 Acc: 0.9762
```

```
Batch 500/5872 - Loss: 0.1024 Acc: 0.9750
Batch 600/5872 - Loss: 0.0989 Acc: 0.9752
Batch 700/5872 - Loss: 0.1058 Acc: 0.9739
Batch 800/5872 - Loss: 0.1035 Acc: 0.9739
Batch 900/5872 - Loss: 0.1085 Acc: 0.9729
Batch 1000/5872 - Loss: 0.1051 Acc: 0.9731
Batch 1100/5872 - Loss: 0.1048 Acc: 0.9732
Batch 1200/5872 - Loss: 0.1033 Acc: 0.9732
Batch 1300/5872 - Loss: 0.0996 Acc: 0.9742
Batch 1400/5872 - Loss: 0.0997 Acc: 0.9743
Batch 1500/5872 - Loss: 0.0995 Acc: 0.9747
Batch 1600/5872 - Loss: 0.0991 Acc: 0.9748
Batch 1700/5872 - Loss: 0.0999 Acc: 0.9748
Batch 1800/5872 - Loss: 0.1000 Acc: 0.9744
Batch 1900/5872 - Loss: 0.1007 Acc: 0.9744
Batch 2000/5872 - Loss: 0.1003 Acc: 0.9743
Batch 2100/5872 - Loss: 0.0995 Acc: 0.9746
Batch 2200/5872 - Loss: 0.0998 Acc: 0.9745
Batch 2300/5872 - Loss: 0.0972 Acc: 0.9751
Batch 2400/5872 - Loss: 0.0963 Acc: 0.9754
Batch 2500/5872 - Loss: 0.0946 Acc: 0.9760
Batch 2600/5872 - Loss: 0.0934 Acc: 0.9762
Batch 2700/5872 - Loss: 0.0925 Acc: 0.9764
Batch 2800/5872 - Loss: 0.0925 Acc: 0.9763
Batch 2900/5872 - Loss: 0.0926 Acc: 0.9764
Batch 3000/5872 - Loss: 0.0929 Acc: 0.9763
Batch 3100/5872 - Loss: 0.0912 Acc: 0.9768
Batch 3200/5872 - Loss: 0.0917 Acc: 0.9766
Batch 3300/5872 - Loss: 0.0914 Acc: 0.9765
Batch 3400/5872 - Loss: 0.0904 Acc: 0.9767
Batch 3500/5872 - Loss: 0.0911 Acc: 0.9767
Batch 3600/5872 - Loss: 0.0920 Acc: 0.9765
Batch 3700/5872 - Loss: 0.0922 Acc: 0.9764
Batch 3800/5872 - Loss: 0.0920 Acc: 0.9764
Batch 3900/5872 - Loss: 0.0908 Acc: 0.9767
Batch 4000/5872 - Loss: 0.0904 Acc: 0.9768
Batch 4100/5872 - Loss: 0.0900 Acc: 0.9769
Batch 4200/5872 - Loss: 0.0898 Acc: 0.9768
Batch 4300/5872 - Loss: 0.0903 Acc: 0.9767
Batch 4400/5872 - Loss: 0.0898 Acc: 0.9768
Batch 4500/5872 - Loss: 0.0898 Acc: 0.9768
Batch 4600/5872 - Loss: 0.0900 Acc: 0.9768
Batch 4700/5872 - Loss: 0.0903 Acc: 0.9765
Batch 4800/5872 - Loss: 0.0899 Acc: 0.9767
Batch 4900/5872 - Loss: 0.0897 Acc: 0.9767
Batch 5000/5872 - Loss: 0.0894 Acc: 0.9766
Batch 5100/5872 - Loss: 0.0899 Acc: 0.9765
Batch 5200/5872 - Loss: 0.0898 Acc: 0.9763
```

```
Batch 5300/5872 - Loss: 0.0898 Acc: 0.9764
Batch 5400/5872 - Loss: 0.0894 Acc: 0.9764
Batch 5500/5872 - Loss: 0.0888 Acc: 0.9766
Batch 5600/5872 - Loss: 0.0884 Acc: 0.9767
Batch 5700/5872 - Loss: 0.0883 Acc: 0.9768
Batch 5800/5872 - Loss: 0.0877 Acc: 0.9769
Epoch 3 - Train Loss: 0.0877 Train Acc: 0.9768 Val Loss: 0.8136 Val Acc: 0.8745

Epoch 4/8
Batch 100/5872 - Loss: 0.0836 Acc: 0.9750
Batch 200/5872 - Loss: 0.0655 Acc: 0.9819
Batch 300/5872 - Loss: 0.0610 Acc: 0.9817
Batch 400/5872 - Loss: 0.0610 Acc: 0.9812
Batch 500/5872 - Loss: 0.0598 Acc: 0.9812
Batch 600/5872 - Loss: 0.0590 Acc: 0.9817
Batch 700/5872 - Loss: 0.0561 Acc: 0.9823
Batch 800/5872 - Loss: 0.0548 Acc: 0.9828
Batch 900/5872 - Loss: 0.0540 Acc: 0.9833
Batch 1000/5872 - Loss: 0.0573 Acc: 0.9828
Batch 1100/5872 - Loss: 0.0588 Acc: 0.9824
Batch 1200/5872 - Loss: 0.0625 Acc: 0.9811
Batch 1300/5872 - Loss: 0.0617 Acc: 0.9811
Batch 1400/5872 - Loss: 0.0633 Acc: 0.9809
Batch 1500/5872 - Loss: 0.0635 Acc: 0.9809
Batch 1600/5872 - Loss: 0.0663 Acc: 0.9802
Batch 1700/5872 - Loss: 0.0657 Acc: 0.9804
Batch 1800/5872 - Loss: 0.0651 Acc: 0.9808
Batch 1900/5872 - Loss: 0.0657 Acc: 0.9803
Batch 2000/5872 - Loss: 0.0659 Acc: 0.9806
Batch 2100/5872 - Loss: 0.0650 Acc: 0.9809
Batch 2200/5872 - Loss: 0.0660 Acc: 0.9807
Batch 2300/5872 - Loss: 0.0660 Acc: 0.9807
Batch 2400/5872 - Loss: 0.0679 Acc: 0.9804
Batch 2500/5872 - Loss: 0.0683 Acc: 0.9803
Batch 2600/5872 - Loss: 0.0677 Acc: 0.9805
Batch 2700/5872 - Loss: 0.0670 Acc: 0.9806
Batch 2800/5872 - Loss: 0.0673 Acc: 0.9804
Batch 2900/5872 - Loss: 0.0674 Acc: 0.9802
Batch 3000/5872 - Loss: 0.0676 Acc: 0.9798
Batch 3100/5872 - Loss: 0.0678 Acc: 0.9797
Batch 3200/5872 - Loss: 0.0683 Acc: 0.9795
Batch 3300/5872 - Loss: 0.0683 Acc: 0.9796
Batch 3400/5872 - Loss: 0.0680 Acc: 0.9796
Batch 3500/5872 - Loss: 0.0677 Acc: 0.9797
Batch 3600/5872 - Loss: 0.0679 Acc: 0.9796
Batch 3700/5872 - Loss: 0.0684 Acc: 0.9796
Batch 3800/5872 - Loss: 0.0683 Acc: 0.9796
Batch 3900/5872 - Loss: 0.0682 Acc: 0.9796
```

```
Batch 4000/5872 - Loss: 0.0676 Acc: 0.9798
Batch 4100/5872 - Loss: 0.0680 Acc: 0.9798
Batch 4200/5872 - Loss: 0.0677 Acc: 0.9798
Batch 4300/5872 - Loss: 0.0683 Acc: 0.9797
Batch 4400/5872 - Loss: 0.0678 Acc: 0.9798
Batch 4500/5872 - Loss: 0.0677 Acc: 0.9797
Batch 4600/5872 - Loss: 0.0675 Acc: 0.9798
Batch 4700/5872 - Loss: 0.0671 Acc: 0.9799
Batch 4800/5872 - Loss: 0.0661 Acc: 0.9802
Batch 4900/5872 - Loss: 0.0662 Acc: 0.9802
Batch 5000/5872 - Loss: 0.0669 Acc: 0.9801
Batch 5100/5872 - Loss: 0.0667 Acc: 0.9801
Batch 5200/5872 - Loss: 0.0663 Acc: 0.9802
Batch 5300/5872 - Loss: 0.0663 Acc: 0.9801
Batch 5400/5872 - Loss: 0.0671 Acc: 0.9800
Batch 5500/5872 - Loss: 0.0666 Acc: 0.9801
Batch 5600/5872 - Loss: 0.0667 Acc: 0.9801
Batch 5700/5872 - Loss: 0.0670 Acc: 0.9800
Batch 5800/5872 - Loss: 0.0676 Acc: 0.9799
Epoch 4 - Train Loss: 0.0673 Train Acc: 0.9799 Val Loss: 0.7939 Val Acc: 0.8822

Epoch 5/8
Batch 100/5872 - Loss: 0.0740 Acc: 0.9738
Batch 200/5872 - Loss: 0.0608 Acc: 0.9806
Batch 300/5872 - Loss: 0.0597 Acc: 0.9804
Batch 400/5872 - Loss: 0.0615 Acc: 0.9803
Batch 500/5872 - Loss: 0.0637 Acc: 0.9798
Batch 600/5872 - Loss: 0.0629 Acc: 0.9806
Batch 700/5872 - Loss: 0.0605 Acc: 0.9812
Batch 800/5872 - Loss: 0.0599 Acc: 0.9816
Batch 900/5872 - Loss: 0.0613 Acc: 0.9821
Batch 1000/5872 - Loss: 0.0610 Acc: 0.9821
Batch 1100/5872 - Loss: 0.0607 Acc: 0.9822
Batch 1200/5872 - Loss: 0.0610 Acc: 0.9821
Batch 1300/5872 - Loss: 0.0604 Acc: 0.9820
Batch 1400/5872 - Loss: 0.0601 Acc: 0.9823
Batch 1500/5872 - Loss: 0.0597 Acc: 0.9826
Batch 1600/5872 - Loss: 0.0585 Acc: 0.9828
Batch 1700/5872 - Loss: 0.0576 Acc: 0.9829
Batch 1800/5872 - Loss: 0.0574 Acc: 0.9829
Batch 1900/5872 - Loss: 0.0573 Acc: 0.9828
Batch 2000/5872 - Loss: 0.0571 Acc: 0.9827
Batch 2100/5872 - Loss: 0.0584 Acc: 0.9825
Batch 2200/5872 - Loss: 0.0584 Acc: 0.9827
Batch 2300/5872 - Loss: 0.0580 Acc: 0.9828
Batch 2400/5872 - Loss: 0.0582 Acc: 0.9828
Batch 2500/5872 - Loss: 0.0582 Acc: 0.9827
Batch 2600/5872 - Loss: 0.0578 Acc: 0.9828
```

```
Batch 2700/5872 - Loss: 0.0578 Acc: 0.9828
Batch 2800/5872 - Loss: 0.0573 Acc: 0.9827
Batch 2900/5872 - Loss: 0.0568 Acc: 0.9829
Batch 3000/5872 - Loss: 0.0564 Acc: 0.9829
Batch 3100/5872 - Loss: 0.0576 Acc: 0.9827
Batch 3200/5872 - Loss: 0.0581 Acc: 0.9826
Batch 3300/5872 - Loss: 0.0580 Acc: 0.9826
Batch 3400/5872 - Loss: 0.0576 Acc: 0.9826
Batch 3500/5872 - Loss: 0.0566 Acc: 0.9829
Batch 3600/5872 - Loss: 0.0567 Acc: 0.9827
Batch 3700/5872 - Loss: 0.0563 Acc: 0.9828
Batch 3800/5872 - Loss: 0.0565 Acc: 0.9827
Batch 3900/5872 - Loss: 0.0561 Acc: 0.9828
Batch 4000/5872 - Loss: 0.0568 Acc: 0.9824
Batch 4100/5872 - Loss: 0.0570 Acc: 0.9823
Batch 4200/5872 - Loss: 0.0572 Acc: 0.9822
Batch 4300/5872 - Loss: 0.0572 Acc: 0.9821
Batch 4400/5872 - Loss: 0.0573 Acc: 0.9820
Batch 4500/5872 - Loss: 0.0571 Acc: 0.9820
Batch 4600/5872 - Loss: 0.0568 Acc: 0.9820
Batch 4700/5872 - Loss: 0.0569 Acc: 0.9819
Batch 4800/5872 - Loss: 0.0564 Acc: 0.9821
Batch 4900/5872 - Loss: 0.0563 Acc: 0.9821
Batch 5000/5872 - Loss: 0.0562 Acc: 0.9821
Batch 5100/5872 - Loss: 0.0563 Acc: 0.9821
Batch 5200/5872 - Loss: 0.0561 Acc: 0.9821
Batch 5300/5872 - Loss: 0.0561 Acc: 0.9822
Batch 5400/5872 - Loss: 0.0562 Acc: 0.9822
Batch 5500/5872 - Loss: 0.0564 Acc: 0.9822
Batch 5600/5872 - Loss: 0.0568 Acc: 0.9820
Batch 5700/5872 - Loss: 0.0568 Acc: 0.9820
Batch 5800/5872 - Loss: 0.0565 Acc: 0.9820
Epoch 5 - Train Loss: 0.0566 Train Acc: 0.9820 Val Loss: 0.8480 Val Acc: 0.8892

Epoch 6/8
Batch 100/5872 - Loss: 0.0616 Acc: 0.9775
Batch 200/5872 - Loss: 0.0551 Acc: 0.9806
Batch 300/5872 - Loss: 0.0532 Acc: 0.9817
Batch 400/5872 - Loss: 0.0522 Acc: 0.9822
Batch 500/5872 - Loss: 0.0496 Acc: 0.9838
Batch 600/5872 - Loss: 0.0487 Acc: 0.9842
Batch 700/5872 - Loss: 0.0484 Acc: 0.9841
Batch 800/5872 - Loss: 0.0479 Acc: 0.9841
Batch 900/5872 - Loss: 0.0474 Acc: 0.9844
Batch 1000/5872 - Loss: 0.0457 Acc: 0.9849
Batch 1100/5872 - Loss: 0.0459 Acc: 0.9849
Batch 1200/5872 - Loss: 0.0457 Acc: 0.9850
Batch 1300/5872 - Loss: 0.0465 Acc: 0.9848
```

```
Batch 1400/5872 - Loss: 0.0486 Acc: 0.9842
Batch 1500/5872 - Loss: 0.0493 Acc: 0.9841
Batch 1600/5872 - Loss: 0.0511 Acc: 0.9837
Batch 1700/5872 - Loss: 0.0511 Acc: 0.9836
Batch 1800/5872 - Loss: 0.0515 Acc: 0.9835
Batch 1900/5872 - Loss: 0.0523 Acc: 0.9831
Batch 2000/5872 - Loss: 0.0516 Acc: 0.9831
Batch 2100/5872 - Loss: 0.0519 Acc: 0.9830
Batch 2200/5872 - Loss: 0.0507 Acc: 0.9834
Batch 2300/5872 - Loss: 0.0526 Acc: 0.9827
Batch 2400/5872 - Loss: 0.0522 Acc: 0.9828
Batch 2500/5872 - Loss: 0.0513 Acc: 0.9830
Batch 2600/5872 - Loss: 0.0507 Acc: 0.9832
Batch 2700/5872 - Loss: 0.0502 Acc: 0.9834
Batch 2800/5872 - Loss: 0.0498 Acc: 0.9834
Batch 2900/5872 - Loss: 0.0502 Acc: 0.9834
Batch 3000/5872 - Loss: 0.0499 Acc: 0.9835
Batch 3100/5872 - Loss: 0.0497 Acc: 0.9835
Batch 3200/5872 - Loss: 0.0498 Acc: 0.9834
Batch 3300/5872 - Loss: 0.0491 Acc: 0.9837
Batch 3400/5872 - Loss: 0.0489 Acc: 0.9837
Batch 3500/5872 - Loss: 0.0492 Acc: 0.9835
Batch 3600/5872 - Loss: 0.0492 Acc: 0.9835
Batch 3700/5872 - Loss: 0.0492 Acc: 0.9835
Batch 3800/5872 - Loss: 0.0489 Acc: 0.9837
Batch 3900/5872 - Loss: 0.0490 Acc: 0.9836
Batch 4000/5872 - Loss: 0.0496 Acc: 0.9834
Batch 4100/5872 - Loss: 0.0500 Acc: 0.9832
Batch 4200/5872 - Loss: 0.0503 Acc: 0.9832
Batch 4300/5872 - Loss: 0.0503 Acc: 0.9832
Batch 4400/5872 - Loss: 0.0507 Acc: 0.9831
Batch 4500/5872 - Loss: 0.0513 Acc: 0.9830
Batch 4600/5872 - Loss: 0.0515 Acc: 0.9830
Batch 4700/5872 - Loss: 0.0517 Acc: 0.9828
Batch 4800/5872 - Loss: 0.0515 Acc: 0.9829
Batch 4900/5872 - Loss: 0.0515 Acc: 0.9829
Batch 5000/5872 - Loss: 0.0517 Acc: 0.9828
Batch 5100/5872 - Loss: 0.0515 Acc: 0.9828
Batch 5200/5872 - Loss: 0.0515 Acc: 0.9827
Batch 5300/5872 - Loss: 0.0515 Acc: 0.9827
Batch 5400/5872 - Loss: 0.0513 Acc: 0.9829
Batch 5500/5872 - Loss: 0.0512 Acc: 0.9829
Batch 5600/5872 - Loss: 0.0508 Acc: 0.9830
Batch 5700/5872 - Loss: 0.0507 Acc: 0.9830
Batch 5800/5872 - Loss: 0.0505 Acc: 0.9831
Epoch 6 - Train Loss: 0.0505 Train Acc: 0.9831 Val Loss: 0.7875 Val Acc: 0.8969

Epoch 7/8
```

```
Batch 100/5872 - Loss: 0.0493 Acc: 0.9788
Batch 200/5872 - Loss: 0.0405 Acc: 0.9838
Batch 300/5872 - Loss: 0.0443 Acc: 0.9829
Batch 400/5872 - Loss: 0.0457 Acc: 0.9822
Batch 500/5872 - Loss: 0.0472 Acc: 0.9822
Batch 600/5872 - Loss: 0.0445 Acc: 0.9831
Batch 700/5872 - Loss: 0.0460 Acc: 0.9825
Batch 800/5872 - Loss: 0.0437 Acc: 0.9833
Batch 900/5872 - Loss: 0.0455 Acc: 0.9825
Batch 1000/5872 - Loss: 0.0445 Acc: 0.9828
Batch 1100/5872 - Loss: 0.0441 Acc: 0.9830
Batch 1200/5872 - Loss: 0.0442 Acc: 0.9830
Batch 1300/5872 - Loss: 0.0453 Acc: 0.9828
Batch 1400/5872 - Loss: 0.0441 Acc: 0.9832
Batch 1500/5872 - Loss: 0.0444 Acc: 0.9832
Batch 1600/5872 - Loss: 0.0440 Acc: 0.9834
Batch 1700/5872 - Loss: 0.0450 Acc: 0.9832
Batch 1800/5872 - Loss: 0.0453 Acc: 0.9831
Batch 1900/5872 - Loss: 0.0455 Acc: 0.9830
Batch 2000/5872 - Loss: 0.0454 Acc: 0.9829
Batch 2100/5872 - Loss: 0.0462 Acc: 0.9826
Batch 2200/5872 - Loss: 0.0458 Acc: 0.9828
Batch 2300/5872 - Loss: 0.0459 Acc: 0.9828
Batch 2400/5872 - Loss: 0.0465 Acc: 0.9826
Batch 2500/5872 - Loss: 0.0457 Acc: 0.9830
Batch 2600/5872 - Loss: 0.0453 Acc: 0.9832
Batch 2700/5872 - Loss: 0.0456 Acc: 0.9831
Batch 2800/5872 - Loss: 0.0456 Acc: 0.9831
Batch 2900/5872 - Loss: 0.0452 Acc: 0.9833
Batch 3000/5872 - Loss: 0.0448 Acc: 0.9834
Batch 3100/5872 - Loss: 0.0443 Acc: 0.9836
Batch 3200/5872 - Loss: 0.0444 Acc: 0.9836
Batch 3300/5872 - Loss: 0.0442 Acc: 0.9837
Batch 3400/5872 - Loss: 0.0444 Acc: 0.9837
Batch 3500/5872 - Loss: 0.0443 Acc: 0.9837
Batch 3600/5872 - Loss: 0.0447 Acc: 0.9835
Batch 3700/5872 - Loss: 0.0447 Acc: 0.9834
Batch 3800/5872 - Loss: 0.0449 Acc: 0.9834
Batch 3900/5872 - Loss: 0.0447 Acc: 0.9836
Batch 4000/5872 - Loss: 0.0444 Acc: 0.9837
Batch 4100/5872 - Loss: 0.0445 Acc: 0.9837
Batch 4200/5872 - Loss: 0.0444 Acc: 0.9837
Batch 4300/5872 - Loss: 0.0447 Acc: 0.9837
Batch 4400/5872 - Loss: 0.0445 Acc: 0.9838
Batch 4500/5872 - Loss: 0.0444 Acc: 0.9839
Batch 4600/5872 - Loss: 0.0442 Acc: 0.9839
Batch 4700/5872 - Loss: 0.0446 Acc: 0.9838
Batch 4800/5872 - Loss: 0.0449 Acc: 0.9836
```

```
Batch 4900/5872 - Loss: 0.0456 Acc: 0.9835
Batch 5000/5872 - Loss: 0.0455 Acc: 0.9835
Batch 5100/5872 - Loss: 0.0456 Acc: 0.9835
Batch 5200/5872 - Loss: 0.0454 Acc: 0.9837
Batch 5300/5872 - Loss: 0.0451 Acc: 0.9838
Batch 5400/5872 - Loss: 0.0448 Acc: 0.9839
Batch 5500/5872 - Loss: 0.0452 Acc: 0.9838
Batch 5600/5872 - Loss: 0.0450 Acc: 0.9839
Batch 5700/5872 - Loss: 0.0449 Acc: 0.9839
Batch 5800/5872 - Loss: 0.0446 Acc: 0.9840
Epoch 7 - Train Loss: 0.0448 Train Acc: 0.9839 Val Loss: 0.8129 Val Acc: 0.8987

Epoch 8/8
Batch 100/5872 - Loss: 0.0659 Acc: 0.9750
Batch 200/5872 - Loss: 0.0463 Acc: 0.9838
Batch 300/5872 - Loss: 0.0456 Acc: 0.9829
Batch 400/5872 - Loss: 0.0490 Acc: 0.9819
Batch 500/5872 - Loss: 0.0497 Acc: 0.9812
Batch 600/5872 - Loss: 0.0466 Acc: 0.9823
Batch 700/5872 - Loss: 0.0463 Acc: 0.9823
Batch 800/5872 - Loss: 0.0474 Acc: 0.9817
Batch 900/5872 - Loss: 0.0455 Acc: 0.9825
Batch 1000/5872 - Loss: 0.0452 Acc: 0.9825
Batch 1100/5872 - Loss: 0.0454 Acc: 0.9826
Batch 1200/5872 - Loss: 0.0443 Acc: 0.9829
Batch 1300/5872 - Loss: 0.0442 Acc: 0.9831
Batch 1400/5872 - Loss: 0.0439 Acc: 0.9832
Batch 1500/5872 - Loss: 0.0442 Acc: 0.9830
Batch 1600/5872 - Loss: 0.0438 Acc: 0.9832
Batch 1700/5872 - Loss: 0.0426 Acc: 0.9837
Batch 1800/5872 - Loss: 0.0429 Acc: 0.9834
Batch 1900/5872 - Loss: 0.0427 Acc: 0.9835
Batch 2000/5872 - Loss: 0.0425 Acc: 0.9835
Batch 2100/5872 - Loss: 0.0420 Acc: 0.9838
Batch 2200/5872 - Loss: 0.0425 Acc: 0.9836
Batch 2300/5872 - Loss: 0.0426 Acc: 0.9836
Batch 2400/5872 - Loss: 0.0422 Acc: 0.9838
Batch 2500/5872 - Loss: 0.0418 Acc: 0.9840
Batch 2600/5872 - Loss: 0.0425 Acc: 0.9837
Batch 2700/5872 - Loss: 0.0420 Acc: 0.9838
Batch 2800/5872 - Loss: 0.0423 Acc: 0.9838
Batch 2900/5872 - Loss: 0.0426 Acc: 0.9837
Batch 3000/5872 - Loss: 0.0424 Acc: 0.9838
Batch 3100/5872 - Loss: 0.0425 Acc: 0.9839
Batch 3200/5872 - Loss: 0.0417 Acc: 0.9842
Batch 3300/5872 - Loss: 0.0420 Acc: 0.9840
Batch 3400/5872 - Loss: 0.0427 Acc: 0.9838
Batch 3500/5872 - Loss: 0.0429 Acc: 0.9838
```

```
Batch 3600/5872 - Loss: 0.0428 Acc: 0.9839
Batch 3700/5872 - Loss: 0.0425 Acc: 0.9840
Batch 3800/5872 - Loss: 0.0421 Acc: 0.9841
Batch 3900/5872 - Loss: 0.0421 Acc: 0.9841
Batch 4000/5872 - Loss: 0.0426 Acc: 0.9840
Batch 4100/5872 - Loss: 0.0425 Acc: 0.9841
Batch 4200/5872 - Loss: 0.0426 Acc: 0.9840
Batch 4300/5872 - Loss: 0.0427 Acc: 0.9840
Batch 4400/5872 - Loss: 0.0429 Acc: 0.9839
Batch 4500/5872 - Loss: 0.0432 Acc: 0.9838
Batch 4600/5872 - Loss: 0.0431 Acc: 0.9839
Batch 4700/5872 - Loss: 0.0433 Acc: 0.9838
Batch 4800/5872 - Loss: 0.0433 Acc: 0.9838
Batch 4900/5872 - Loss: 0.0428 Acc: 0.9840
Batch 5000/5872 - Loss: 0.0433 Acc: 0.9838
Batch 5100/5872 - Loss: 0.0430 Acc: 0.9838
Batch 5200/5872 - Loss: 0.0428 Acc: 0.9839
Batch 5300/5872 - Loss: 0.0426 Acc: 0.9839
Batch 5400/5872 - Loss: 0.0424 Acc: 0.9840
Batch 5500/5872 - Loss: 0.0422 Acc: 0.9841
Batch 5600/5872 - Loss: 0.0422 Acc: 0.9841
Batch 5700/5872 - Loss: 0.0421 Acc: 0.9841
Batch 5800/5872 - Loss: 0.0421 Acc: 0.9842
Epoch 8 - Train Loss: 0.0421 Train Acc: 0.9842 Val Loss: 0.8310 Val Acc: 0.8987
→ Fold 4 Best Val Acc: 0.8987

==============================
Fold 5
==============================

Some weights of BertForSequenceClassification were not initialized from the
model checkpoint at bert-base-uncased and are newly initialized:
['classifier.bias', 'classifier.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.


Epoch 1/8
Batch 100/5872 - Loss: 2.5118 Acc: 0.1575
Batch 200/5872 - Loss: 2.3120 Acc: 0.2537
Batch 300/5872 - Loss: 2.0902 Acc: 0.3308
Batch 400/5872 - Loss: 1.9078 Acc: 0.3928
Batch 500/5872 - Loss: 1.7536 Acc: 0.4432
Batch 600/5872 - Loss: 1.6182 Acc: 0.4879
Batch 700/5872 - Loss: 1.5171 Acc: 0.5212
Batch 800/5872 - Loss: 1.4295 Acc: 0.5480
Batch 900/5872 - Loss: 1.3658 Acc: 0.5693
Batch 1000/5872 - Loss: 1.3011 Acc: 0.5893
Batch 1100/5872 - Loss: 1.2400 Acc: 0.6100
```

```
Batch 1200/5872 - Loss: 1.1879 Acc: 0.6249
Batch 1300/5872 - Loss: 1.1404 Acc: 0.6402
Batch 1400/5872 - Loss: 1.0966 Acc: 0.6538
Batch 1500/5872 - Loss: 1.0597 Acc: 0.6652
Batch 1600/5872 - Loss: 1.0243 Acc: 0.6763
Batch 1700/5872 - Loss: 0.9945 Acc: 0.6860
Batch 1800/5872 - Loss: 0.9654 Acc: 0.6954
Batch 1900/5872 - Loss: 0.9376 Acc: 0.7041
Batch 2000/5872 - Loss: 0.9144 Acc: 0.7118
Batch 2100/5872 - Loss: 0.8904 Acc: 0.7194
Batch 2200/5872 - Loss: 0.8681 Acc: 0.7265
Batch 2300/5872 - Loss: 0.8450 Acc: 0.7337
Batch 2400/5872 - Loss: 0.8269 Acc: 0.7398
Batch 2500/5872 - Loss: 0.8089 Acc: 0.7452
Batch 2600/5872 - Loss: 0.7923 Acc: 0.7512
Batch 2700/5872 - Loss: 0.7754 Acc: 0.7569
Batch 2800/5872 - Loss: 0.7566 Acc: 0.7630
Batch 2900/5872 - Loss: 0.7405 Acc: 0.7683
Batch 3000/5872 - Loss: 0.7265 Acc: 0.7733
Batch 3100/5872 - Loss: 0.7135 Acc: 0.7782
Batch 3200/5872 - Loss: 0.7025 Acc: 0.7822
Batch 3300/5872 - Loss: 0.6895 Acc: 0.7865
Batch 3400/5872 - Loss: 0.6758 Acc: 0.7908
Batch 3500/5872 - Loss: 0.6652 Acc: 0.7946
Batch 3600/5872 - Loss: 0.6542 Acc: 0.7982
Batch 3700/5872 - Loss: 0.6440 Acc: 0.8019
Batch 3800/5872 - Loss: 0.6331 Acc: 0.8056
Batch 3900/5872 - Loss: 0.6221 Acc: 0.8090
Batch 4000/5872 - Loss: 0.6128 Acc: 0.8122
Batch 4100/5872 - Loss: 0.6042 Acc: 0.8152
Batch 4200/5872 - Loss: 0.5957 Acc: 0.8182
Batch 4300/5872 - Loss: 0.5894 Acc: 0.8207
Batch 4400/5872 - Loss: 0.5819 Acc: 0.8233
Batch 4500/5872 - Loss: 0.5743 Acc: 0.8257
Batch 4600/5872 - Loss: 0.5660 Acc: 0.8283
Batch 4700/5872 - Loss: 0.5570 Acc: 0.8310
Batch 4800/5872 - Loss: 0.5507 Acc: 0.8332
Batch 4900/5872 - Loss: 0.5436 Acc: 0.8355
Batch 5000/5872 - Loss: 0.5374 Acc: 0.8376
Batch 5100/5872 - Loss: 0.5314 Acc: 0.8396
Batch 5200/5872 - Loss: 0.5256 Acc: 0.8417
Batch 5300/5872 - Loss: 0.5199 Acc: 0.8436
Batch 5400/5872 - Loss: 0.5137 Acc: 0.8456
Batch 5500/5872 - Loss: 0.5076 Acc: 0.8475
Batch 5600/5872 - Loss: 0.5023 Acc: 0.8494
Batch 5700/5872 - Loss: 0.4974 Acc: 0.8510
Batch 5800/5872 - Loss: 0.4909 Acc: 0.8531
Epoch 1 - Train Loss: 0.4873 Train Acc: 0.8543 Val Loss: 0.8527 Val Acc: 0.8437
```

```
Epoch 2/8
Batch 100/5872 - Loss: 0.1197 Acc: 0.9688
Batch 200/5872 - Loss: 0.1416 Acc: 0.9681
Batch 300/5872 - Loss: 0.1645 Acc: 0.9621
Batch 400/5872 - Loss: 0.1560 Acc: 0.9631
Batch 500/5872 - Loss: 0.1519 Acc: 0.9637
Batch 600/5872 - Loss: 0.1487 Acc: 0.9654
Batch 700/5872 - Loss: 0.1430 Acc: 0.9663
Batch 800/5872 - Loss: 0.1375 Acc: 0.9673
Batch 900/5872 - Loss: 0.1399 Acc: 0.9675
Batch 1000/5872 - Loss: 0.1403 Acc: 0.9669
Batch 1100/5872 - Loss: 0.1424 Acc: 0.9661
Batch 1200/5872 - Loss: 0.1412 Acc: 0.9668
Batch 1300/5872 - Loss: 0.1396 Acc: 0.9672
Batch 1400/5872 - Loss: 0.1377 Acc: 0.9674
Batch 1500/5872 - Loss: 0.1369 Acc: 0.9678
Batch 1600/5872 - Loss: 0.1364 Acc: 0.9677
Batch 1700/5872 - Loss: 0.1355 Acc: 0.9679
Batch 1800/5872 - Loss: 0.1341 Acc: 0.9678
Batch 1900/5872 - Loss: 0.1340 Acc: 0.9680
Batch 2000/5872 - Loss: 0.1341 Acc: 0.9679
Batch 2100/5872 - Loss: 0.1327 Acc: 0.9679
Batch 2200/5872 - Loss: 0.1321 Acc: 0.9683
Batch 2300/5872 - Loss: 0.1322 Acc: 0.9685
Batch 2400/5872 - Loss: 0.1311 Acc: 0.9686
Batch 2500/5872 - Loss: 0.1309 Acc: 0.9686
Batch 2600/5872 - Loss: 0.1301 Acc: 0.9689
Batch 2700/5872 - Loss: 0.1305 Acc: 0.9691
Batch 2800/5872 - Loss: 0.1302 Acc: 0.9691
Batch 2900/5872 - Loss: 0.1306 Acc: 0.9688
Batch 3000/5872 - Loss: 0.1317 Acc: 0.9683
Batch 3100/5872 - Loss: 0.1335 Acc: 0.9680
Batch 3200/5872 - Loss: 0.1333 Acc: 0.9680
Batch 3300/5872 - Loss: 0.1332 Acc: 0.9679
Batch 3400/5872 - Loss: 0.1322 Acc: 0.9682
Batch 3500/5872 - Loss: 0.1327 Acc: 0.9683
Batch 3600/5872 - Loss: 0.1328 Acc: 0.9683
Batch 3700/5872 - Loss: 0.1321 Acc: 0.9684
Batch 3800/5872 - Loss: 0.1313 Acc: 0.9686
Batch 3900/5872 - Loss: 0.1311 Acc: 0.9687
Batch 4000/5872 - Loss: 0.1299 Acc: 0.9689
Batch 4100/5872 - Loss: 0.1288 Acc: 0.9692
Batch 4200/5872 - Loss: 0.1281 Acc: 0.9694
Batch 4300/5872 - Loss: 0.1283 Acc: 0.9692
Batch 4400/5872 - Loss: 0.1280 Acc: 0.9692
Batch 4500/5872 - Loss: 0.1268 Acc: 0.9694
Batch 4600/5872 - Loss: 0.1266 Acc: 0.9694
```

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Batch 4700/5872 - Loss: 0.1262 Acc: 0.9695
Batch 4800/5872 - Loss: 0.1263 Acc: 0.9696
Batch 4900/5872 - Loss: 0.1261 Acc: 0.9696
Batch 5000/5872 - Loss: 0.1273 Acc: 0.9695
Batch 5100/5872 - Loss: 0.1264 Acc: 0.9695
Batch 5200/5872 - Loss: 0.1254 Acc: 0.9696
Batch 5300/5872 - Loss: 0.1247 Acc: 0.9697
Batch 5400/5872 - Loss: 0.1245 Acc: 0.9697
Batch 5500/5872 - Loss: 0.1243 Acc: 0.9697
Batch 5600/5872 - Loss: 0.1235 Acc: 0.9698
Batch 5700/5872 - Loss: 0.1241 Acc: 0.9696
Batch 5800/5872 - Loss: 0.1237 Acc: 0.9698
Epoch 2 - Train Loss: 0.1236 Train Acc: 0.9698 Val Loss: 0.7406 Val Acc: 0.8826

Epoch 3/8
Batch 100/5872 - Loss: 0.0562 Acc: 0.9875
Batch 200/5872 - Loss: 0.0797 Acc: 0.9788
Batch 300/5872 - Loss: 0.0767 Acc: 0.9788
Batch 400/5872 - Loss: 0.0750 Acc: 0.9788
Batch 500/5872 - Loss: 0.0810 Acc: 0.9770
Batch 600/5872 - Loss: 0.0883 Acc: 0.9762
Batch 700/5872 - Loss: 0.0928 Acc: 0.9761
Batch 800/5872 - Loss: 0.0931 Acc: 0.9766
Batch 900/5872 - Loss: 0.0921 Acc: 0.9765
Batch 1000/5872 - Loss: 0.0912 Acc: 0.9766
Batch 1100/5872 - Loss: 0.0937 Acc: 0.9761
Batch 1200/5872 - Loss: 0.0911 Acc: 0.9768
Batch 1300/5872 - Loss: 0.0933 Acc: 0.9760
Batch 1400/5872 - Loss: 0.0933 Acc: 0.9757
Batch 1500/5872 - Loss: 0.0915 Acc: 0.9762
Batch 1600/5872 - Loss: 0.0922 Acc: 0.9761
Batch 1700/5872 - Loss: 0.0905 Acc: 0.9764
Batch 1800/5872 - Loss: 0.0911 Acc: 0.9764
Batch 1900/5872 - Loss: 0.0922 Acc: 0.9762
Batch 2000/5872 - Loss: 0.0911 Acc: 0.9762
Batch 2100/5872 - Loss: 0.0919 Acc: 0.9761
Batch 2200/5872 - Loss: 0.0914 Acc: 0.9762
Batch 2300/5872 - Loss: 0.0924 Acc: 0.9758
Batch 2400/5872 - Loss: 0.0914 Acc: 0.9760
Batch 2500/5872 - Loss: 0.0910 Acc: 0.9759
Batch 2600/5872 - Loss: 0.0908 Acc: 0.9759
Batch 2700/5872 - Loss: 0.0907 Acc: 0.9759
Batch 2800/5872 - Loss: 0.0905 Acc: 0.9757
Batch 2900/5872 - Loss: 0.0899 Acc: 0.9758
Batch 3000/5872 - Loss: 0.0905 Acc: 0.9755
Batch 3100/5872 - Loss: 0.0903 Acc: 0.9753
Batch 3200/5872 - Loss: 0.0893 Acc: 0.9755
Batch 3300/5872 - Loss: 0.0894 Acc: 0.9754
```

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Batch 3400/5872 - Loss: 0.0885 Acc: 0.9755
Batch 3500/5872 - Loss: 0.0883 Acc: 0.9756
Batch 3600/5872 - Loss: 0.0886 Acc: 0.9757
Batch 3700/5872 - Loss: 0.0880 Acc: 0.9759
Batch 3800/5872 - Loss: 0.0876 Acc: 0.9760
Batch 3900/5872 - Loss: 0.0874 Acc: 0.9761
Batch 4000/5872 - Loss: 0.0874 Acc: 0.9761
Batch 4100/5872 - Loss: 0.0862 Acc: 0.9764
Batch 4200/5872 - Loss: 0.0862 Acc: 0.9765
Batch 4300/5872 - Loss: 0.0870 Acc: 0.9765
Batch 4400/5872 - Loss: 0.0881 Acc: 0.9761
Batch 4500/5872 - Loss: 0.0881 Acc: 0.9761
Batch 4600/5872 - Loss: 0.0880 Acc: 0.9763
Batch 4700/5872 - Loss: 0.0875 Acc: 0.9764
Batch 4800/5872 - Loss: 0.0875 Acc: 0.9764
Batch 4900/5872 - Loss: 0.0868 Acc: 0.9764
Batch 5000/5872 - Loss: 0.0867 Acc: 0.9763
Batch 5100/5872 - Loss: 0.0859 Acc: 0.9765
Batch 5200/5872 - Loss: 0.0857 Acc: 0.9765
Batch 5300/5872 - Loss: 0.0853 Acc: 0.9766
Batch 5400/5872 - Loss: 0.0857 Acc: 0.9765
Batch 5500/5872 - Loss: 0.0853 Acc: 0.9765
Batch 5600/5872 - Loss: 0.0853 Acc: 0.9766
Batch 5700/5872 - Loss: 0.0855 Acc: 0.9766
Batch 5800/5872 - Loss: 0.0854 Acc: 0.9766
Epoch 3 - Train Loss: 0.0855 Train Acc: 0.9766 Val Loss: 0.7667 Val Acc: 0.8892

Epoch 4/8
Batch 100/5872 - Loss: 0.0975 Acc: 0.9725
Batch 200/5872 - Loss: 0.0743 Acc: 0.9800
Batch 300/5872 - Loss: 0.0734 Acc: 0.9796
Batch 400/5872 - Loss: 0.0782 Acc: 0.9791
Batch 500/5872 - Loss: 0.0763 Acc: 0.9800
Batch 600/5872 - Loss: 0.0829 Acc: 0.9785
Batch 700/5872 - Loss: 0.0857 Acc: 0.9773
Batch 800/5872 - Loss: 0.0866 Acc: 0.9773
Batch 900/5872 - Loss: 0.0834 Acc: 0.9779
Batch 1000/5872 - Loss: 0.0845 Acc: 0.9774
Batch 1100/5872 - Loss: 0.0818 Acc: 0.9782
Batch 1200/5872 - Loss: 0.0808 Acc: 0.9783
Batch 1300/5872 - Loss: 0.0796 Acc: 0.9785
Batch 1400/5872 - Loss: 0.0777 Acc: 0.9788
Batch 1500/5872 - Loss: 0.0771 Acc: 0.9792
Batch 1600/5872 - Loss: 0.0758 Acc: 0.9793
Batch 1700/5872 - Loss: 0.0742 Acc: 0.9798
Batch 1800/5872 - Loss: 0.0727 Acc: 0.9802
Batch 1900/5872 - Loss: 0.0738 Acc: 0.9802
Batch 2000/5872 - Loss: 0.0739 Acc: 0.9801
```

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Batch 2100/5872 - Loss: 0.0741 Acc: 0.9799
Batch 2200/5872 - Loss: 0.0732 Acc: 0.9801
Batch 2300/5872 - Loss: 0.0741 Acc: 0.9798
Batch 2400/5872 - Loss: 0.0732 Acc: 0.9800
Batch 2500/5872 - Loss: 0.0740 Acc: 0.9795
Batch 2600/5872 - Loss: 0.0743 Acc: 0.9794
Batch 2700/5872 - Loss: 0.0735 Acc: 0.9794
Batch 2800/5872 - Loss: 0.0725 Acc: 0.9796
Batch 2900/5872 - Loss: 0.0719 Acc: 0.9796
Batch 3000/5872 - Loss: 0.0709 Acc: 0.9799
Batch 3100/5872 - Loss: 0.0705 Acc: 0.9799
Batch 3200/5872 - Loss: 0.0697 Acc: 0.9800
Batch 3300/5872 - Loss: 0.0695 Acc: 0.9799
Batch 3400/5872 - Loss: 0.0690 Acc: 0.9801
Batch 3500/5872 - Loss: 0.0688 Acc: 0.9801
Batch 3600/5872 - Loss: 0.0681 Acc: 0.9803
Batch 3700/5872 - Loss: 0.0685 Acc: 0.9802
Batch 3800/5872 - Loss: 0.0698 Acc: 0.9799
Batch 3900/5872 - Loss: 0.0697 Acc: 0.9799
Batch 4000/5872 - Loss: 0.0704 Acc: 0.9797
Batch 4100/5872 - Loss: 0.0699 Acc: 0.9798
Batch 4200/5872 - Loss: 0.0694 Acc: 0.9799
Batch 4300/5872 - Loss: 0.0690 Acc: 0.9801
Batch 4400/5872 - Loss: 0.0693 Acc: 0.9800
Batch 4500/5872 - Loss: 0.0692 Acc: 0.9800
Batch 4600/5872 - Loss: 0.0691 Acc: 0.9800
Batch 4700/5872 - Loss: 0.0691 Acc: 0.9799
Batch 4800/5872 - Loss: 0.0682 Acc: 0.9801
Batch 4900/5872 - Loss: 0.0682 Acc: 0.9801
Batch 5000/5872 - Loss: 0.0688 Acc: 0.9801
Batch 5100/5872 - Loss: 0.0687 Acc: 0.9801
Batch 5200/5872 - Loss: 0.0682 Acc: 0.9802
Batch 5300/5872 - Loss: 0.0678 Acc: 0.9803
Batch 5400/5872 - Loss: 0.0678 Acc: 0.9803
Batch 5500/5872 - Loss: 0.0680 Acc: 0.9802
Batch 5600/5872 - Loss: 0.0676 Acc: 0.9803
Batch 5700/5872 - Loss: 0.0672 Acc: 0.9804
Batch 5800/5872 - Loss: 0.0675 Acc: 0.9804
Epoch 4 - Train Loss: 0.0673 Train Acc: 0.9804 Val Loss: 0.8310 Val Acc: 0.8767

Epoch 5/8
Batch 100/5872 - Loss: 0.0355 Acc: 0.9862
Batch 200/5872 - Loss: 0.0435 Acc: 0.9838
Batch 300/5872 - Loss: 0.0458 Acc: 0.9842
Batch 400/5872 - Loss: 0.0438 Acc: 0.9853
Batch 500/5872 - Loss: 0.0525 Acc: 0.9835
Batch 600/5872 - Loss: 0.0513 Acc: 0.9838
Batch 700/5872 - Loss: 0.0517 Acc: 0.9836
```

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Batch 800/5872 - Loss: 0.0553 Acc: 0.9830
Batch 900/5872 - Loss: 0.0556 Acc: 0.9831
Batch 1000/5872 - Loss: 0.0582 Acc: 0.9825
Batch 1100/5872 - Loss: 0.0570 Acc: 0.9828
Batch 1200/5872 - Loss: 0.0575 Acc: 0.9826
Batch 1300/5872 - Loss: 0.0580 Acc: 0.9824
Batch 1400/5872 - Loss: 0.0584 Acc: 0.9825
Batch 1500/5872 - Loss: 0.0586 Acc: 0.9822
Batch 1600/5872 - Loss: 0.0600 Acc: 0.9818
Batch 1700/5872 - Loss: 0.0588 Acc: 0.9821
Batch 1800/5872 - Loss: 0.0580 Acc: 0.9824
Batch 1900/5872 - Loss: 0.0587 Acc: 0.9821
Batch 2000/5872 - Loss: 0.0588 Acc: 0.9821
Batch 2100/5872 - Loss: 0.0591 Acc: 0.9821
Batch 2200/5872 - Loss: 0.0587 Acc: 0.9820
Batch 2300/5872 - Loss: 0.0591 Acc: 0.9818
Batch 2400/5872 - Loss: 0.0589 Acc: 0.9819
Batch 2500/5872 - Loss: 0.0579 Acc: 0.9821
Batch 2600/5872 - Loss: 0.0579 Acc: 0.9821
Batch 2700/5872 - Loss: 0.0576 Acc: 0.9820
Batch 2800/5872 - Loss: 0.0582 Acc: 0.9821
Batch 2900/5872 - Loss: 0.0579 Acc: 0.9822
Batch 3000/5872 - Loss: 0.0587 Acc: 0.9820
Batch 3100/5872 - Loss: 0.0586 Acc: 0.9820
Batch 3200/5872 - Loss: 0.0592 Acc: 0.9817
Batch 3300/5872 - Loss: 0.0604 Acc: 0.9814
Batch 3400/5872 - Loss: 0.0603 Acc: 0.9814
Batch 3500/5872 - Loss: 0.0600 Acc: 0.9814
Batch 3600/5872 - Loss: 0.0600 Acc: 0.9812
Batch 3700/5872 - Loss: 0.0598 Acc: 0.9812
Batch 3800/5872 - Loss: 0.0598 Acc: 0.9812
Batch 3900/5872 - Loss: 0.0597 Acc: 0.9813
Batch 4000/5872 - Loss: 0.0592 Acc: 0.9814
Batch 4100/5872 - Loss: 0.0586 Acc: 0.9816
Batch 4200/5872 - Loss: 0.0583 Acc: 0.9817
Batch 4300/5872 - Loss: 0.0577 Acc: 0.9819
Batch 4400/5872 - Loss: 0.0580 Acc: 0.9817
Batch 4500/5872 - Loss: 0.0576 Acc: 0.9818
Batch 4600/5872 - Loss: 0.0577 Acc: 0.9817
Batch 4700/5872 - Loss: 0.0572 Acc: 0.9818
Batch 4800/5872 - Loss: 0.0570 Acc: 0.9818
Batch 4900/5872 - Loss: 0.0569 Acc: 0.9819
Batch 5000/5872 - Loss: 0.0571 Acc: 0.9819
Batch 5100/5872 - Loss: 0.0569 Acc: 0.9819
Batch 5200/5872 - Loss: 0.0567 Acc: 0.9820
Batch 5300/5872 - Loss: 0.0565 Acc: 0.9821
Batch 5400/5872 - Loss: 0.0563 Acc: 0.9822
Batch 5500/5872 - Loss: 0.0559 Acc: 0.9823
```

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Batch 5600/5872 - Loss: 0.0560 Acc: 0.9822
Batch 5700/5872 - Loss: 0.0556 Acc: 0.9823
Batch 5800/5872 - Loss: 0.0555 Acc: 0.9823
Epoch 5 - Train Loss: 0.0552 Train Acc: 0.9824 Val Loss: 0.8192 Val Acc: 0.8925

Epoch 6/8
Batch 100/5872 - Loss: 0.0693 Acc: 0.9800
Batch 200/5872 - Loss: 0.0628 Acc: 0.9806
Batch 300/5872 - Loss: 0.0568 Acc: 0.9825
Batch 400/5872 - Loss: 0.0554 Acc: 0.9831
Batch 500/5872 - Loss: 0.0573 Acc: 0.9822
Batch 600/5872 - Loss: 0.0577 Acc: 0.9817
Batch 700/5872 - Loss: 0.0533 Acc: 0.9829
Batch 800/5872 - Loss: 0.0528 Acc: 0.9830
Batch 900/5872 - Loss: 0.0517 Acc: 0.9831
Batch 1000/5872 - Loss: 0.0515 Acc: 0.9825
Batch 1100/5872 - Loss: 0.0513 Acc: 0.9824
Batch 1200/5872 - Loss: 0.0505 Acc: 0.9824
Batch 1300/5872 - Loss: 0.0499 Acc: 0.9827
Batch 1400/5872 - Loss: 0.0501 Acc: 0.9829
Batch 1500/5872 - Loss: 0.0505 Acc: 0.9824
Batch 1600/5872 - Loss: 0.0505 Acc: 0.9825
Batch 1700/5872 - Loss: 0.0506 Acc: 0.9825
Batch 1800/5872 - Loss: 0.0506 Acc: 0.9824
Batch 1900/5872 - Loss: 0.0514 Acc: 0.9820
Batch 2000/5872 - Loss: 0.0517 Acc: 0.9820
Batch 2100/5872 - Loss: 0.0517 Acc: 0.9820
Batch 2200/5872 - Loss: 0.0520 Acc: 0.9822
Batch 2300/5872 - Loss: 0.0518 Acc: 0.9822
Batch 2400/5872 - Loss: 0.0515 Acc: 0.9824
Batch 2500/5872 - Loss: 0.0519 Acc: 0.9821
Batch 2600/5872 - Loss: 0.0514 Acc: 0.9825
Batch 2700/5872 - Loss: 0.0511 Acc: 0.9826
Batch 2800/5872 - Loss: 0.0516 Acc: 0.9825
Batch 2900/5872 - Loss: 0.0508 Acc: 0.9828
Batch 3000/5872 - Loss: 0.0509 Acc: 0.9828
Batch 3100/5872 - Loss: 0.0506 Acc: 0.9828
Batch 3200/5872 - Loss: 0.0503 Acc: 0.9829
Batch 3300/5872 - Loss: 0.0495 Acc: 0.9832
Batch 3400/5872 - Loss: 0.0495 Acc: 0.9832
Batch 3500/5872 - Loss: 0.0496 Acc: 0.9831
Batch 3600/5872 - Loss: 0.0504 Acc: 0.9828
Batch 3700/5872 - Loss: 0.0506 Acc: 0.9829
Batch 3800/5872 - Loss: 0.0511 Acc: 0.9828
Batch 3900/5872 - Loss: 0.0505 Acc: 0.9829
Batch 4000/5872 - Loss: 0.0502 Acc: 0.9830
Batch 4100/5872 - Loss: 0.0506 Acc: 0.9828
Batch 4200/5872 - Loss: 0.0503 Acc: 0.9828
```

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Batch 4300/5872 - Loss: 0.0504 Acc: 0.9828
Batch 4400/5872 - Loss: 0.0505 Acc: 0.9828
Batch 4500/5872 - Loss: 0.0506 Acc: 0.9829
Batch 4600/5872 - Loss: 0.0505 Acc: 0.9829
Batch 4700/5872 - Loss: 0.0506 Acc: 0.9829
Batch 4800/5872 - Loss: 0.0510 Acc: 0.9829
Batch 4900/5872 - Loss: 0.0509 Acc: 0.9829
Batch 5000/5872 - Loss: 0.0503 Acc: 0.9831
Batch 5100/5872 - Loss: 0.0499 Acc: 0.9832
Batch 5200/5872 - Loss: 0.0496 Acc: 0.9833
Batch 5300/5872 - Loss: 0.0493 Acc: 0.9834
Batch 5400/5872 - Loss: 0.0500 Acc: 0.9832
Batch 5500/5872 - Loss: 0.0498 Acc: 0.9832
Batch 5600/5872 - Loss: 0.0495 Acc: 0.9834
Batch 5700/5872 - Loss: 0.0495 Acc: 0.9833
Batch 5800/5872 - Loss: 0.0496 Acc: 0.9833
Epoch 6 - Train Loss: 0.0497 Train Acc: 0.9832 Val Loss: 0.8046 Val Acc: 0.8950

Epoch 7/8
Batch 100/5872 - Loss: 0.0288 Acc: 0.9888
Batch 200/5872 - Loss: 0.0387 Acc: 0.9856
Batch 300/5872 - Loss: 0.0480 Acc: 0.9838
Batch 400/5872 - Loss: 0.0473 Acc: 0.9834
Batch 500/5872 - Loss: 0.0431 Acc: 0.9845
Batch 600/5872 - Loss: 0.0405 Acc: 0.9852
Batch 700/5872 - Loss: 0.0444 Acc: 0.9848
Batch 800/5872 - Loss: 0.0430 Acc: 0.9853
Batch 900/5872 - Loss: 0.0420 Acc: 0.9854
Batch 1000/5872 - Loss: 0.0416 Acc: 0.9858
Batch 1100/5872 - Loss: 0.0439 Acc: 0.9850
Batch 1200/5872 - Loss: 0.0433 Acc: 0.9852
Batch 1300/5872 - Loss: 0.0438 Acc: 0.9852
Batch 1400/5872 - Loss: 0.0436 Acc: 0.9854
Batch 1500/5872 - Loss: 0.0428 Acc: 0.9857
Batch 1600/5872 - Loss: 0.0428 Acc: 0.9857
Batch 1700/5872 - Loss: 0.0442 Acc: 0.9854
Batch 1800/5872 - Loss: 0.0449 Acc: 0.9851
Batch 1900/5872 - Loss: 0.0446 Acc: 0.9853
Batch 2000/5872 - Loss: 0.0463 Acc: 0.9848
Batch 2100/5872 - Loss: 0.0464 Acc: 0.9847
Batch 2200/5872 - Loss: 0.0475 Acc: 0.9843
Batch 2300/5872 - Loss: 0.0480 Acc: 0.9842
Batch 2400/5872 - Loss: 0.0478 Acc: 0.9842
Batch 2500/5872 - Loss: 0.0476 Acc: 0.9841
Batch 2600/5872 - Loss: 0.0473 Acc: 0.9841
Batch 2700/5872 - Loss: 0.0471 Acc: 0.9843
Batch 2800/5872 - Loss: 0.0468 Acc: 0.9843
Batch 2900/5872 - Loss: 0.0467 Acc: 0.9842
```

```
Batch 3000/5872 - Loss: 0.0466 Acc: 0.9842
Batch 3100/5872 - Loss: 0.0466 Acc: 0.9842
Batch 3200/5872 - Loss: 0.0461 Acc: 0.9843
Batch 3300/5872 - Loss: 0.0460 Acc: 0.9843
Batch 3400/5872 - Loss: 0.0458 Acc: 0.9843
Batch 3500/5872 - Loss: 0.0461 Acc: 0.9841
Batch 3600/5872 - Loss: 0.0465 Acc: 0.9840
Batch 3700/5872 - Loss: 0.0468 Acc: 0.9839
Batch 3800/5872 - Loss: 0.0463 Acc: 0.9840
Batch 3900/5872 - Loss: 0.0464 Acc: 0.9840
Batch 4000/5872 - Loss: 0.0461 Acc: 0.9841
Batch 4100/5872 - Loss: 0.0459 Acc: 0.9842
Batch 4200/5872 - Loss: 0.0459 Acc: 0.9842
Batch 4300/5872 - Loss: 0.0460 Acc: 0.9842
Batch 4400/5872 - Loss: 0.0460 Acc: 0.9841
Batch 4500/5872 - Loss: 0.0460 Acc: 0.9841
Batch 4600/5872 - Loss: 0.0458 Acc: 0.9841
Batch 4700/5872 - Loss: 0.0460 Acc: 0.9841
Batch 4800/5872 - Loss: 0.0461 Acc: 0.9840
Batch 4900/5872 - Loss: 0.0461 Acc: 0.9840
Batch 5000/5872 - Loss: 0.0459 Acc: 0.9840
Batch 5100/5872 - Loss: 0.0458 Acc: 0.9839
Batch 5200/5872 - Loss: 0.0455 Acc: 0.9840
Batch 5300/5872 - Loss: 0.0457 Acc: 0.9839
Batch 5400/5872 - Loss: 0.0457 Acc: 0.9839
Batch 5500/5872 - Loss: 0.0456 Acc: 0.9839
Batch 5600/5872 - Loss: 0.0456 Acc: 0.9840
Batch 5700/5872 - Loss: 0.0456 Acc: 0.9840
Batch 5800/5872 - Loss: 0.0460 Acc: 0.9839
Epoch 7 - Train Loss: 0.0459 Train Acc: 0.9839 Val Loss: 0.7824 Val Acc: 0.8994

Epoch 8/8
Batch 100/5872 - Loss: 0.0365 Acc: 0.9825
Batch 200/5872 - Loss: 0.0300 Acc: 0.9875
Batch 300/5872 - Loss: 0.0336 Acc: 0.9858
Batch 400/5872 - Loss: 0.0345 Acc: 0.9859
Batch 500/5872 - Loss: 0.0360 Acc: 0.9855
Batch 600/5872 - Loss: 0.0364 Acc: 0.9860
Batch 700/5872 - Loss: 0.0366 Acc: 0.9862
Batch 800/5872 - Loss: 0.0361 Acc: 0.9861
Batch 900/5872 - Loss: 0.0379 Acc: 0.9854
Batch 1000/5872 - Loss: 0.0371 Acc: 0.9856
Batch 1100/5872 - Loss: 0.0368 Acc: 0.9858
Batch 1200/5872 - Loss: 0.0381 Acc: 0.9854
Batch 1300/5872 - Loss: 0.0406 Acc: 0.9845
Batch 1400/5872 - Loss: 0.0415 Acc: 0.9845
Batch 1500/5872 - Loss: 0.0420 Acc: 0.9842
Batch 1600/5872 - Loss: 0.0432 Acc: 0.9839
```

```
Batch 1700/5872 - Loss: 0.0429 Acc: 0.9839
Batch 1800/5872 - Loss: 0.0429 Acc: 0.9839
Batch 1900/5872 - Loss: 0.0419 Acc: 0.9842
Batch 2000/5872 - Loss: 0.0419 Acc: 0.9842
Batch 2100/5872 - Loss: 0.0427 Acc: 0.9839
Batch 2200/5872 - Loss: 0.0436 Acc: 0.9835
Batch 2300/5872 - Loss: 0.0429 Acc: 0.9838
Batch 2400/5872 - Loss: 0.0428 Acc: 0.9837
Batch 2500/5872 - Loss: 0.0425 Acc: 0.9839
Batch 2600/5872 - Loss: 0.0428 Acc: 0.9836
Batch 2700/5872 - Loss: 0.0428 Acc: 0.9835
Batch 2800/5872 - Loss: 0.0433 Acc: 0.9834
Batch 2900/5872 - Loss: 0.0430 Acc: 0.9834
Batch 3000/5872 - Loss: 0.0425 Acc: 0.9837
Batch 3100/5872 - Loss: 0.0422 Acc: 0.9837
Batch 3200/5872 - Loss: 0.0426 Acc: 0.9836
Batch 3300/5872 - Loss: 0.0422 Acc: 0.9838
Batch 3400/5872 - Loss: 0.0424 Acc: 0.9836
Batch 3500/5872 - Loss: 0.0427 Acc: 0.9836
Batch 3600/5872 - Loss: 0.0419 Acc: 0.9839
Batch 3700/5872 - Loss: 0.0414 Acc: 0.9841
Batch 3800/5872 - Loss: 0.0417 Acc: 0.9841
Batch 3900/5872 - Loss: 0.0418 Acc: 0.9841
Batch 4000/5872 - Loss: 0.0415 Acc: 0.9843
Batch 4100/5872 - Loss: 0.0414 Acc: 0.9843
Batch 4200/5872 - Loss: 0.0416 Acc: 0.9842
Batch 4300/5872 - Loss: 0.0418 Acc: 0.9841
Batch 4400/5872 - Loss: 0.0414 Acc: 0.9843
Batch 4500/5872 - Loss: 0.0416 Acc: 0.9843
Batch 4600/5872 - Loss: 0.0416 Acc: 0.9843
Batch 4700/5872 - Loss: 0.0416 Acc: 0.9844
Batch 4800/5872 - Loss: 0.0414 Acc: 0.9844
Batch 4900/5872 - Loss: 0.0417 Acc: 0.9843
Batch 5000/5872 - Loss: 0.0417 Acc: 0.9843
Batch 5100/5872 - Loss: 0.0415 Acc: 0.9844
Batch 5200/5872 - Loss: 0.0415 Acc: 0.9844
Batch 5300/5872 - Loss: 0.0417 Acc: 0.9843
Batch 5400/5872 - Loss: 0.0418 Acc: 0.9843
Batch 5500/5872 - Loss: 0.0415 Acc: 0.9844
Batch 5600/5872 - Loss: 0.0416 Acc: 0.9844
Batch 5700/5872 - Loss: 0.0417 Acc: 0.9843
Batch 5800/5872 - Loss: 0.0417 Acc: 0.9844
Epoch 8 - Train Loss: 0.0418 Train Acc: 0.9843 Val Loss: 0.7899 Val Acc: 0.9002
→ Fold 5 Best Val Acc: 0.9002

==============================
5-Fold CV Results
==============================
```

```
   fold   val_acc
0     1  0.888481
1     2  0.907190
2     3  0.896552
3     4  0.898716
4     5  0.900183

 5-Fold CV Accuracy: 0.8982 ± 0.0067

Retraining final BERT on full TRAIN set

Some weights of BertForSequenceClassification were not initialized from the
model checkpoint at bert-base-uncased and are newly initialized:
['classifier.bias', 'classifier.weight']
You should probably TRAIN this model on a down-stream task to be able to use it
for predictions and inference.


Epoch 1/8
Batch 100/7339 - Loss: 2.4804 Acc: 0.1525
Batch 200/7339 - Loss: 2.2277 Acc: 0.2881
Batch 300/7339 - Loss: 2.0178 Acc: 0.3517
Batch 400/7339 - Loss: 1.8257 Acc: 0.4156
Batch 500/7339 - Loss: 1.6907 Acc: 0.4587
Batch 600/7339 - Loss: 1.5760 Acc: 0.4977
Batch 700/7339 - Loss: 1.4866 Acc: 0.5238
Batch 800/7339 - Loss: 1.4119 Acc: 0.5487
Batch 900/7339 - Loss: 1.3444 Acc: 0.5694
Batch 1000/7339 - Loss: 1.2805 Acc: 0.5901
Batch 1100/7339 - Loss: 1.2222 Acc: 0.6080
Batch 1200/7339 - Loss: 1.1727 Acc: 0.6236
Batch 1300/7339 - Loss: 1.1310 Acc: 0.6360
Batch 1400/7339 - Loss: 1.0917 Acc: 0.6489
Batch 1500/7339 - Loss: 1.0554 Acc: 0.6603
Batch 1600/7339 - Loss: 1.0190 Acc: 0.6720
Batch 1700/7339 - Loss: 0.9862 Acc: 0.6825
Batch 1800/7339 - Loss: 0.9561 Acc: 0.6922
Batch 1900/7339 - Loss: 0.9318 Acc: 0.7008
Batch 2000/7339 - Loss: 0.9082 Acc: 0.7084
Batch 2100/7339 - Loss: 0.8841 Acc: 0.7160
Batch 2200/7339 - Loss: 0.8609 Acc: 0.7238
Batch 2300/7339 - Loss: 0.8401 Acc: 0.7305
Batch 2400/7339 - Loss: 0.8227 Acc: 0.7363
Batch 2500/7339 - Loss: 0.8058 Acc: 0.7424
Batch 2600/7339 - Loss: 0.7891 Acc: 0.7485
Batch 2700/7339 - Loss: 0.7711 Acc: 0.7545
Batch 2800/7339 - Loss: 0.7559 Acc: 0.7593
Batch 2900/7339 - Loss: 0.7405 Acc: 0.7646
Batch 3000/7339 - Loss: 0.7239 Acc: 0.7700
```

```
Batch 3100/7339 - Loss: 0.7110 Acc: 0.7744
Batch 3200/7339 - Loss: 0.6982 Acc: 0.7789
Batch 3300/7339 - Loss: 0.6871 Acc: 0.7829
Batch 3400/7339 - Loss: 0.6746 Acc: 0.7874
Batch 3500/7339 - Loss: 0.6657 Acc: 0.7909
Batch 3600/7339 - Loss: 0.6565 Acc: 0.7943
Batch 3700/7339 - Loss: 0.6484 Acc: 0.7975
Batch 3800/7339 - Loss: 0.6382 Acc: 0.8009
Batch 3900/7339 - Loss: 0.6300 Acc: 0.8040
Batch 4000/7339 - Loss: 0.6211 Acc: 0.8072
Batch 4100/7339 - Loss: 0.6137 Acc: 0.8099
Batch 4200/7339 - Loss: 0.6057 Acc: 0.8128
Batch 4300/7339 - Loss: 0.5987 Acc: 0.8153
Batch 4400/7339 - Loss: 0.5904 Acc: 0.8181
Batch 4500/7339 - Loss: 0.5831 Acc: 0.8207
Batch 4600/7339 - Loss: 0.5759 Acc: 0.8233
Batch 4700/7339 - Loss: 0.5692 Acc: 0.8256
Batch 4800/7339 - Loss: 0.5627 Acc: 0.8278
Batch 4900/7339 - Loss: 0.5571 Acc: 0.8299
Batch 5000/7339 - Loss: 0.5513 Acc: 0.8320
Batch 5100/7339 - Loss: 0.5446 Acc: 0.8343
Batch 5200/7339 - Loss: 0.5388 Acc: 0.8361
Batch 5300/7339 - Loss: 0.5337 Acc: 0.8380
Batch 5400/7339 - Loss: 0.5273 Acc: 0.8399
Batch 5500/7339 - Loss: 0.5215 Acc: 0.8417
Batch 5600/7339 - Loss: 0.5160 Acc: 0.8437
Batch 5700/7339 - Loss: 0.5116 Acc: 0.8453
Batch 5800/7339 - Loss: 0.5065 Acc: 0.8472
Batch 5900/7339 - Loss: 0.5014 Acc: 0.8488
Batch 6000/7339 - Loss: 0.4960 Acc: 0.8505
Batch 6100/7339 - Loss: 0.4910 Acc: 0.8522
Batch 6200/7339 - Loss: 0.4861 Acc: 0.8538
Batch 6300/7339 - Loss: 0.4823 Acc: 0.8552
Batch 6400/7339 - Loss: 0.4784 Acc: 0.8565
Batch 6500/7339 - Loss: 0.4739 Acc: 0.8579
Batch 6600/7339 - Loss: 0.4703 Acc: 0.8593
Batch 6700/7339 - Loss: 0.4665 Acc: 0.8606
Batch 6800/7339 - Loss: 0.4624 Acc: 0.8620
Batch 6900/7339 - Loss: 0.4588 Acc: 0.8633
Batch 7000/7339 - Loss: 0.4543 Acc: 0.8647
Batch 7100/7339 - Loss: 0.4511 Acc: 0.8659
Batch 7200/7339 - Loss: 0.4481 Acc: 0.8669
Batch 7300/7339 - Loss: 0.4441 Acc: 0.8683
Epoch 1 - Train Loss: 0.4425 Train Acc: 0.8688 Val Loss: 0.2413 Val Acc: 0.9520

Epoch 2/8
Batch 100/7339 - Loss: 0.1772 Acc: 0.9613
Batch 200/7339 - Loss: 0.1362 Acc: 0.9694
```

```
Batch 300/7339 - Loss: 0.1215 Acc: 0.9717
Batch 400/7339 - Loss: 0.1324 Acc: 0.9700
Batch 500/7339 - Loss: 0.1470 Acc: 0.9650
Batch 600/7339 - Loss: 0.1443 Acc: 0.9656
Batch 700/7339 - Loss: 0.1416 Acc: 0.9664
Batch 800/7339 - Loss: 0.1418 Acc: 0.9664
Batch 900/7339 - Loss: 0.1471 Acc: 0.9653
Batch 1000/7339 - Loss: 0.1447 Acc: 0.9651
Batch 1100/7339 - Loss: 0.1439 Acc: 0.9649
Batch 1200/7339 - Loss: 0.1447 Acc: 0.9644
Batch 1300/7339 - Loss: 0.1431 Acc: 0.9648
Batch 1400/7339 - Loss: 0.1445 Acc: 0.9647
Batch 1500/7339 - Loss: 0.1435 Acc: 0.9646
Batch 1600/7339 - Loss: 0.1420 Acc: 0.9644
Batch 1700/7339 - Loss: 0.1411 Acc: 0.9648
Batch 1800/7339 - Loss: 0.1393 Acc: 0.9651
Batch 1900/7339 - Loss: 0.1420 Acc: 0.9647
Batch 2000/7339 - Loss: 0.1411 Acc: 0.9651
Batch 2100/7339 - Loss: 0.1408 Acc: 0.9652
Batch 2200/7339 - Loss: 0.1398 Acc: 0.9653
Batch 2300/7339 - Loss: 0.1382 Acc: 0.9654
Batch 2400/7339 - Loss: 0.1390 Acc: 0.9653
Batch 2500/7339 - Loss: 0.1380 Acc: 0.9656
Batch 2600/7339 - Loss: 0.1392 Acc: 0.9652
Batch 2700/7339 - Loss: 0.1384 Acc: 0.9656
Batch 2800/7339 - Loss: 0.1364 Acc: 0.9662
Batch 2900/7339 - Loss: 0.1372 Acc: 0.9659
Batch 3000/7339 - Loss: 0.1377 Acc: 0.9656
Batch 3100/7339 - Loss: 0.1382 Acc: 0.9656
Batch 3200/7339 - Loss: 0.1378 Acc: 0.9656
Batch 3300/7339 - Loss: 0.1374 Acc: 0.9658
Batch 3400/7339 - Loss: 0.1361 Acc: 0.9661
Batch 3500/7339 - Loss: 0.1364 Acc: 0.9659
Batch 3600/7339 - Loss: 0.1358 Acc: 0.9660
Batch 3700/7339 - Loss: 0.1347 Acc: 0.9663
Batch 3800/7339 - Loss: 0.1345 Acc: 0.9663
Batch 3900/7339 - Loss: 0.1340 Acc: 0.9664
Batch 4000/7339 - Loss: 0.1338 Acc: 0.9663
Batch 4100/7339 - Loss: 0.1340 Acc: 0.9663
Batch 4200/7339 - Loss: 0.1345 Acc: 0.9661
Batch 4300/7339 - Loss: 0.1338 Acc: 0.9662
Batch 4400/7339 - Loss: 0.1333 Acc: 0.9665
Batch 4500/7339 - Loss: 0.1338 Acc: 0.9663
Batch 4600/7339 - Loss: 0.1325 Acc: 0.9667
Batch 4700/7339 - Loss: 0.1317 Acc: 0.9668
Batch 4800/7339 - Loss: 0.1320 Acc: 0.9667
Batch 4900/7339 - Loss: 0.1315 Acc: 0.9669
Batch 5000/7339 - Loss: 0.1314 Acc: 0.9670
```

```
Batch 5100/7339 - Loss: 0.1313 Acc: 0.9669
Batch 5200/7339 - Loss: 0.1311 Acc: 0.9670
Batch 5300/7339 - Loss: 0.1298 Acc: 0.9672
Batch 5400/7339 - Loss: 0.1297 Acc: 0.9672
Batch 5500/7339 - Loss: 0.1289 Acc: 0.9673
Batch 5600/7339 - Loss: 0.1286 Acc: 0.9675
Batch 5700/7339 - Loss: 0.1299 Acc: 0.9672
Batch 5800/7339 - Loss: 0.1294 Acc: 0.9674
Batch 5900/7339 - Loss: 0.1286 Acc: 0.9675
Batch 6000/7339 - Loss: 0.1288 Acc: 0.9674
Batch 6100/7339 - Loss: 0.1288 Acc: 0.9674
Batch 6200/7339 - Loss: 0.1278 Acc: 0.9674
Batch 6300/7339 - Loss: 0.1277 Acc: 0.9675
Batch 6400/7339 - Loss: 0.1270 Acc: 0.9677
Batch 6500/7339 - Loss: 0.1263 Acc: 0.9679
Batch 6600/7339 - Loss: 0.1258 Acc: 0.9680
Batch 6700/7339 - Loss: 0.1255 Acc: 0.9680
Batch 6800/7339 - Loss: 0.1253 Acc: 0.9681
Batch 6900/7339 - Loss: 0.1255 Acc: 0.9681
Batch 7000/7339 - Loss: 0.1252 Acc: 0.9681
Batch 7100/7339 - Loss: 0.1245 Acc: 0.9683
Batch 7200/7339 - Loss: 0.1245 Acc: 0.9683
Batch 7300/7339 - Loss: 0.1244 Acc: 0.9683
Epoch 2 - Train Loss: 0.1246 Train Acc: 0.9683 Val Loss: 0.0825 Val Acc: 0.9780

Epoch 3/8
Batch 100/7339 - Loss: 0.0819 Acc: 0.9762
Batch 200/7339 - Loss: 0.0836 Acc: 0.9781
Batch 300/7339 - Loss: 0.0868 Acc: 0.9762
Batch 400/7339 - Loss: 0.0872 Acc: 0.9756
Batch 500/7339 - Loss: 0.0877 Acc: 0.9760
Batch 600/7339 - Loss: 0.0924 Acc: 0.9750
Batch 700/7339 - Loss: 0.0904 Acc: 0.9755
Batch 800/7339 - Loss: 0.0967 Acc: 0.9739
Batch 900/7339 - Loss: 0.0952 Acc: 0.9746
Batch 1000/7339 - Loss: 0.0937 Acc: 0.9748
Batch 1100/7339 - Loss: 0.0952 Acc: 0.9743
Batch 1200/7339 - Loss: 0.0953 Acc: 0.9741
Batch 1300/7339 - Loss: 0.0979 Acc: 0.9735
Batch 1400/7339 - Loss: 0.0956 Acc: 0.9740
Batch 1500/7339 - Loss: 0.0935 Acc: 0.9742
Batch 1600/7339 - Loss: 0.0917 Acc: 0.9744
Batch 1700/7339 - Loss: 0.0944 Acc: 0.9742
Batch 1800/7339 - Loss: 0.0934 Acc: 0.9744
Batch 1900/7339 - Loss: 0.0940 Acc: 0.9745
Batch 2000/7339 - Loss: 0.0936 Acc: 0.9744
Batch 2100/7339 - Loss: 0.0937 Acc: 0.9742
Batch 2200/7339 - Loss: 0.0933 Acc: 0.9744
```

```
Batch 2300/7339 - Loss: 0.0925 Acc: 0.9745
Batch 2400/7339 - Loss: 0.0919 Acc: 0.9746
Batch 2500/7339 - Loss: 0.0917 Acc: 0.9747
Batch 2600/7339 - Loss: 0.0911 Acc: 0.9747
Batch 2700/7339 - Loss: 0.0903 Acc: 0.9749
Batch 2800/7339 - Loss: 0.0921 Acc: 0.9745
Batch 2900/7339 - Loss: 0.0926 Acc: 0.9742
Batch 3000/7339 - Loss: 0.0914 Acc: 0.9746
Batch 3100/7339 - Loss: 0.0911 Acc: 0.9745
Batch 3200/7339 - Loss: 0.0900 Acc: 0.9748
Batch 3300/7339 - Loss: 0.0898 Acc: 0.9747
Batch 3400/7339 - Loss: 0.0896 Acc: 0.9746
Batch 3500/7339 - Loss: 0.0898 Acc: 0.9746
Batch 3600/7339 - Loss: 0.0889 Acc: 0.9749
Batch 3700/7339 - Loss: 0.0888 Acc: 0.9749
Batch 3800/7339 - Loss: 0.0883 Acc: 0.9751
Batch 3900/7339 - Loss: 0.0877 Acc: 0.9752
Batch 4000/7339 - Loss: 0.0876 Acc: 0.9752
Batch 4100/7339 - Loss: 0.0874 Acc: 0.9754
Batch 4200/7339 - Loss: 0.0872 Acc: 0.9753
Batch 4300/7339 - Loss: 0.0867 Acc: 0.9753
Batch 4400/7339 - Loss: 0.0862 Acc: 0.9755
Batch 4500/7339 - Loss: 0.0864 Acc: 0.9753
Batch 4600/7339 - Loss: 0.0865 Acc: 0.9754
Batch 4700/7339 - Loss: 0.0862 Acc: 0.9755
Batch 4800/7339 - Loss: 0.0867 Acc: 0.9753
Batch 4900/7339 - Loss: 0.0866 Acc: 0.9754
Batch 5000/7339 - Loss: 0.0863 Acc: 0.9754
Batch 5100/7339 - Loss: 0.0871 Acc: 0.9753
Batch 5200/7339 - Loss: 0.0870 Acc: 0.9752
Batch 5300/7339 - Loss: 0.0868 Acc: 0.9753
Batch 5400/7339 - Loss: 0.0872 Acc: 0.9752
Batch 5500/7339 - Loss: 0.0871 Acc: 0.9752
Batch 5600/7339 - Loss: 0.0870 Acc: 0.9752
Batch 5700/7339 - Loss: 0.0868 Acc: 0.9752
Batch 5800/7339 - Loss: 0.0871 Acc: 0.9751
Batch 5900/7339 - Loss: 0.0866 Acc: 0.9752
Batch 6000/7339 - Loss: 0.0865 Acc: 0.9752
Batch 6100/7339 - Loss: 0.0865 Acc: 0.9750
Batch 6200/7339 - Loss: 0.0869 Acc: 0.9751
Batch 6300/7339 - Loss: 0.0866 Acc: 0.9750
Batch 6400/7339 - Loss: 0.0865 Acc: 0.9751
Batch 6500/7339 - Loss: 0.0858 Acc: 0.9752
Batch 6600/7339 - Loss: 0.0855 Acc: 0.9753
Batch 6700/7339 - Loss: 0.0849 Acc: 0.9754
Batch 6800/7339 - Loss: 0.0849 Acc: 0.9754
Batch 6900/7339 - Loss: 0.0848 Acc: 0.9754
Batch 7000/7339 - Loss: 0.0853 Acc: 0.9753
```

```
Batch 7100/7339 - Loss: 0.0854 Acc: 0.9752
Batch 7200/7339 - Loss: 0.0849 Acc: 0.9752
Batch 7300/7339 - Loss: 0.0846 Acc: 0.9753
Epoch 3 - Train Loss: 0.0847 Train Acc: 0.9752 Val Loss: 0.0672 Val Acc: 0.9780

Epoch 4/8
Batch 100/7339 - Loss: 0.0913 Acc: 0.9750
Batch 200/7339 - Loss: 0.0943 Acc: 0.9744
Batch 300/7339 - Loss: 0.0848 Acc: 0.9775
Batch 400/7339 - Loss: 0.0928 Acc: 0.9753
Batch 500/7339 - Loss: 0.0873 Acc: 0.9760
Batch 600/7339 - Loss: 0.0884 Acc: 0.9754
Batch 700/7339 - Loss: 0.0859 Acc: 0.9754
Batch 800/7339 - Loss: 0.0867 Acc: 0.9747
Batch 900/7339 - Loss: 0.0867 Acc: 0.9742
Batch 1000/7339 - Loss: 0.0871 Acc: 0.9742
Batch 1100/7339 - Loss: 0.0885 Acc: 0.9739
Batch 1200/7339 - Loss: 0.0898 Acc: 0.9739
Batch 1300/7339 - Loss: 0.0899 Acc: 0.9739
Batch 1400/7339 - Loss: 0.0884 Acc: 0.9740
Batch 1500/7339 - Loss: 0.0854 Acc: 0.9748
Batch 1600/7339 - Loss: 0.0831 Acc: 0.9754
Batch 1700/7339 - Loss: 0.0849 Acc: 0.9751
Batch 1800/7339 - Loss: 0.0837 Acc: 0.9752
Batch 1900/7339 - Loss: 0.0829 Acc: 0.9754
Batch 2000/7339 - Loss: 0.0819 Acc: 0.9758
Batch 2100/7339 - Loss: 0.0821 Acc: 0.9759
Batch 2200/7339 - Loss: 0.0815 Acc: 0.9760
Batch 2300/7339 - Loss: 0.0816 Acc: 0.9759
Batch 2400/7339 - Loss: 0.0822 Acc: 0.9757
Batch 2500/7339 - Loss: 0.0802 Acc: 0.9762
Batch 2600/7339 - Loss: 0.0805 Acc: 0.9762
Batch 2700/7339 - Loss: 0.0806 Acc: 0.9762
Batch 2800/7339 - Loss: 0.0800 Acc: 0.9763
Batch 2900/7339 - Loss: 0.0801 Acc: 0.9763
Batch 3000/7339 - Loss: 0.0816 Acc: 0.9760
Batch 3100/7339 - Loss: 0.0804 Acc: 0.9762
Batch 3200/7339 - Loss: 0.0803 Acc: 0.9762
Batch 3300/7339 - Loss: 0.0795 Acc: 0.9764
Batch 3400/7339 - Loss: 0.0793 Acc: 0.9764
Batch 3500/7339 - Loss: 0.0790 Acc: 0.9764
Batch 3600/7339 - Loss: 0.0788 Acc: 0.9764
Batch 3700/7339 - Loss: 0.0783 Acc: 0.9765
Batch 3800/7339 - Loss: 0.0781 Acc: 0.9763
Batch 3900/7339 - Loss: 0.0777 Acc: 0.9763
Batch 4000/7339 - Loss: 0.0774 Acc: 0.9763
Batch 4100/7339 - Loss: 0.0770 Acc: 0.9764
Batch 4200/7339 - Loss: 0.0769 Acc: 0.9766
```

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Batch 4300/7339 - Loss: 0.0769 Acc: 0.9766
Batch 4400/7339 - Loss: 0.0764 Acc: 0.9766
Batch 4500/7339 - Loss: 0.0763 Acc: 0.9766
Batch 4600/7339 - Loss: 0.0759 Acc: 0.9767
Batch 4700/7339 - Loss: 0.0759 Acc: 0.9767
Batch 4800/7339 - Loss: 0.0764 Acc: 0.9765
Batch 4900/7339 - Loss: 0.0771 Acc: 0.9763
Batch 5000/7339 - Loss: 0.0766 Acc: 0.9765
Batch 5100/7339 - Loss: 0.0766 Acc: 0.9765
Batch 5200/7339 - Loss: 0.0767 Acc: 0.9764
Batch 5300/7339 - Loss: 0.0767 Acc: 0.9764
Batch 5400/7339 - Loss: 0.0762 Acc: 0.9765
Batch 5500/7339 - Loss: 0.0759 Acc: 0.9765
Batch 5600/7339 - Loss: 0.0763 Acc: 0.9765
Batch 5700/7339 - Loss: 0.0764 Acc: 0.9764
Batch 5800/7339 - Loss: 0.0769 Acc: 0.9764
Batch 5900/7339 - Loss: 0.0765 Acc: 0.9765
Batch 6000/7339 - Loss: 0.0761 Acc: 0.9765
Batch 6100/7339 - Loss: 0.0761 Acc: 0.9765
Batch 6200/7339 - Loss: 0.0759 Acc: 0.9765
Batch 6300/7339 - Loss: 0.0757 Acc: 0.9765
Batch 6400/7339 - Loss: 0.0753 Acc: 0.9767
Batch 6500/7339 - Loss: 0.0748 Acc: 0.9768
Batch 6600/7339 - Loss: 0.0748 Acc: 0.9768
Batch 6700/7339 - Loss: 0.0747 Acc: 0.9769
Batch 6800/7339 - Loss: 0.0745 Acc: 0.9770
Batch 6900/7339 - Loss: 0.0742 Acc: 0.9771
Batch 7000/7339 - Loss: 0.0740 Acc: 0.9772
Batch 7100/7339 - Loss: 0.0740 Acc: 0.9772
Batch 7200/7339 - Loss: 0.0736 Acc: 0.9773
Batch 7300/7339 - Loss: 0.0735 Acc: 0.9773
Epoch 4 - Train Loss: 0.0735 Train Acc: 0.9773 Val Loss: 0.0614 Val Acc: 0.9760

Epoch 5/8
Batch 100/7339 - Loss: 0.0621 Acc: 0.9800
Batch 200/7339 - Loss: 0.0591 Acc: 0.9806
Batch 300/7339 - Loss: 0.0581 Acc: 0.9800
Batch 400/7339 - Loss: 0.0625 Acc: 0.9794
Batch 500/7339 - Loss: 0.0636 Acc: 0.9790
Batch 600/7339 - Loss: 0.0629 Acc: 0.9785
Batch 700/7339 - Loss: 0.0657 Acc: 0.9770
Batch 800/7339 - Loss: 0.0641 Acc: 0.9762
Batch 900/7339 - Loss: 0.0604 Acc: 0.9776
Batch 1000/7339 - Loss: 0.0645 Acc: 0.9768
Batch 1100/7339 - Loss: 0.0637 Acc: 0.9772
Batch 1200/7339 - Loss: 0.0642 Acc: 0.9770
Batch 1300/7339 - Loss: 0.0629 Acc: 0.9774
Batch 1400/7339 - Loss: 0.0635 Acc: 0.9772
```

```
Batch 1500/7339 - Loss: 0.0652 Acc: 0.9768
Batch 1600/7339 - Loss: 0.0642 Acc: 0.9769
Batch 1700/7339 - Loss: 0.0636 Acc: 0.9771
Batch 1800/7339 - Loss: 0.0627 Acc: 0.9774
Batch 1900/7339 - Loss: 0.0629 Acc: 0.9776
Batch 2000/7339 - Loss: 0.0625 Acc: 0.9776
Batch 2100/7339 - Loss: 0.0627 Acc: 0.9773
Batch 2200/7339 - Loss: 0.0627 Acc: 0.9772
Batch 2300/7339 - Loss: 0.0623 Acc: 0.9776
Batch 2400/7339 - Loss: 0.0619 Acc: 0.9777
Batch 2500/7339 - Loss: 0.0621 Acc: 0.9776
Batch 2600/7339 - Loss: 0.0619 Acc: 0.9775
Batch 2700/7339 - Loss: 0.0617 Acc: 0.9775
Batch 2800/7339 - Loss: 0.0616 Acc: 0.9774
Batch 2900/7339 - Loss: 0.0618 Acc: 0.9775
Batch 3000/7339 - Loss: 0.0614 Acc: 0.9776
Batch 3100/7339 - Loss: 0.0616 Acc: 0.9774
Batch 3200/7339 - Loss: 0.0615 Acc: 0.9775
Batch 3300/7339 - Loss: 0.0610 Acc: 0.9778
Batch 3400/7339 - Loss: 0.0625 Acc: 0.9774
Batch 3500/7339 - Loss: 0.0626 Acc: 0.9775
Batch 3600/7339 - Loss: 0.0624 Acc: 0.9775
Batch 3700/7339 - Loss: 0.0620 Acc: 0.9776
Batch 3800/7339 - Loss: 0.0617 Acc: 0.9777
Batch 3900/7339 - Loss: 0.0621 Acc: 0.9777
Batch 4000/7339 - Loss: 0.0623 Acc: 0.9776
Batch 4100/7339 - Loss: 0.0617 Acc: 0.9777
Batch 4200/7339 - Loss: 0.0616 Acc: 0.9777
Batch 4300/7339 - Loss: 0.0614 Acc: 0.9779
Batch 4400/7339 - Loss: 0.0616 Acc: 0.9779
Batch 4500/7339 - Loss: 0.0615 Acc: 0.9780
Batch 4600/7339 - Loss: 0.0613 Acc: 0.9782
Batch 4700/7339 - Loss: 0.0611 Acc: 0.9783
Batch 4800/7339 - Loss: 0.0607 Acc: 0.9785
Batch 4900/7339 - Loss: 0.0602 Acc: 0.9786
Batch 5000/7339 - Loss: 0.0600 Acc: 0.9787
Batch 5100/7339 - Loss: 0.0601 Acc: 0.9786
Batch 5200/7339 - Loss: 0.0600 Acc: 0.9786
Batch 5300/7339 - Loss: 0.0599 Acc: 0.9786
Batch 5400/7339 - Loss: 0.0596 Acc: 0.9787
Batch 5500/7339 - Loss: 0.0596 Acc: 0.9788
Batch 5600/7339 - Loss: 0.0602 Acc: 0.9786
Batch 5700/7339 - Loss: 0.0601 Acc: 0.9786
Batch 5800/7339 - Loss: 0.0595 Acc: 0.9789
Batch 5900/7339 - Loss: 0.0594 Acc: 0.9790
Batch 6000/7339 - Loss: 0.0593 Acc: 0.9790
Batch 6100/7339 - Loss: 0.0599 Acc: 0.9788
Batch 6200/7339 - Loss: 0.0596 Acc: 0.9789
```

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Batch 6300/7339 - Loss: 0.0594 Acc: 0.9790
Batch 6400/7339 - Loss: 0.0597 Acc: 0.9788
Batch 6500/7339 - Loss: 0.0593 Acc: 0.9789
Batch 6600/7339 - Loss: 0.0597 Acc: 0.9788
Batch 6700/7339 - Loss: 0.0597 Acc: 0.9789
Batch 6800/7339 - Loss: 0.0595 Acc: 0.9790
Batch 6900/7339 - Loss: 0.0593 Acc: 0.9790
Batch 7000/7339 - Loss: 0.0594 Acc: 0.9790
Batch 7100/7339 - Loss: 0.0595 Acc: 0.9791
Batch 7200/7339 - Loss: 0.0594 Acc: 0.9791
Batch 7300/7339 - Loss: 0.0591 Acc: 0.9792
Epoch 5 - Train Loss: 0.0591 Train Acc: 0.9792 Val Loss: 0.0827 Val Acc: 0.9760

Epoch 6/8
Batch 100/7339 - Loss: 0.0421 Acc: 0.9838
Batch 200/7339 - Loss: 0.0610 Acc: 0.9775
Batch 300/7339 - Loss: 0.0580 Acc: 0.9788
Batch 400/7339 - Loss: 0.0532 Acc: 0.9803
Batch 500/7339 - Loss: 0.0492 Acc: 0.9820
Batch 600/7339 - Loss: 0.0517 Acc: 0.9810
Batch 700/7339 - Loss: 0.0559 Acc: 0.9802
Batch 800/7339 - Loss: 0.0572 Acc: 0.9802
Batch 900/7339 - Loss: 0.0569 Acc: 0.9797
Batch 1000/7339 - Loss: 0.0563 Acc: 0.9801
Batch 1100/7339 - Loss: 0.0577 Acc: 0.9800
Batch 1200/7339 - Loss: 0.0578 Acc: 0.9799
Batch 1300/7339 - Loss: 0.0570 Acc: 0.9804
Batch 1400/7339 - Loss: 0.0555 Acc: 0.9808
Batch 1500/7339 - Loss: 0.0549 Acc: 0.9808
Batch 1600/7339 - Loss: 0.0547 Acc: 0.9809
Batch 1700/7339 - Loss: 0.0542 Acc: 0.9808
Batch 1800/7339 - Loss: 0.0530 Acc: 0.9811
Batch 1900/7339 - Loss: 0.0531 Acc: 0.9811
Batch 2000/7339 - Loss: 0.0532 Acc: 0.9810
Batch 2100/7339 - Loss: 0.0524 Acc: 0.9812
Batch 2200/7339 - Loss: 0.0515 Acc: 0.9815
Batch 2300/7339 - Loss: 0.0519 Acc: 0.9814
Batch 2400/7339 - Loss: 0.0522 Acc: 0.9813
Batch 2500/7339 - Loss: 0.0516 Acc: 0.9815
Batch 2600/7339 - Loss: 0.0511 Acc: 0.9816
Batch 2700/7339 - Loss: 0.0512 Acc: 0.9816
Batch 2800/7339 - Loss: 0.0515 Acc: 0.9815
Batch 2900/7339 - Loss: 0.0516 Acc: 0.9814
Batch 3000/7339 - Loss: 0.0508 Acc: 0.9817
Batch 3100/7339 - Loss: 0.0507 Acc: 0.9818
Batch 3200/7339 - Loss: 0.0512 Acc: 0.9818
Batch 3300/7339 - Loss: 0.0520 Acc: 0.9816
Batch 3400/7339 - Loss: 0.0528 Acc: 0.9812
```

```
Batch 3500/7339 - Loss: 0.0528 Acc: 0.9812
Batch 3600/7339 - Loss: 0.0532 Acc: 0.9810
Batch 3700/7339 - Loss: 0.0521 Acc: 0.9813
Batch 3800/7339 - Loss: 0.0519 Acc: 0.9814
Batch 3900/7339 - Loss: 0.0522 Acc: 0.9813
Batch 4000/7339 - Loss: 0.0519 Acc: 0.9814
Batch 4100/7339 - Loss: 0.0520 Acc: 0.9814
Batch 4200/7339 - Loss: 0.0523 Acc: 0.9813
Batch 4300/7339 - Loss: 0.0532 Acc: 0.9811
Batch 4400/7339 - Loss: 0.0532 Acc: 0.9811
Batch 4500/7339 - Loss: 0.0531 Acc: 0.9812
Batch 4600/7339 - Loss: 0.0536 Acc: 0.9810
Batch 4700/7339 - Loss: 0.0539 Acc: 0.9810
Batch 4800/7339 - Loss: 0.0540 Acc: 0.9808
Batch 4900/7339 - Loss: 0.0541 Acc: 0.9807
Batch 5000/7339 - Loss: 0.0543 Acc: 0.9807
Batch 5100/7339 - Loss: 0.0541 Acc: 0.9807
Batch 5200/7339 - Loss: 0.0542 Acc: 0.9807
Batch 5300/7339 - Loss: 0.0539 Acc: 0.9808
Batch 5400/7339 - Loss: 0.0539 Acc: 0.9807
Batch 5500/7339 - Loss: 0.0539 Acc: 0.9807
Batch 5600/7339 - Loss: 0.0535 Acc: 0.9808
Batch 5700/7339 - Loss: 0.0536 Acc: 0.9807
Batch 5800/7339 - Loss: 0.0536 Acc: 0.9807
Batch 5900/7339 - Loss: 0.0536 Acc: 0.9806
Batch 6000/7339 - Loss: 0.0536 Acc: 0.9806
Batch 6100/7339 - Loss: 0.0539 Acc: 0.9805
Batch 6200/7339 - Loss: 0.0541 Acc: 0.9804
Batch 6300/7339 - Loss: 0.0544 Acc: 0.9803
Batch 6400/7339 - Loss: 0.0538 Acc: 0.9805
Batch 6500/7339 - Loss: 0.0536 Acc: 0.9806
Batch 6600/7339 - Loss: 0.0537 Acc: 0.9805
Batch 6700/7339 - Loss: 0.0541 Acc: 0.9805
Batch 6800/7339 - Loss: 0.0542 Acc: 0.9805
Batch 6900/7339 - Loss: 0.0541 Acc: 0.9805
Batch 7000/7339 - Loss: 0.0542 Acc: 0.9805
Batch 7100/7339 - Loss: 0.0541 Acc: 0.9806
Batch 7200/7339 - Loss: 0.0540 Acc: 0.9806
Batch 7300/7339 - Loss: 0.0542 Acc: 0.9805
Epoch 6 - Train Loss: 0.0542 Train Acc: 0.9805 Val Loss: 0.0648 Val Acc: 0.9740

Epoch 7/8
Batch 100/7339 - Loss: 0.0624 Acc: 0.9788
Batch 200/7339 - Loss: 0.0525 Acc: 0.9812
Batch 300/7339 - Loss: 0.0517 Acc: 0.9812
Batch 400/7339 - Loss: 0.0521 Acc: 0.9816
Batch 500/7339 - Loss: 0.0527 Acc: 0.9812
Batch 600/7339 - Loss: 0.0505 Acc: 0.9821
```

```
Batch 700/7339 - Loss: 0.0556 Acc: 0.9804
Batch 800/7339 - Loss: 0.0541 Acc: 0.9806
Batch 900/7339 - Loss: 0.0535 Acc: 0.9804
Batch 1000/7339 - Loss: 0.0539 Acc: 0.9804
Batch 1100/7339 - Loss: 0.0536 Acc: 0.9805
Batch 1200/7339 - Loss: 0.0520 Acc: 0.9810
Batch 1300/7339 - Loss: 0.0514 Acc: 0.9813
Batch 1400/7339 - Loss: 0.0515 Acc: 0.9814
Batch 1500/7339 - Loss: 0.0515 Acc: 0.9814
Batch 1600/7339 - Loss: 0.0516 Acc: 0.9812
Batch 1700/7339 - Loss: 0.0510 Acc: 0.9815
Batch 1800/7339 - Loss: 0.0513 Acc: 0.9813
Batch 1900/7339 - Loss: 0.0513 Acc: 0.9813
Batch 2000/7339 - Loss: 0.0528 Acc: 0.9808
Batch 2100/7339 - Loss: 0.0531 Acc: 0.9807
Batch 2200/7339 - Loss: 0.0538 Acc: 0.9805
Batch 2300/7339 - Loss: 0.0533 Acc: 0.9808
Batch 2400/7339 - Loss: 0.0527 Acc: 0.9809
Batch 2500/7339 - Loss: 0.0526 Acc: 0.9809
Batch 2600/7339 - Loss: 0.0531 Acc: 0.9806
Batch 2700/7339 - Loss: 0.0533 Acc: 0.9805
Batch 2800/7339 - Loss: 0.0530 Acc: 0.9804
Batch 2900/7339 - Loss: 0.0527 Acc: 0.9806
Batch 3000/7339 - Loss: 0.0523 Acc: 0.9806
Batch 3100/7339 - Loss: 0.0518 Acc: 0.9809
Batch 3200/7339 - Loss: 0.0517 Acc: 0.9809
Batch 3300/7339 - Loss: 0.0516 Acc: 0.9809
Batch 3400/7339 - Loss: 0.0513 Acc: 0.9810
Batch 3500/7339 - Loss: 0.0514 Acc: 0.9811
Batch 3600/7339 - Loss: 0.0516 Acc: 0.9810
Batch 3700/7339 - Loss: 0.0513 Acc: 0.9811
Batch 3800/7339 - Loss: 0.0509 Acc: 0.9812
Batch 3900/7339 - Loss: 0.0508 Acc: 0.9812
Batch 4000/7339 - Loss: 0.0503 Acc: 0.9814
Batch 4100/7339 - Loss: 0.0502 Acc: 0.9814
Batch 4200/7339 - Loss: 0.0502 Acc: 0.9814
Batch 4300/7339 - Loss: 0.0498 Acc: 0.9815
Batch 4400/7339 - Loss: 0.0498 Acc: 0.9814
Batch 4500/7339 - Loss: 0.0499 Acc: 0.9814
Batch 4600/7339 - Loss: 0.0494 Acc: 0.9816
Batch 4700/7339 - Loss: 0.0496 Acc: 0.9815
Batch 4800/7339 - Loss: 0.0495 Acc: 0.9815
Batch 4900/7339 - Loss: 0.0496 Acc: 0.9815
Batch 5000/7339 - Loss: 0.0491 Acc: 0.9817
Batch 5100/7339 - Loss: 0.0495 Acc: 0.9815
Batch 5200/7339 - Loss: 0.0497 Acc: 0.9814
Batch 5300/7339 - Loss: 0.0496 Acc: 0.9814
Batch 5400/7339 - Loss: 0.0498 Acc: 0.9814
```

```
Batch 5500/7339 - Loss: 0.0502 Acc: 0.9813
Batch 5600/7339 - Loss: 0.0501 Acc: 0.9814
Batch 5700/7339 - Loss: 0.0503 Acc: 0.9812
Batch 5800/7339 - Loss: 0.0504 Acc: 0.9813
Batch 5900/7339 - Loss: 0.0504 Acc: 0.9812
Batch 6000/7339 - Loss: 0.0504 Acc: 0.9812
Batch 6100/7339 - Loss: 0.0503 Acc: 0.9812
Batch 6200/7339 - Loss: 0.0506 Acc: 0.9812
Batch 6300/7339 - Loss: 0.0507 Acc: 0.9812
Batch 6400/7339 - Loss: 0.0504 Acc: 0.9812
Batch 6500/7339 - Loss: 0.0505 Acc: 0.9812
Batch 6600/7339 - Loss: 0.0506 Acc: 0.9812
Batch 6700/7339 - Loss: 0.0508 Acc: 0.9811
Batch 6800/7339 - Loss: 0.0507 Acc: 0.9811
Batch 6900/7339 - Loss: 0.0507 Acc: 0.9811
Batch 7000/7339 - Loss: 0.0507 Acc: 0.9811
Batch 7100/7339 - Loss: 0.0507 Acc: 0.9812
Batch 7200/7339 - Loss: 0.0504 Acc: 0.9813
Batch 7300/7339 - Loss: 0.0504 Acc: 0.9813
Epoch 7 - Train Loss: 0.0502 Train Acc: 0.9813 Val Loss: 0.0690 Val Acc: 0.9760

Epoch 8/8
Batch 100/7339 - Loss: 0.0485 Acc: 0.9838
Batch 200/7339 - Loss: 0.0540 Acc: 0.9825
Batch 300/7339 - Loss: 0.0563 Acc: 0.9792
Batch 400/7339 - Loss: 0.0513 Acc: 0.9812
Batch 500/7339 - Loss: 0.0527 Acc: 0.9802
Batch 600/7339 - Loss: 0.0514 Acc: 0.9800
Batch 700/7339 - Loss: 0.0495 Acc: 0.9811
Batch 800/7339 - Loss: 0.0516 Acc: 0.9803
Batch 900/7339 - Loss: 0.0503 Acc: 0.9808
Batch 1000/7339 - Loss: 0.0499 Acc: 0.9811
Batch 1100/7339 - Loss: 0.0490 Acc: 0.9814
Batch 1200/7339 - Loss: 0.0494 Acc: 0.9811
Batch 1300/7339 - Loss: 0.0482 Acc: 0.9816
Batch 1400/7339 - Loss: 0.0472 Acc: 0.9818
Batch 1500/7339 - Loss: 0.0482 Acc: 0.9818
Batch 1600/7339 - Loss: 0.0490 Acc: 0.9813
Batch 1700/7339 - Loss: 0.0478 Acc: 0.9818
Batch 1800/7339 - Loss: 0.0470 Acc: 0.9822
Batch 1900/7339 - Loss: 0.0473 Acc: 0.9821
Batch 2000/7339 - Loss: 0.0468 Acc: 0.9822
Batch 2100/7339 - Loss: 0.0470 Acc: 0.9822
Batch 2200/7339 - Loss: 0.0468 Acc: 0.9823
Batch 2300/7339 - Loss: 0.0471 Acc: 0.9823
Batch 2400/7339 - Loss: 0.0464 Acc: 0.9825
Batch 2500/7339 - Loss: 0.0462 Acc: 0.9827
Batch 2600/7339 - Loss: 0.0463 Acc: 0.9826
```

```
Batch 2700/7339 - Loss: 0.0463 Acc: 0.9826
Batch 2800/7339 - Loss: 0.0465 Acc: 0.9825
Batch 2900/7339 - Loss: 0.0464 Acc: 0.9826
Batch 3000/7339 - Loss: 0.0467 Acc: 0.9825
Batch 3100/7339 - Loss: 0.0465 Acc: 0.9826
Batch 3200/7339 - Loss: 0.0465 Acc: 0.9826
Batch 3300/7339 - Loss: 0.0462 Acc: 0.9827
Batch 3400/7339 - Loss: 0.0458 Acc: 0.9828
Batch 3500/7339 - Loss: 0.0460 Acc: 0.9828
Batch 3600/7339 - Loss: 0.0466 Acc: 0.9825
Batch 3700/7339 - Loss: 0.0467 Acc: 0.9825
Batch 3800/7339 - Loss: 0.0466 Acc: 0.9825
Batch 3900/7339 - Loss: 0.0468 Acc: 0.9824
Batch 4000/7339 - Loss: 0.0465 Acc: 0.9824
Batch 4100/7339 - Loss: 0.0466 Acc: 0.9825
Batch 4200/7339 - Loss: 0.0464 Acc: 0.9826
Batch 4300/7339 - Loss: 0.0462 Acc: 0.9826
Batch 4400/7339 - Loss: 0.0459 Acc: 0.9828
Batch 4500/7339 - Loss: 0.0462 Acc: 0.9825
Batch 4600/7339 - Loss: 0.0458 Acc: 0.9827
Batch 4700/7339 - Loss: 0.0455 Acc: 0.9828
Batch 4800/7339 - Loss: 0.0461 Acc: 0.9826
Batch 4900/7339 - Loss: 0.0463 Acc: 0.9826
Batch 5000/7339 - Loss: 0.0465 Acc: 0.9825
Batch 5100/7339 - Loss: 0.0465 Acc: 0.9825
Batch 5200/7339 - Loss: 0.0470 Acc: 0.9823
Batch 5300/7339 - Loss: 0.0470 Acc: 0.9823
Batch 5400/7339 - Loss: 0.0471 Acc: 0.9823
Batch 5500/7339 - Loss: 0.0473 Acc: 0.9822
Batch 5600/7339 - Loss: 0.0472 Acc: 0.9822
Batch 5700/7339 - Loss: 0.0473 Acc: 0.9821
Batch 5800/7339 - Loss: 0.0473 Acc: 0.9821
Batch 5900/7339 - Loss: 0.0477 Acc: 0.9820
Batch 6000/7339 - Loss: 0.0479 Acc: 0.9819
Batch 6100/7339 - Loss: 0.0482 Acc: 0.9818
Batch 6200/7339 - Loss: 0.0480 Acc: 0.9818
Batch 6300/7339 - Loss: 0.0482 Acc: 0.9817
Batch 6400/7339 - Loss: 0.0480 Acc: 0.9817
Batch 6500/7339 - Loss: 0.0479 Acc: 0.9818
Batch 6600/7339 - Loss: 0.0480 Acc: 0.9817
Batch 6700/7339 - Loss: 0.0482 Acc: 0.9817
Batch 6800/7339 - Loss: 0.0485 Acc: 0.9817
Batch 6900/7339 - Loss: 0.0487 Acc: 0.9816
Batch 7000/7339 - Loss: 0.0485 Acc: 0.9817
Batch 7100/7339 - Loss: 0.0484 Acc: 0.9817
Batch 7200/7339 - Loss: 0.0485 Acc: 0.9817
Batch 7300/7339 - Loss: 0.0484 Acc: 0.9817
Epoch 8 - Train Loss: 0.0484 Train Acc: 0.9817 Val Loss: 0.0639 Val Acc: 0.9760
```

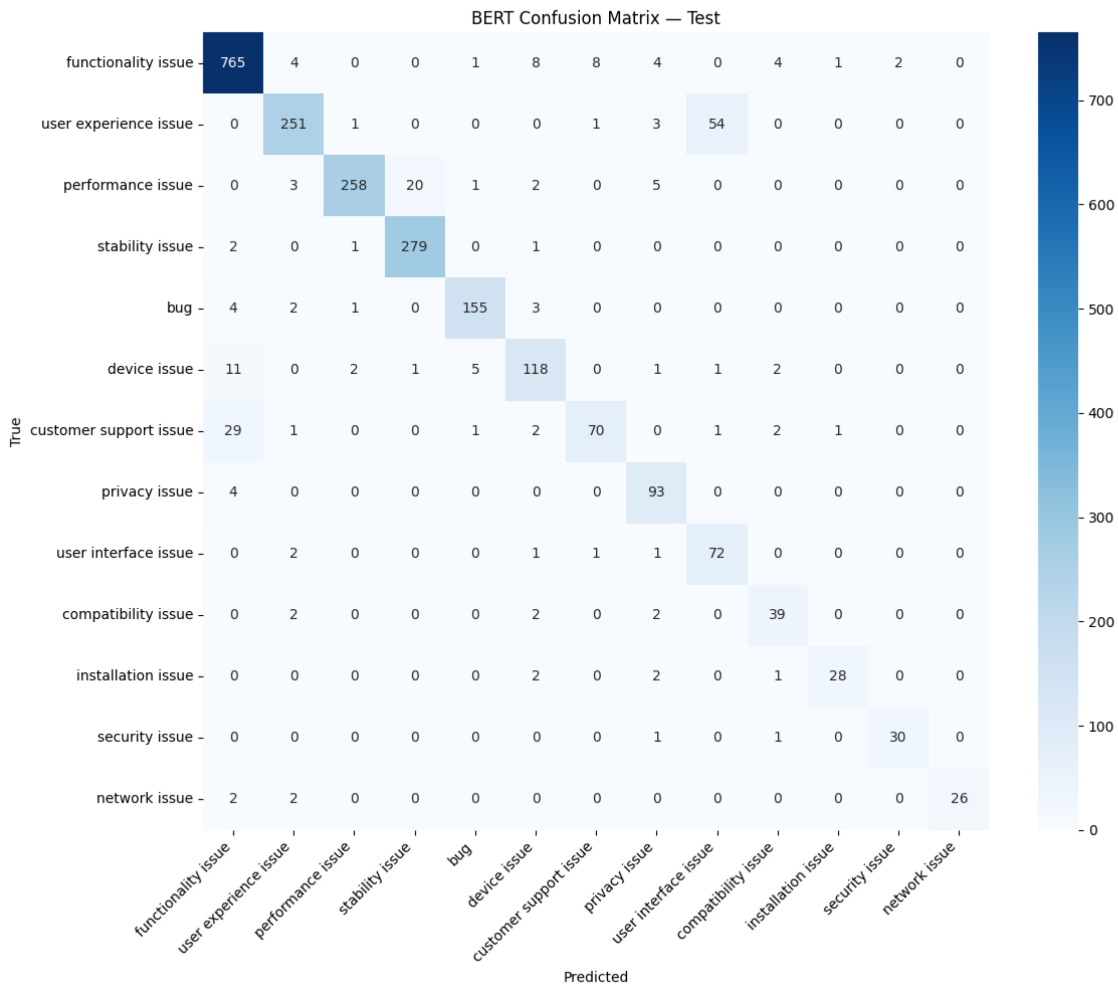
```
 Saved BERT model to: ./bert_best_final_model

==============================
FINAL TEST RESULTS
==============================
 FINAL TEST ACCURACY: 0.9077
 FINAL TEST LOSS: 0.5262

=== FINAL CLASSIFICATION REPORT ===
                        precision    recall  f1-score   support

   functionality issue     0.9364    0.9598    0.9480       797
 user experience issue     0.9401    0.8097    0.8700       310
     performance issue     0.9810    0.8927    0.9348       289
       stability issue     0.9300    0.9859    0.9571       283
                   bug     0.9509    0.9394    0.9451       165
          device issue     0.8489    0.8369    0.8429       141
customer support issue     0.8750    0.6542    0.7487       107
         privacy issue     0.8304    0.9588    0.8900        97
  user interface issue     0.5625    0.9351    0.7024        77
   compatibility issue     0.7959    0.8667    0.8298        45
    installation issue     0.9333    0.8485    0.8889        33
        security issue     0.9375    0.9375    0.9375        32
         network issue     1.0000    0.8667    0.9286        30

              accuracy                         0.9077      2406
             macro avg     0.8863    0.8840    0.8787      2406
          weighted avg     0.9165    0.9077    0.9086      2406
```




```
Test predictions saved to: bert_test_predictions.csv
CV results saved to: bert_cv_results.csv
Classification report saved to: bert_classification_report.csv
```